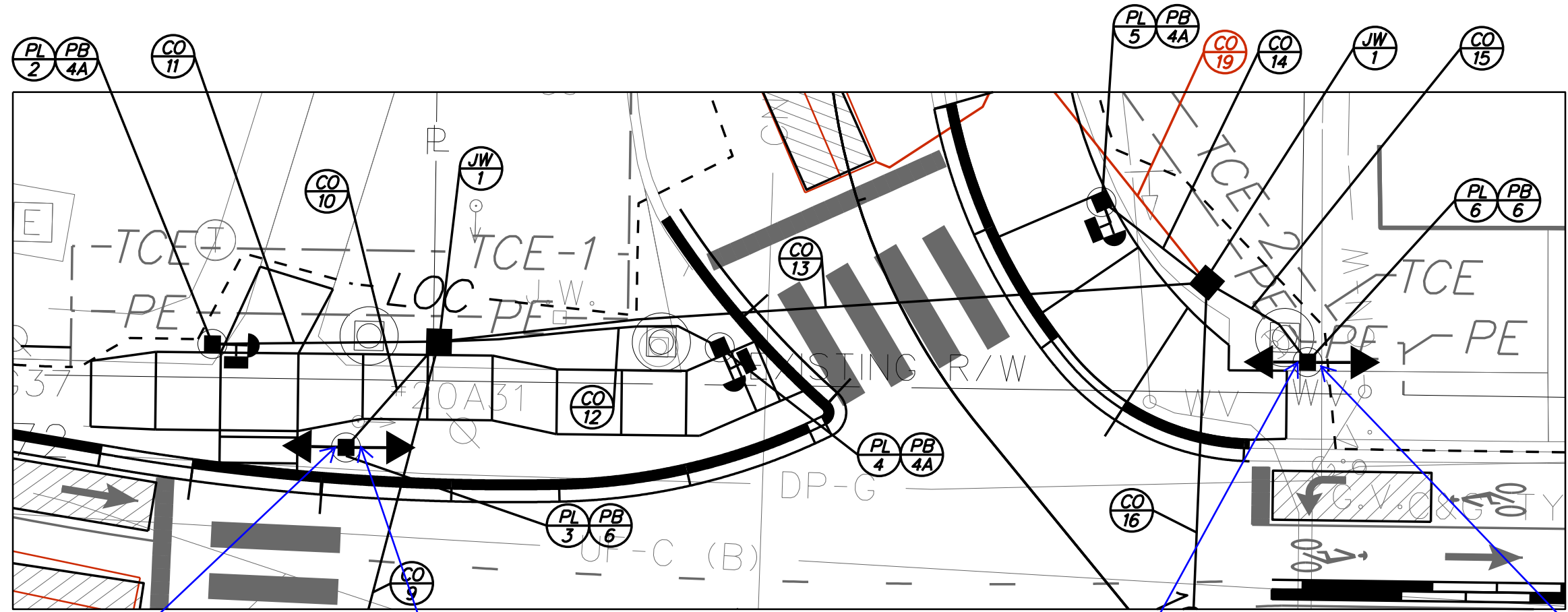
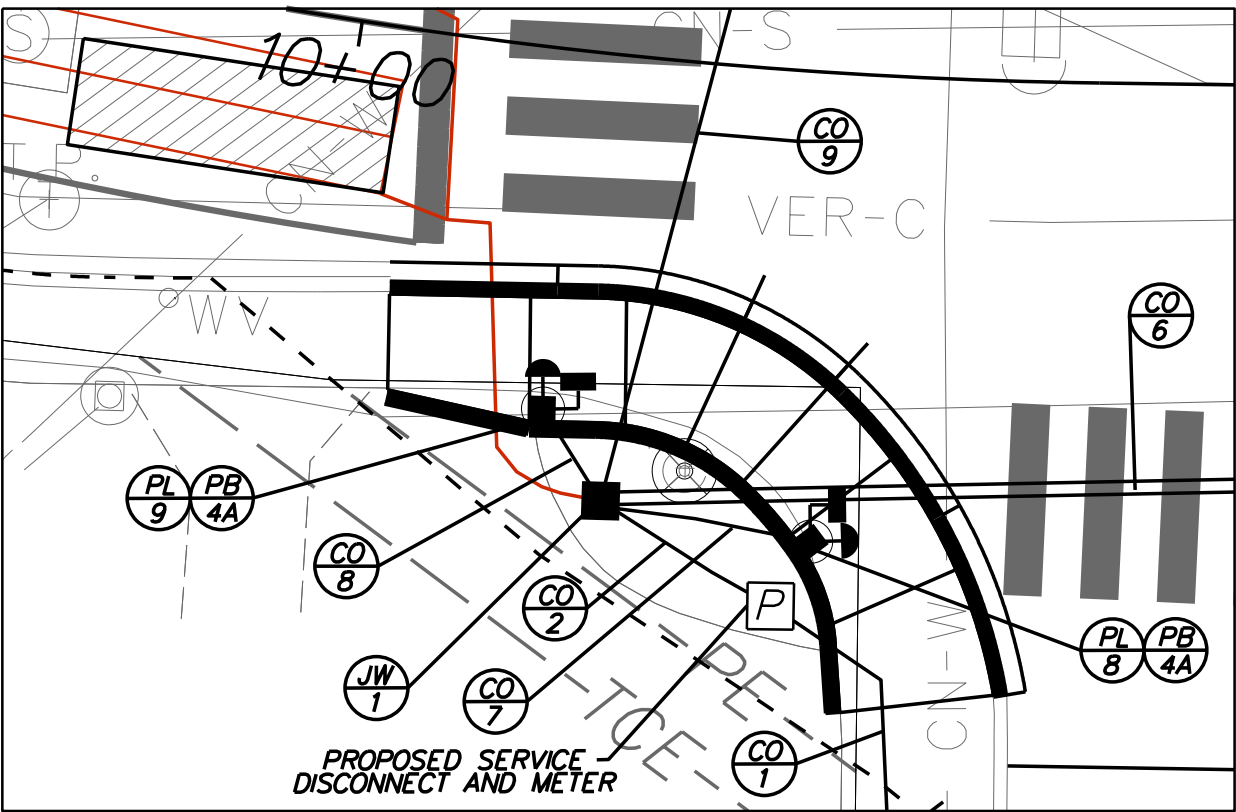


CONDUIT RUN SCHEDULE					
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
1	1 (GALV. STEEL)	2"	60'	B/T	(3)*8 THHN
2	1 (GALV. STEEL)	2"	10'	B/T	(1)2/*8 U.F. W/ GROUND
3	4	4"	20'	T	(1)4/*14, (6)5/*14, (4)9/*14, (1)16/*14, (3)4/*18, GRDSMART CAT5E CABLES, (4)1/*6 GRD, (6)2/*14
	1 (GALV. STEEL)	2"	20'	T	(1)2/*8 U.F. W/ GROUND
4	1	3"	10'	T	(1)4/*14, (1)16/*14, (3)4/*18, GRDSMART CAT5E CABLES, (1)1/*6 GRD
5	1	2.5"	15'	T	(1)5/*14, (1)1/*6 GRD
6	1 (SCH 80 HDPE)	4"	75'	B	(4)5/*14, (2)9/*14, (1)1/*6 GRD, (2)2/*14
	1 (GALV. STEEL)	2"	75'	B	(1)2/*8 U.F. W/ GROUND
7	1	2.5"	15'	O	(1)5/*14, (1)1/*6 GRD
8	1	2.5"	10'	O	(1)5/*14, (1)1/*6 GRD
9	1 (SCH 80 HDPE)	4"	55'	B	(2)5/*14, (2)9/*14, (1)1/*6 GRD
10	1	2.5"	15'	T	(2)9/*14, (1)1/*6 GRD
11	1	2.5"	20'	T	(1)5/*14, (1)1/*6 GRD
12	1	2.5"	25'	T	(1)5/*14, (1)1/*6 GRD
13	1 (SCH 80 HDPE)	4"	60'	B	EMPTY
14	1	2.5"	15'	T	(1)5/*14, (1)1/*6 GRD
15	1	2.5"	10'	T	(2)9/*14, (1)1/*6 GRD
16	1 (SCH 80 HDPE)	4"	55'	B	(1)5/*14, (2)9/*14, (1)1/*6 GRD, (1)2/*14
17	1	4"	10'	T	DELDOT FORCES
18	1	4"	25'	T	(3)2/*14, (1)1/*6 GRD
19	1	4"	25'	T	(1)2/*14, (1)1/*6 GRD

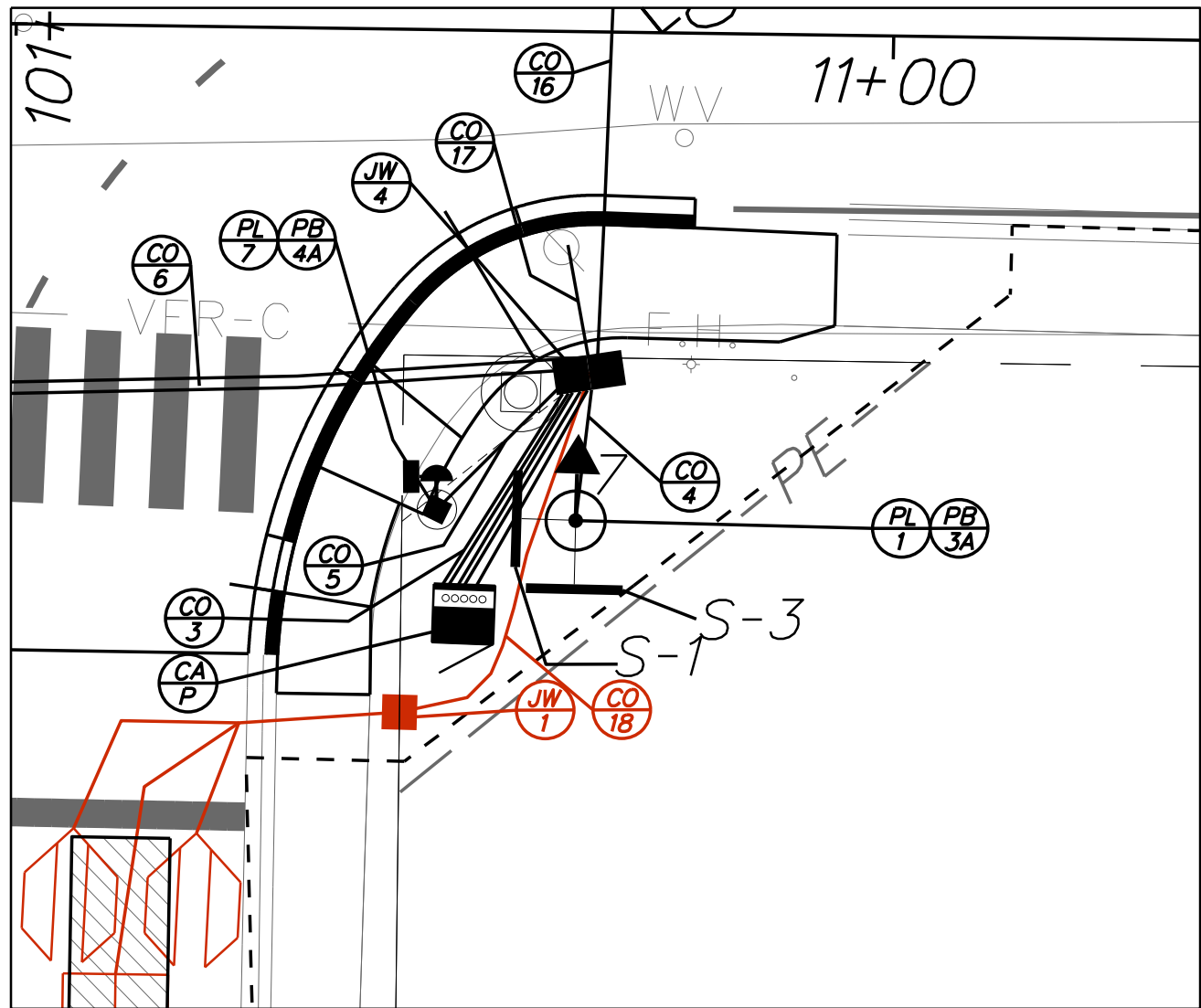
NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT



DETAIL 1 - NORTH SIDE
SCALE: 1" = 10'

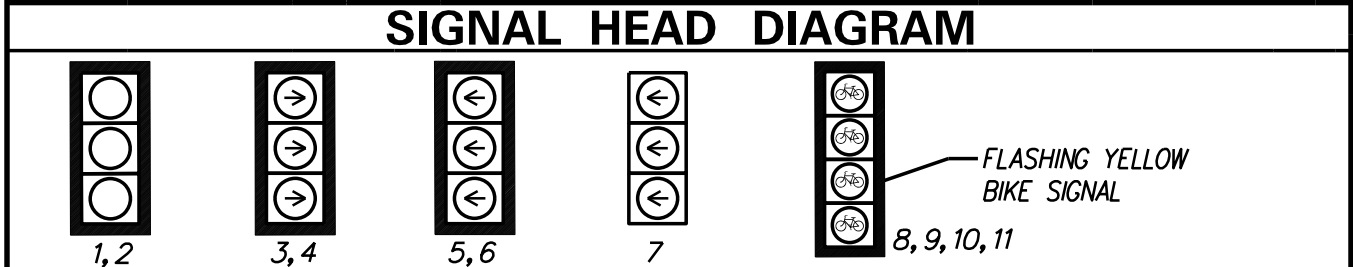
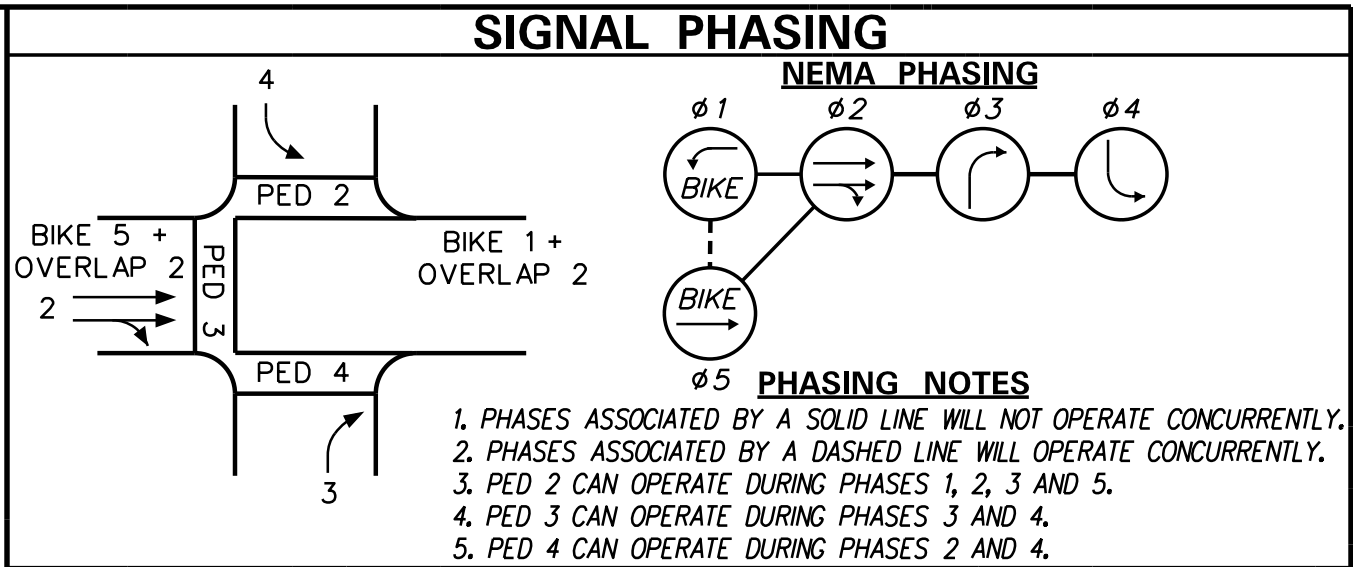


DETAIL 2 - SOUTHWEST CORNER
SCALE: 1" = 10'



DETAIL 3 - SOUTHEAST CORNER
SCALE: 1" = 10'

NOTE: FOR CLARITY OF THE DETAILS, MAST ARMS ARE NOT SHOWN.
ONLY THE POLE BASE LOCATION IS SHOWN ON THE DETAIL.



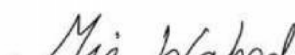
LEGEND

	ABANDON		EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)		PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)		EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)		PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)		EXISTING POLE IDENTIFIER (* OF POLE)
	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)		PROPOSED POLE IDENTIFIER (* OF POLE)
	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)		REMOVE BY CONTRACTOR
	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)		REMOVE BY OTHERS
	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)		REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	⬢	⬢
MAST ARM	⬢	⬢
MICROWAVE DETECTION	⬢	⬢
OPTICOM RECEIVER	⬢	⬢
OVERHEAD SIGNING	⬢	⬢
PEDESTRIAN POLE/BASE	⬢	⬢
PEDESTRIAN PUSHBUTTON	⬢	⬢
PEDESTRIAN SIGNAL HEAD	⬢	⬢
RIGHT-OF-WAY	---	---
SERVICE PEDESTAL	⬢	⬢
SIGNAL CABINET	⬢	⬢
SIGNAL HEAD	⬢	⬢
SIGNAL POLE/BASE	⬢	⬢
CCTV CAMERA	⬢	⬢
SPAN WIRE	XX	XX
UTILITY POLE	⬢	⬢
VIDEO DETECTION	⬢	⬢

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY		
JMT		
	07/20/2021	
THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.	DATE	SEAL
CONCURRENCE FOR INSTALLATION		
	7/28/2021	
CHIEF TRAFFIC-ENGINEER	DATE	

ADDENDA / REVISIONS		N.T.S.	DELAWARE AVENUE SEPARATED BIKEWAY- ORCHARD ROAD TO LIBRARY AVENUE	CONTRACT T201801501 COUNTY NEW CASTLE	PERMIT NO. N569T DESIGNED BY: J.SMITH CHECKED BY: M.WAHED	SIGNAL PLAN DELAWARE AVE AND ORCHARD RD	SECTION JMT SHEET NO. 170
ADDED LOOP DETECTORS	03-07-2022 (JDS)						

ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.
4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS, SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.
5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42 TO 48 INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.
6. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BOTH VEHICLES AND BICYCLES, AS WELL AS PROVIDE INTERSECTION BIKE COUNTS.
6. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BICYCLES ONLY AND WILL PROVIDE INTERSECTION BIKE COUNTS. LOOP DETECTORS WILL PROVIDE DETECTION FOR VEHICLES ONLY.
7. THE FISHEYE DETECTOR SHALL BE PLACED ON THE MAST ARM TO ACHIEVE OPTIMUM DETECTION PER MANUFACTURER'S SPECIFICATION. THE DESIRED DETECTION ZONES SHALL BE SET AND APPROVED BY DELDOT.
8. THE EXISTING CONTROLLER CABINET SHALL BE MAINTAINED UNTIL THE PROPOSED CONTROLLER CABINET IS OPERATIONAL.
9. REMOVAL OF POLE BASES SHALL BE PAID BY ITEM #211000.
10. R10-3E SIGNS SHALL BE INSTALLED ON POLES WITH ONE PEDESTRIAN SIGNAL HEAD. R10-3E-DE SIGNS SHALL BE INSTALLED ON POLES WITH TWO PEDESTRIAN SIGNAL HEADS.
11. BACKPLATES WILL BE INSTALLED FOR ALL SIGNAL HEADS. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATES.
12. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10'. AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.
13. THE APS SIGNAL SHALL BE MAINTAINED DURING CONSTRUCTION. THE CONTRACTOR SHALL INSTALL APS STATION WITH THE PUSHBUTTONS ON POLE #4, #5, #6 AND #7, ROTATED SO THAT THE FACE OF THE PUSHBUTTON IS PARALLEL WITH THE CROSSING.
14. DUE TO THE DEPTH OF THE WALL ALONG THE SIDEWALK ALONG SOUTH COLLEGE AVENUE, CONDUIT #1 WILL NEED TO BE HAND TRENCHED IN THE SPACE BETWEEN THE SIDEWALK AND THE WALL. ELBOW CONNECTORS WILL NEED TO BE USED TO BEND THE CONDUIT AT THE POWER SOURCE AND AROUND THE NORTHERN END OF THE WALL TO THE SERVICE DISCONNECT PEDESTAL.
15. INSTALL TYPE 4 ANCHOR BOLTS ON TYPE 6 POLE BASE FOUNDATIONS.
16. INSTALL THE TYPE P CABINET ON A 58"x58" TYPE M FOUNDATION.
17. CONTACT NICK MOGLE (302-760-2589) BEFORE POURING THE CABINET BASE TO DETERMINE THE CONDUIT SIZE NEEDED TO CONNECT INTO THE CABINET.
18. PLACE PEDESTRIAN SIGNAL HEAD FOR PEDESTRIAN SCRAMBLE PHASE ABOVE THE OTHER PEDESTRIAN SIGNAL HEADS.
19. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO ANY CHANGES TO THE TRAFFIC SIGNAL OPERATION. THE CONTRACTOR SHALL HAVE CORRESPONDING SIGNAL TIMESHEET ISSUED VIA DELDOT TRAFFIC/TMC PRIOR TO MODIFYING THE TRAFFIC SIGNAL OPERATION.
20. COORDINATE WITH BHADRESH PATEL (302-366-7000) AT THE CITY OF NEWARK TO ADJUST THE SECONDARY ELECTRICAL CABLES PRIOR TO INSTALLING MAST ARM #1.
21. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM #837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.
22. INSTALL ALL SIGNAL POLE AND PEDESTAL POLE FOUNDATIONS WITHIN OR ADJACENT TO BRICK SIDEWALK FLUSH WITH FINAL PROPOSED BRICK SIDEWALK GRADE.

PREPARED BY
JMT



James D. Smith Jr.

04/04/2022

THIS SEAL APPLIES TO ONLY THIS SHEET

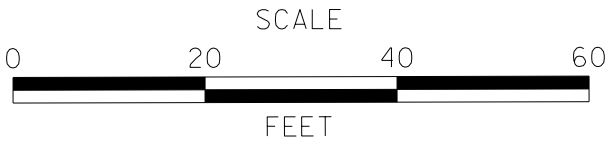
DATE

SEAL

MAST ARM SCHEDULE			
NO.	LENGTH	NO. OF SIGNAL HEADS	S. F. OF SIGNAGE
1	40' MAST ARM	1 - 3 SECTION, 1 - 5 SECTION	23.25 SF
2	35' MAST ARM	2 - 3 SECTION	15.75 SF
3	30' MAST ARM	1 - 3 SECTION, 1 - 5 SECTION	17.25 SF

ADDENDA / REVISIONS

ADDED POLE BASE NOTES	JDS 12/09/2021
ADDED LOOP DETECTORS AND REVISED POWER SOURCE	04-04-2022 (JDS)



DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT T201801501	PERMIT NO. N428T
COUNTY NEW CASTLE	DESIGNED BY: J.SMITH
	CHECKED BY: M.WAHED

SIGNAL PLAN
DELAWARE AVE AND
S. COLLEGE AVE

SECTION JMT
SHEET NO. 175

SIGN DETAILS

ALL SIGNS ARE RIGID MOUNT

S College Ave

S College Ave

S-1, S-2
D3-1
66" x 18"
8" / 5" D-SERIES

E Delaware Ave

E Delaware Ave

S-8, S-9
D3-1
72" x 18"
8" / 5" D-SERIES

LEFT TURN
YIELD
ON FLASHING

S-3
R10-CUSTOM
30" x 36"

NO TURN
ON RED

S-4
R10-11
24" x 30"

S-5
R3-2
36" x 36"

NO TURN
ON RED
MON-FRI
8AM-6PM

S-6
R10-11-CUSTOM
30" x 36"

START CROSSING
Watch For Vehicles

DONT START
Flash Crossing
If Shorted
TIME REMAINING
To Finish Crossing

DONT CROSS

PUSH BUTTON
TO CROSS
Delaware Avenue

S-10, S-11
R10-31
9" x 15"

START CROSSING
Watch For Vehicles

DONT START
Flash Crossing
If Shorted
TIME REMAINING
To Finish Crossing

DONT CROSS

PUSH BUTTON
TO CROSS
S College Avenue

S-12, S-13
R10-31
9" x 15"

START CROSSING
Watch For Vehicles

DONT START
Flash Crossing
If Shorted
TIME REMAINING
To Finish Crossing

DONT CROSS

PUSH BUTTON
TO CROSS
Delaware Avenue

S-14, S-15
R10-31
9" x 15"

START CROSSING
Watch For Vehicles

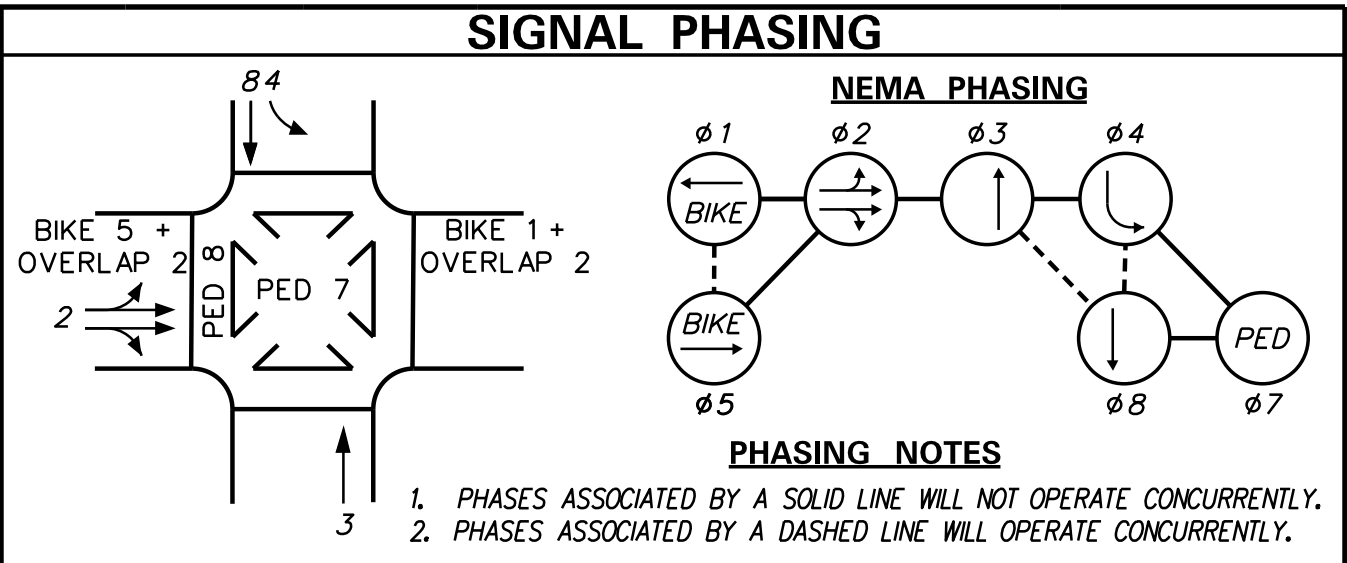
DONT START
Flash Crossing
If Shorted
TIME REMAINING
To Finish Crossing

DONT CROSS

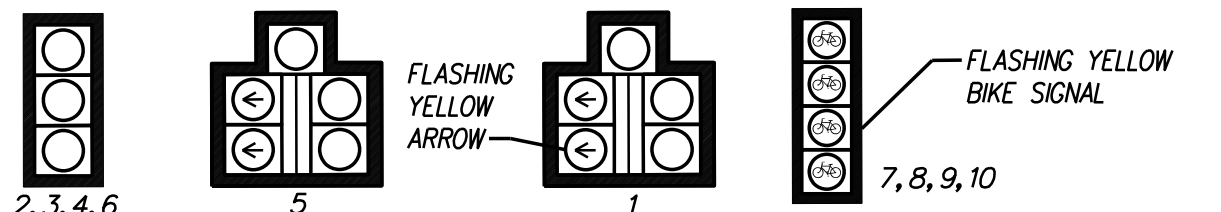
PUSH BUTTON
TO CROSS
Delaware Avenue

S-16, S-17
R10-31
9" x 15"

ULTIMATE



SIGNAL HEAD DIAGRAM



LEGEND

(AB)	ABANDON	(OH)	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	(OH)	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CX)	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	(PB)	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PB)	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PL)	EXISTING POLE IDENTIFIER (* OF POLE)
(JW)	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(PL)	PROPOSED POLE IDENTIFIER (* OF POLE)
(JW)	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(RM)	REMOVE BY CONTRACTOR
(MA)	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY OTHERS
(MA)	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	◊	◊
MAST ARM	➤	➤
MICROWAVE DETECTION	◄	◄
OPTICOM RECEIVER	○	○
OVERHEAD SIGNING	+	+
PEDESTRIAN POLE/BASE	⊙	⊙
PEDESTRIAN PUSHBUTTON	→	→
PEDESTRIAN SIGNAL HEAD	→	→
RIGHT-OF-WAY	---	---
SERVICE PEDESTAL	⌈	⌈
SIGNAL CABINET	⌈	⌈
SIGNAL HEAD	➤	➤
SIGNAL POLE/BASE	⊙	⊙
CCTV CAMERA	◄	◄
SPAN WIRE	XX	XX
UTILITY POLE	⊙	⊙
VIDEO DETECTION	◄	◄

GENERAL SIGNAL NOTES

1. ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
2. ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY

IMT



Mir A. Wahed

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS
BEARING THE "JMT" SECTION DESIGNATION.

SEAL

CONCURRENCE FOR INSTALLATION

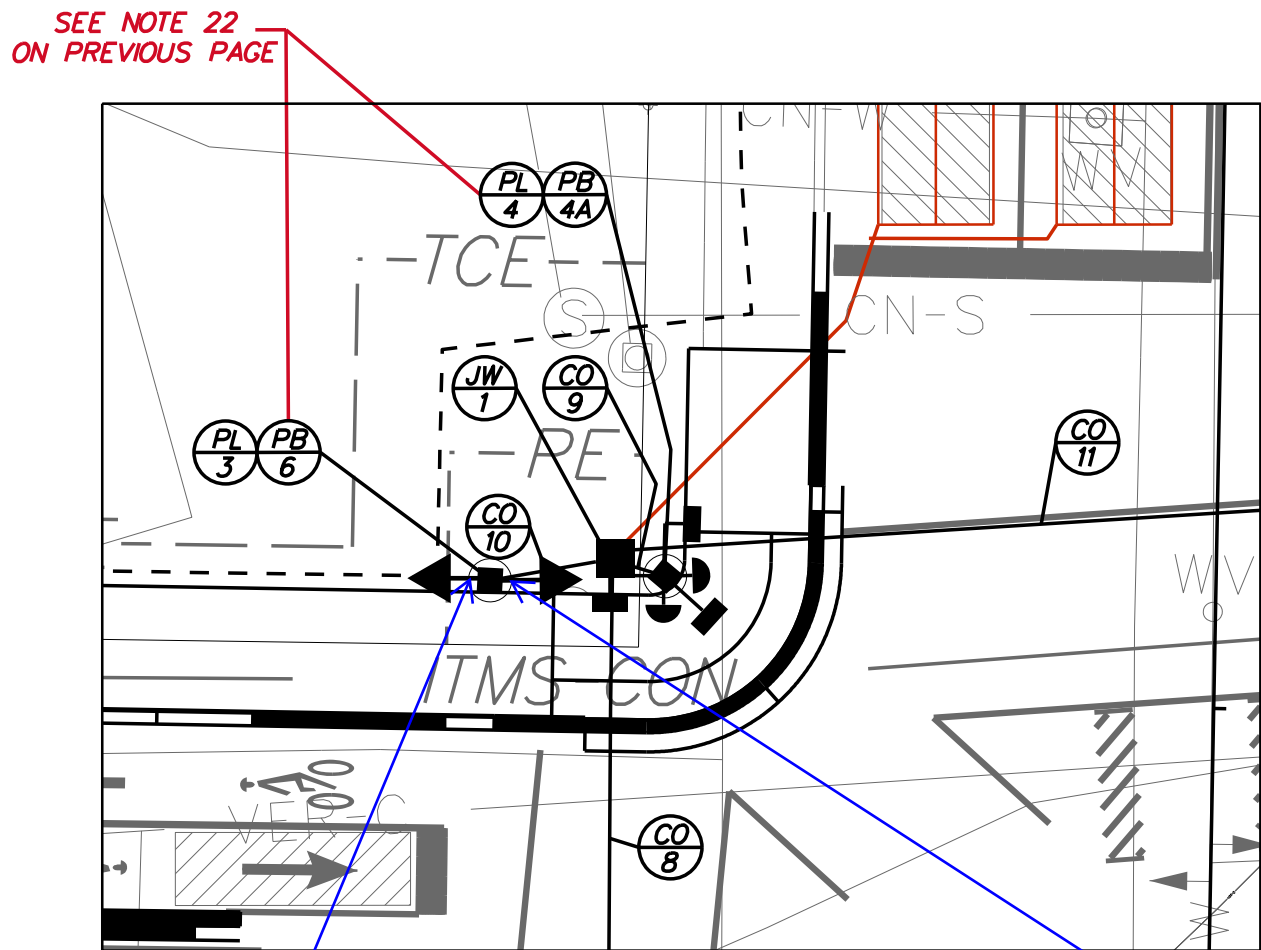
CHIEF TRAFFIC ENGINEER

7/28/2021

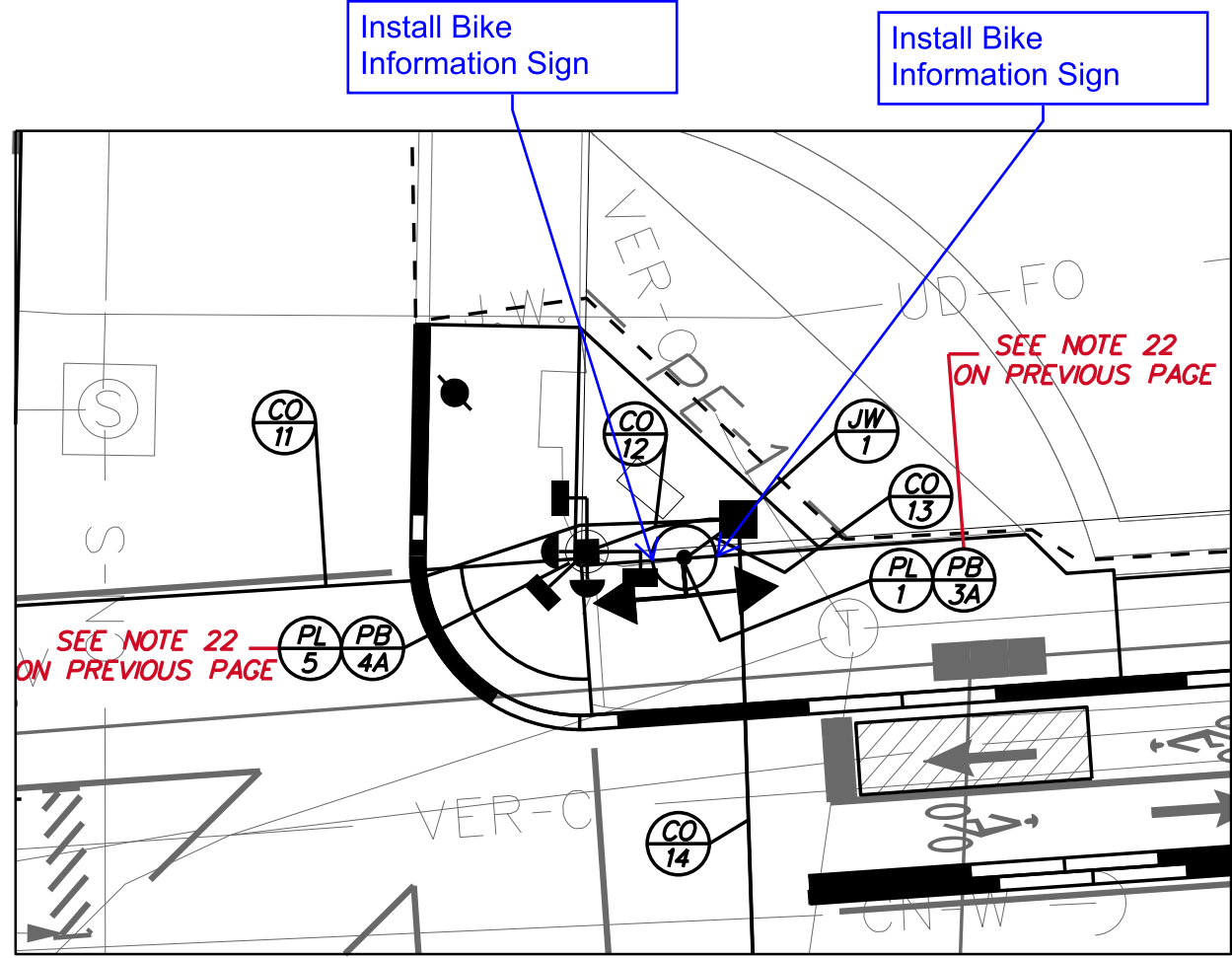
DATE

CONDUIT RUN SCHEDULE						
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE	
1	1 (GALV. STEEL)	2"	115' 40"	*T	3 (#) *8 THHN	
2	1 (GALV. STEEL)	2"	10'	T	(1) 2/#8 U.F. W/GROUND	
3	4	4"	5'	T	(1) 25/#14, (5) 9/#14, (1) 16/#14, (3) 4/#18, GRIDSMART CAT5E CABLE, (4) 1/#6 GRD, (6) 2/#14	
4	1	2.5"	10'	T	(3) 5/#14, (1) 1/#6 GRD	
5	1 (SCH 80 HDPE)	4"	70'	B	(6) 5/#14, (3) 9/#14, (1) 4/#18, (1) 1/#6 GRD, (4) 2/#14	
6	1	3"	10'	O	(1) 9/#14, (1) 4/#18, (1) 1/#6 GRD	
7	1	2.5"	20'	O	(3) 5/#14, (1) 1/#6 GRD	
8	1 (SCH 80 HDPE)	4"	80'	B	(3) 5/#14, (2) 9/#14, (1) 1/#6 GRD, (2) 2/#14	
9	1	2.5"	5'	O	(5) 5/#14, (1) 1/#6 GRD	
10	1	2.5"	10'	O	(2) 9/#14, (1) 1/#6 GRD	
11	1 (SCH 80 HDPE)	4"	75'	B	EMPTY	
12	1	2.5"	10'	O	(3) 5/#14, (1) 1/#6 GRD	
13	1	3"	5'	O	(2) 9/#14, (1) 16/#14, (2) 4/#18, GRIDSMART CAT5E CABLE, (1) 1/#6 GRD	
14	1 (SCH 80 HDPE)	4"	75'	B	(3) 5/#14, (2) 9/#14, (1) 16/#14, (2) 4/#18, GRIDSMART CAT5E CABLE, (1) 1/#6 GRD	
15	1	4"	20'	O	(2) 2/#14, (1) 1/#6 GRD	

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT
* - HAND TRENCHED

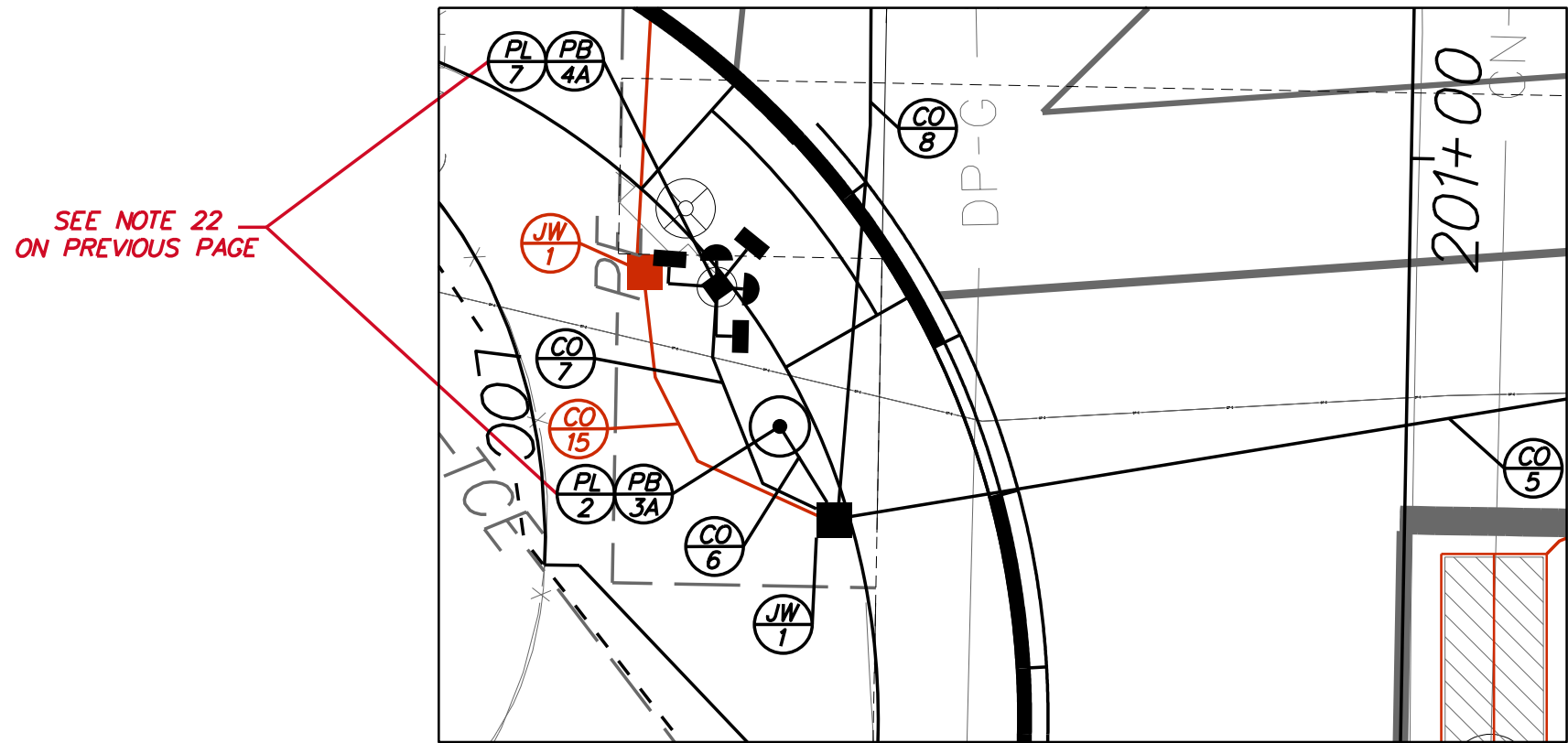


DETAIL 1 - NORTHWEST CORNER
SCALE: 1"=10'



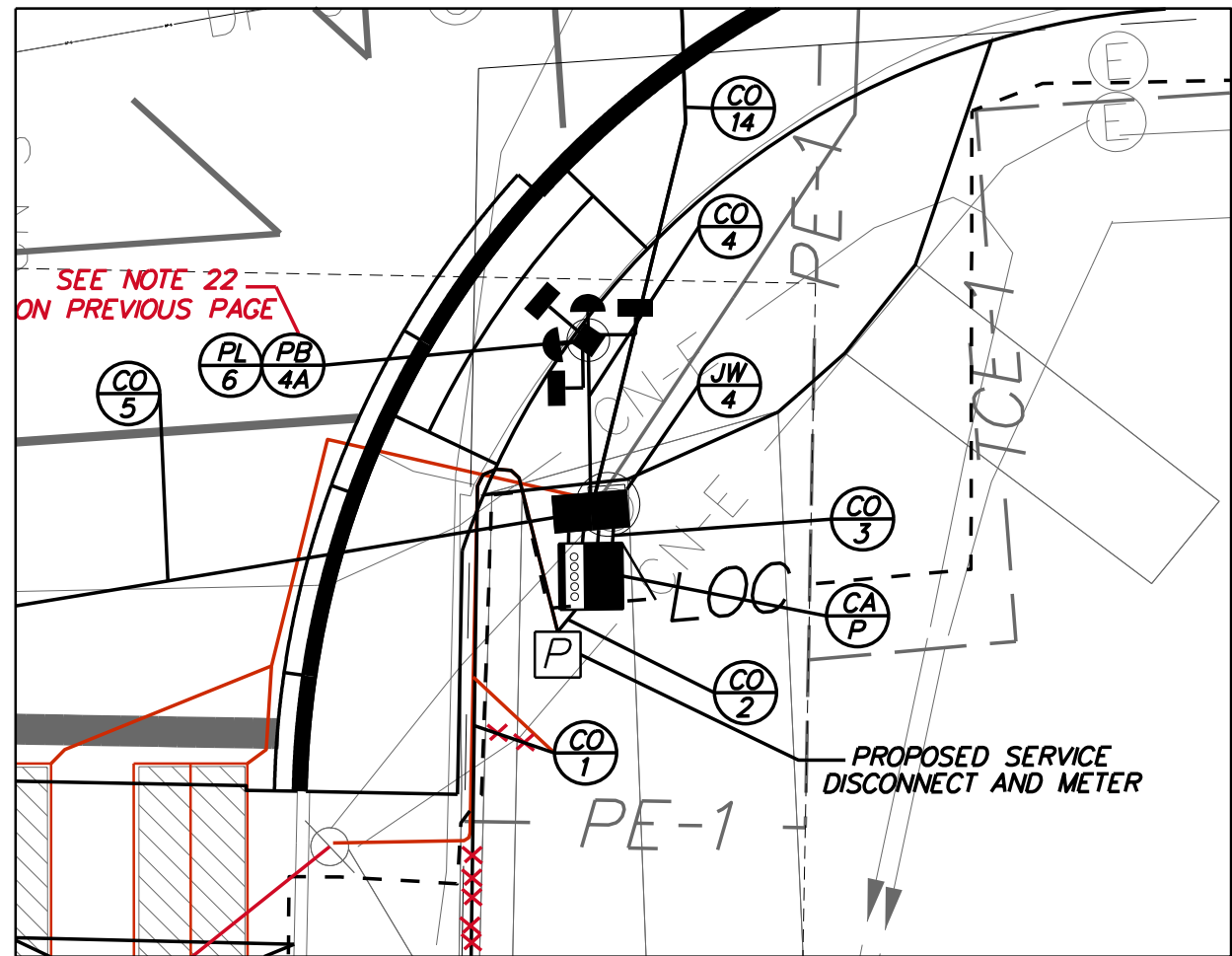
DETAIL 2 - NORTHEAST CORNER
SCALE: 1"=10'

NOTE: FOR CLARITY OF THE DETAILS, MAST ARMS ARE NOT SHOWN. ONLY THE POLE BASE LOCATION IS SHOWN ON THE DETAIL.



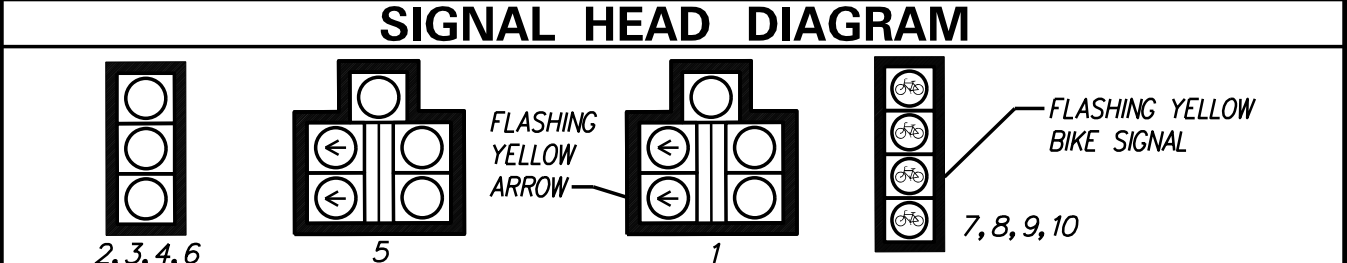
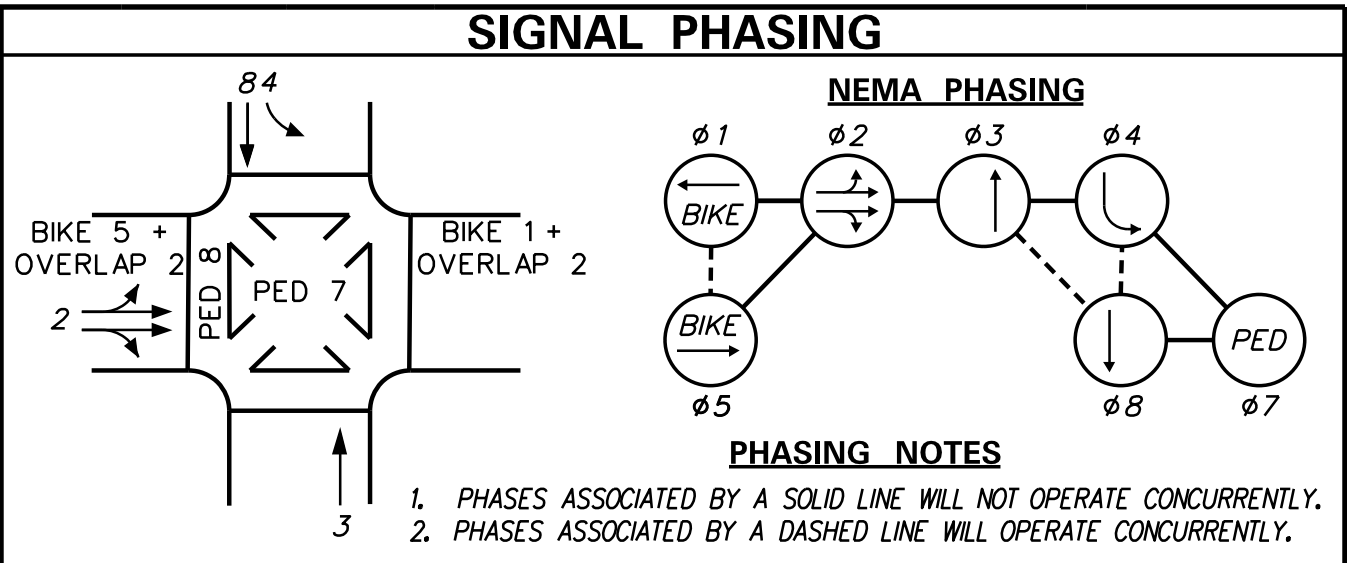
DETAIL 3 - SOUTHWEST CORNER
SCALE: 1"=10'

NOTE: FOR CLARITY OF THE DETAILS, MAST ARMS ARE NOT SHOWN. ONLY THE POLE BASE LOCATION IS SHOWN ON THE DETAIL.



DETAIL 4 - SOUTHEAST CORNER
SCALE: 1"=10'

PREPARED BY JMT	
	04/04/2022
THIS SEAL APPLIES TO ONLY THIS SHEET	DATE
SEAL	



LEGEND		
	ABANDON	 EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	 PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	 EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	 PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	 EXISTING POLE IDENTIFIER (* OF POLE)
	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	 PROPOSED POLE IDENTIFIER (* OF POLE)
	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	 REMOVE BY CONTRACTOR
	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	 REMOVE BY OTHERS
	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	 REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	⬢	⬢
MAST ARM	⬢	⬢
MICROWAVE DETECTION	⬢	⬢
OPTICOM RECEIVER	⬢	⬢
OVERHEAD SIGNING	⬢	⬢
PEDESTRIAN POLE/BASE	⬢	⬢
PEDESTRIAN PUSHBUTTON	⬢	⬢
PEDESTRIAN SIGNAL HEAD	⬢	⬢
RIGHT-OF-WAY	---	R/W
SERVICE PEDESTAL	⬢	⬢
SIGNAL CABINET	⬢	⬢
SIGNAL HEAD	⬢	⬢
SIGNAL POLE/BASE	⬢	⬢
CCTV CAMERA	⬢	⬢
SPAN WIRE	XX	XX
UTILITY POLE	⬢	⬢
VIDEO DETECTION	⬢	⬢

GENERAL SIGNAL NOTES

1. ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.

2. ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY JMT	
	07/20/2021
THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.	DATE
SEAL	

CONCURRENCE FOR INSTALLATION	
	7/28/2021
CHIEF TRAFFIC ENGINEER	DATE

ADDENDA / REVISIONS	
ADDED POLE BASE NOTES	JDS 12/09/2021
ADDED LOOP DETECTORS AND REVISED POWER SOURCE	04-04-2022 (JDS)

N.T.S.

DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

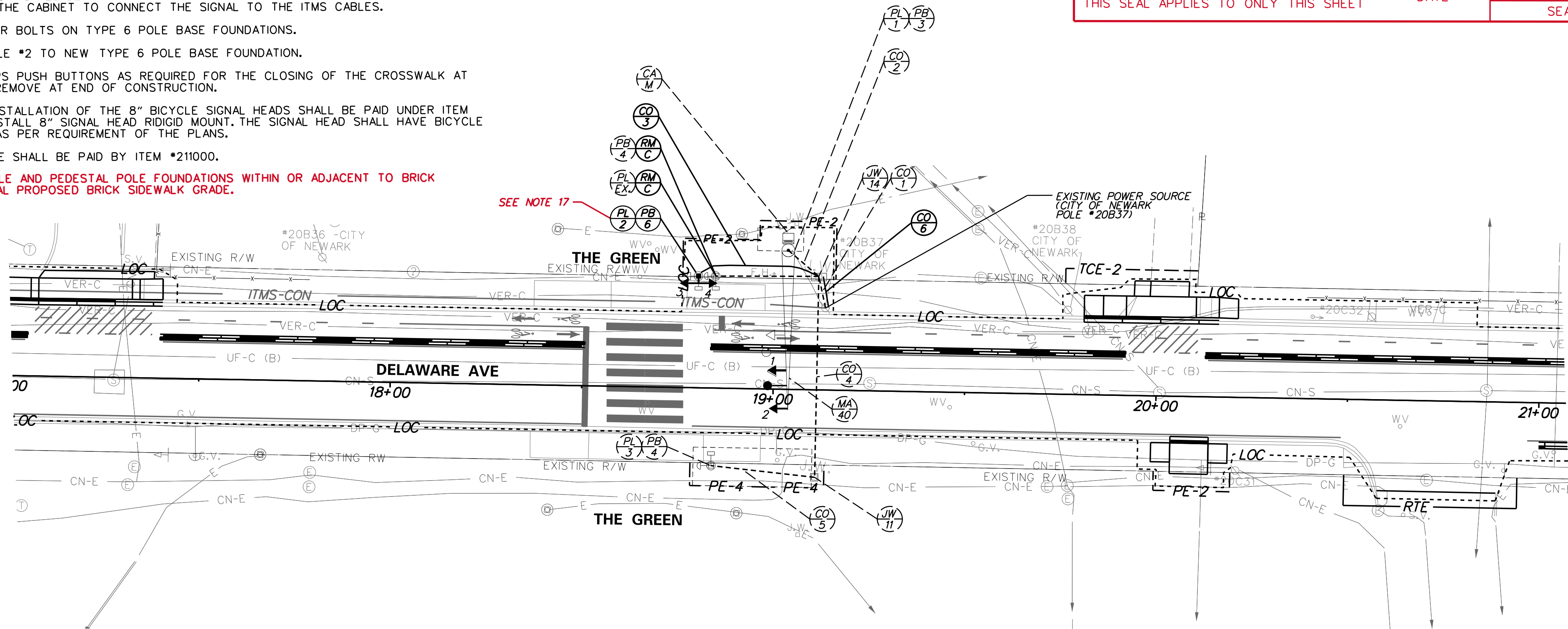
CONTRACT	PERMIT NO.	N428T
T201801501	DESIGNED BY: J.SMITH	
COUNTY	CHECKED BY: M.WAHED	
NEW CASTLE		

SIGNAL PLAN
DELAWARE AVE AND
S. COLLEGE AVE

SECTION
JMT
SHEET NO.
176

ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.
4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.
5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42 TO 48 INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.
6. SLIDE EXISTING SIGNAL HEAD #1 AND #2 AND OPTICOM DETECTOR AS SHOWN ON THE PLANS.
7. MAINTAIN THE EXISTING BACKPLATES ON SIGNAL HEAD #1 AND #2. INSTALL BACKPLATES FOR BICYCLE SIGNAL HEAD #3 AND #4. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE NEW BACKPLATES.
8. ALL PEDESTRIAN POLES WITH PEDESTRIAN SIGNALS ARE 10' AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.
9. MAINTAIN THE EXISTING ELECTRICAL SERVICE CONSISTS OF A CITY OF NEWARK METER ON EXISTING POLE #20B37 WITH AN INLINE FUSE CONDULET.
10. AUDIBLE PEDESTRIAN PUSH BUTTONS WILL BE INSTALLED AT THIS INTERSECTION PRIOR TO DETOURING PEDESTRIANS FROM THE SOUTH COLLEGE AVENUE INTERSECTION. AFTER ALL PEDESTRIAN DETOURS FOR THE PROJECT ARE COMPLETED, THE AUDIBLE PEDESTRIAN PUSH BUTTONS SHALL BE REMOVED.
11. A SEPARATE ITMS CONDUIT SHALL BE RUN FROM UTILITY POLE #20B37 TO THE EXISTING TYPE 14 JUNCTION WELL NEXT TO THE CABINET TO CONNECT THE SIGNAL TO THE ITMS CABLES.
12. INSTALL TYPE 4 ANCHOR BOLTS ON TYPE 6 POLE BASE FOUNDATIONS.
13. RELOCATE EXISTING POLE #2 TO NEW TYPE 6 POLE BASE FOUNDATION.
14. INSTALL TEMPORARY APS PUSH BUTTONS AS REQUIRED FOR THE CLOSING OF THE CROSSWALK AT SOUTH COLLEGE AVENUE. REMOVE AT END OF CONSTRUCTION.
15. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM #837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIDIGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.
16. REMOVAL OF POLE BASE SHALL BE PAID BY ITEM #211000.
17. INSTALL ALL SIGNAL POLE AND PEDESTAL POLE FOUNDATIONS WITHIN OR ADJACENT TO BRICK SIDEWALK FLUSH WITH FINAL PROPOSED BRICK SIDEWALK GRADE.



CONDUIT RUN SCHEDULE					
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
#1	1	3"	25'	-	EX. [(1)2/#8 U.F. W/ GROUND]
#2	1	3"	5'	-	NEW [(2)5/#14], EX. [(2)5/#14, (1)1/#6 GRD]
3	1	2.5"	35'	-	NEW [(3)5/#14, (1)1/#6 GRD]
#4	1	4"	50'	-	EX. [(1)5/#14, (1)1/#6 GRD]
#5	1	2.5"	25'	-	EX. [(1)5/#14, (1)1/#6 GRD]
6	1	4"	10'	T	NEW [DELDOT FORCES]

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT
* EXISTING

MAST ARM SCHEDULE			
NO.	LENGTH	NO. OF SIGNAL HEADS	S.F. OF SIGNAGE
#1	40' MAST ARM	2 - 3 SECTION	0 SF

* EXISTING

POLE SCHEDULE			
NO.	TYPE	STATION	OFFSET
#1	40' MAST ARM W/ TYPE 3 BASE	19+04	34' LT
2	14' PEDESTAL POLE W/ TYPE 6 BASE	18+85	29' LT
#3	14' PEDESTAL POLE W/ TYPE 4 BASE	18+83	20' RT

* EXISTING

ULTIMATE



PREPARED BY
JMT

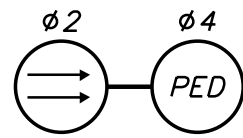
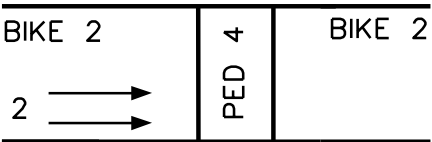
Mir A. Wahed
12/09/2021

DATE

SEAL

SIGNAL PHASING

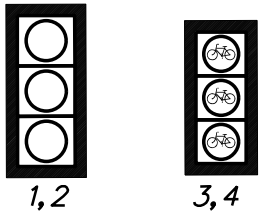
NEMA PHASING



PHASING NOTES

1. PHASES ASSOCIATED BY A SOLID LINE WILL NOT OPERATE CONCURRENTLY.
2. PHASES ASSOCIATED BY A DASHED LINE WILL OPERATE CONCURRENTLY.

SIGNAL HEAD DIAGRAM



LEGEND

(AB)	ABANDON	(OH)	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	(OH)	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	(PB)	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PB)	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PL)	EXISTING POLE IDENTIFIER (* OF POLE)
(JW)	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(PL)	PROPOSED POLE IDENTIFIER (* OF POLE)
(JW)	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(RM)	REMOVE BY CONTRACTOR
(MA)	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY OTHERS
(MA)	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	(TC)	REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	◇	◇
MAST ARM	▶	▶
MICROWAVE DETECTION	▶	▶
OPTICOM RECEIVER	○	○
OVERHEAD SIGNING	+	+
PEDESTRIAN POLE/BASE	⊙	⊙
PEDESTRIAN PUSHBUTTON	→	→
PEDESTRIAN SIGNAL HEAD	→	→
RIGHT-OF-WAY	---	R/W
SERVICE PEDESTAL	□	□
SIGNAL CABINET	□	□
SIGNAL HEAD	→	→
SIGNAL POLE/BASE	⊙	⊙
CCTV CAMERA	◀	◀
SPAN WIRE	XX	XX
UTILITY POLE	○	○
VIDEO DETECTION	◀	◀

GENERAL SIGNAL NOTES

1. ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
2. ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY

JMT



Mir A. Wahed

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

DATE

SEAL

CONCURRENCE FOR INSTALLATION

Chief Traffic Engineer

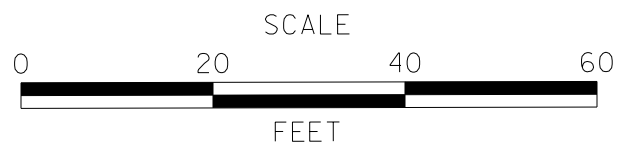
7/28/2021

DATE

ADDENDA / REVISIONS

ADDED POLE BASE NOTES

JDS 12/09/2021



DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT	PERMIT NO.	N831
T201801501	DESIGNED BY: J.SMITH	
COUNTY	CHECKED BY: M.WAHED	
NEW CASTLE		

SIGNAL PLAN
DELAWARE AVE AND
THE GREEN

SECTION	JMT
SHEET NO.	177

ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.

4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.

5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP OR SIDEWALK, IN ACCORDANCE WITH CURRENT WHEN POSSIBLE. WHERE THERE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42" TO 48" INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.

6. FISHEYE VIDEO DETECTOR SHALL BE PLACED ON THE MAST ARM TO ACHIEVE OPTIMUM DETECTION PER MANUFACTURER'S SPECIFICATION. THE DESIRED DETECTION ZONES SHALL BE SET AND APPROVED BY DELDOT.

7. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR VEHICLES AND BICYCLES, AS WELL AS PROVIDE INTERSECTION BIKE COUNTS.

7. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BICYCLES ONLY AND WILL PROVIDE INTERSECTION BIKE COUNTS. LOOP DETECTORS WILL PROVIDE DETECTION FOR VEHICLES ONLY.

8. REMOVAL OF POLE BASES SHALL BE PAID BY ITEM #211000.

9. R10-3E SIGNS SHALL BE INSTALLED ON POLES WITH ONE PEDESTRIAN SIGNAL HEAD. R10-3E-DE SIGNS SHALL BE INSTALLED ON POLES WITH TWO PEDESTRIAN SIGNAL HEADS. INSTALL R10-3E(L) SIGNS ON POLE #2 AND #3.

10. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10' AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAILS.

11. BACKPLATES WILL BE INSTALLED FOR SIGNAL HEADS #7-10. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATES.

12. AN ITMS CONDUIT SHALL RUN FROM UTILITY POLE #20D311 TO THE TYPE 1 JUNCTION WELL DIRECTLY NEXT TO THE POLE. A SEPARATE ITMS CONDUIT SHALL RUN FROM THE TYPE 1 JUNCTION WELL TO THE TYPE 4 JUNCTION WELL NEXT TO THE CABINET TO CONNECT THE SIGNAL TO THE ITMS CABLES.

13. INSTALL TYPE 4 ANCHOR BOLTS ON TYPE 6 POLE BASE FOUNDATIONS.

ADDITIONAL GENERAL SIGNAL NOTES

14. FOR ANY PEDESTAL POLE WITH BIKE AND PEDESTRIAN SIGNALS, BIKE SIGNALS SHOULD BE INSTALLED HIGH ENOUGH FOR A PEDESTRIAN SIGNAL AND PUSHBUTTON TO BE INSTALLED BELOW THE BIKE SIGNALS.

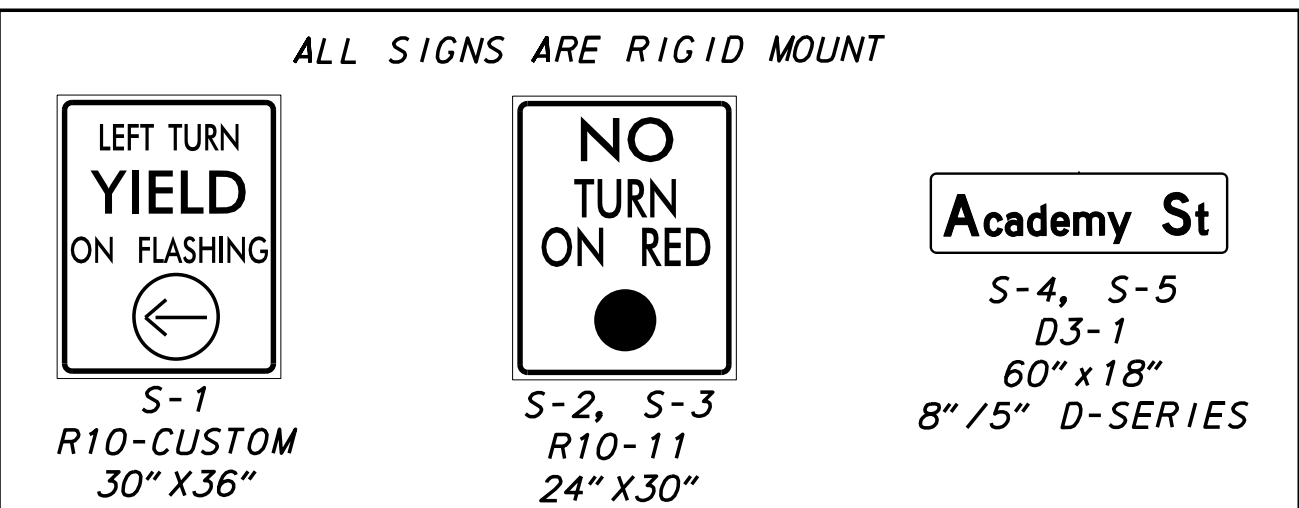
15. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO ANY CHANGES TO THE TRAFFIC SIGNAL OPERATION. THE CONTRACTOR SHALL HAVE CORRESPONDING SIGNAL TIMESHEET ISSUED VIA DELDOT TRAFFIC/TMC PRIOR TO MODIFYING THE TRAFFIC SIGNAL OPERATION.

16. COORDINATE WITH BHADRESH PATEL (302-366-7000) AT THE CITY OF NEWARK TO ADJUST THE SECONDARY ELECTRICAL CABLES PRIOR TO INSTSTALLING THE MAST ARM.

17. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM #837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIDGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.

18. INSTALL ALL SIGNAL POLE AND PEDESTAL POLE FOUNDATIONS WITHIN OR ADJACENT TO BRICK SIDEWALK FLUSH WITH FINAL PROPOSED BRICK SIDEWALK GRADE.

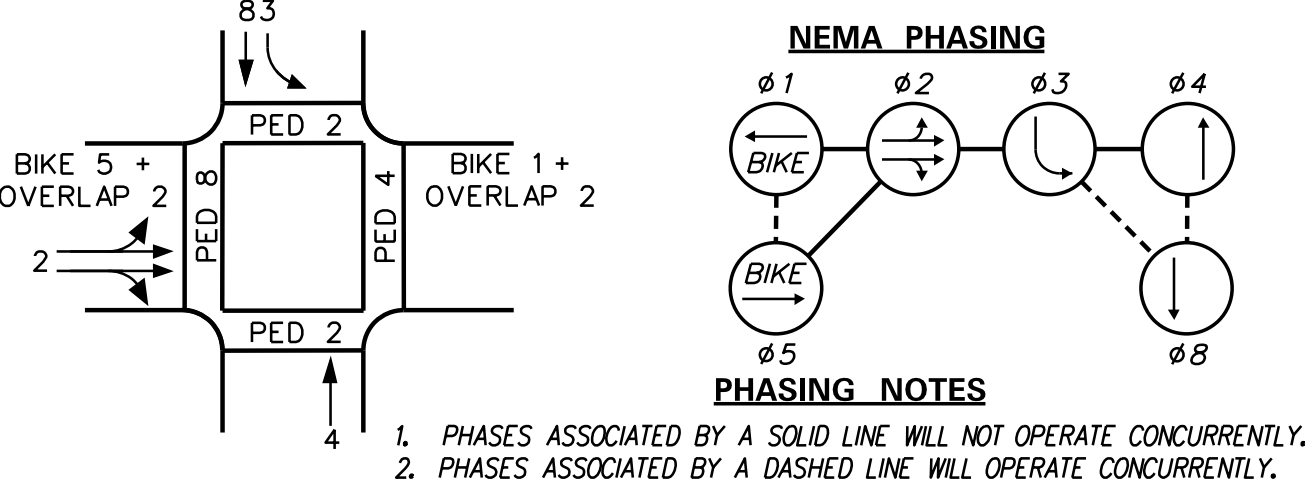
OVERHEAD SIGN DETAILS



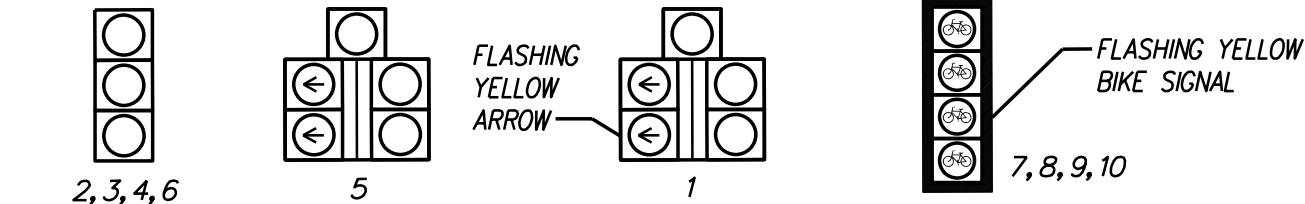
ULTIMATE



SIGNAL PHASING



SIGNAL HEAD DIAGRAM



LEGEND

(AB)	ABANDON	(OH)	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	(OH)	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	(PB)	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PB)	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PL)	EXISTING POLE IDENTIFIER (* OF POLE)
(JW)	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(PL)	PROPOSED POLE IDENTIFIER (* OF POLE)
(JW)	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(RM)	REMOVE BY CONTRACTOR
(MA)	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY OTHERS
(MA)	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	◊	◊
MAST ARM	▶	▶
MICROWAVE DETECTION	◄	◄
OPTICOM RECEIVER	○	○
OVERHEAD SIGNING	+	+
PEDESTRIAN POLE/BASE	⊙	⊙
PEDESTRIAN PUSHBUTTON	→	→
PEDESTRIAN SIGNAL HEAD	↗	↗
RIGHT-OF-WAY	---	R/W
SERVICE PEDESTAL	□	□
SIGNAL CABINET	□	■
SIGNAL HEAD	→	→
SIGNAL POLE/BASE	⊙	⊙
CCTV CAMERA	◄	◄
SPAN WIRE	XX	XX
UTILITY POLE	⊙	⊙
VIDEO DETECTION	◄	◄

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY

JMT



Amir A. Wahed

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

SEAL

CONCURRENCE FOR INSTALLATION

Chief Traffic Engineer

7/28/2021

DATE

DELAWARE AVENUE SEPARATED BIKEWAY- ORCHARD ROAD TO LIBRARY AVENUE

CONTRACT

T201801501

COUNTY

NEW CASTLE

PERMIT NO.

N429T

DESIGNED BY: J.SMITH

CHECKED BY: M.WAHED

SIGNAL PLAN DELAWARE AVENUE @ ACADEMY STREET

SECTION

JMT

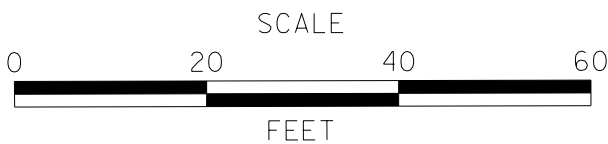
SHEET NO.

180

ADDENDA / REVISIONS

ADDED POLE BASE NOTES JDS 12/09/2021

ADDED LOOP DETECTORS 03-07-2022 (JDS)



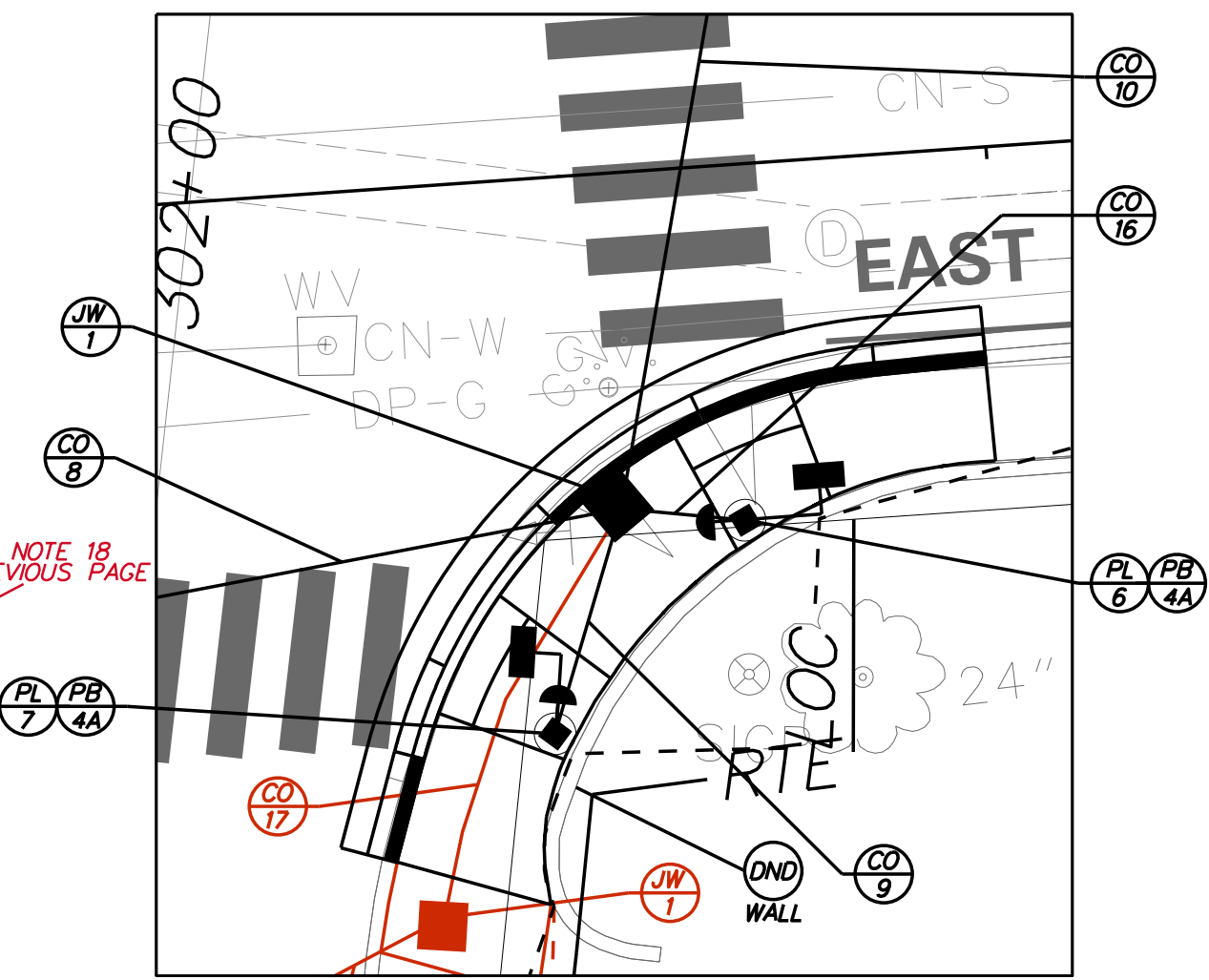
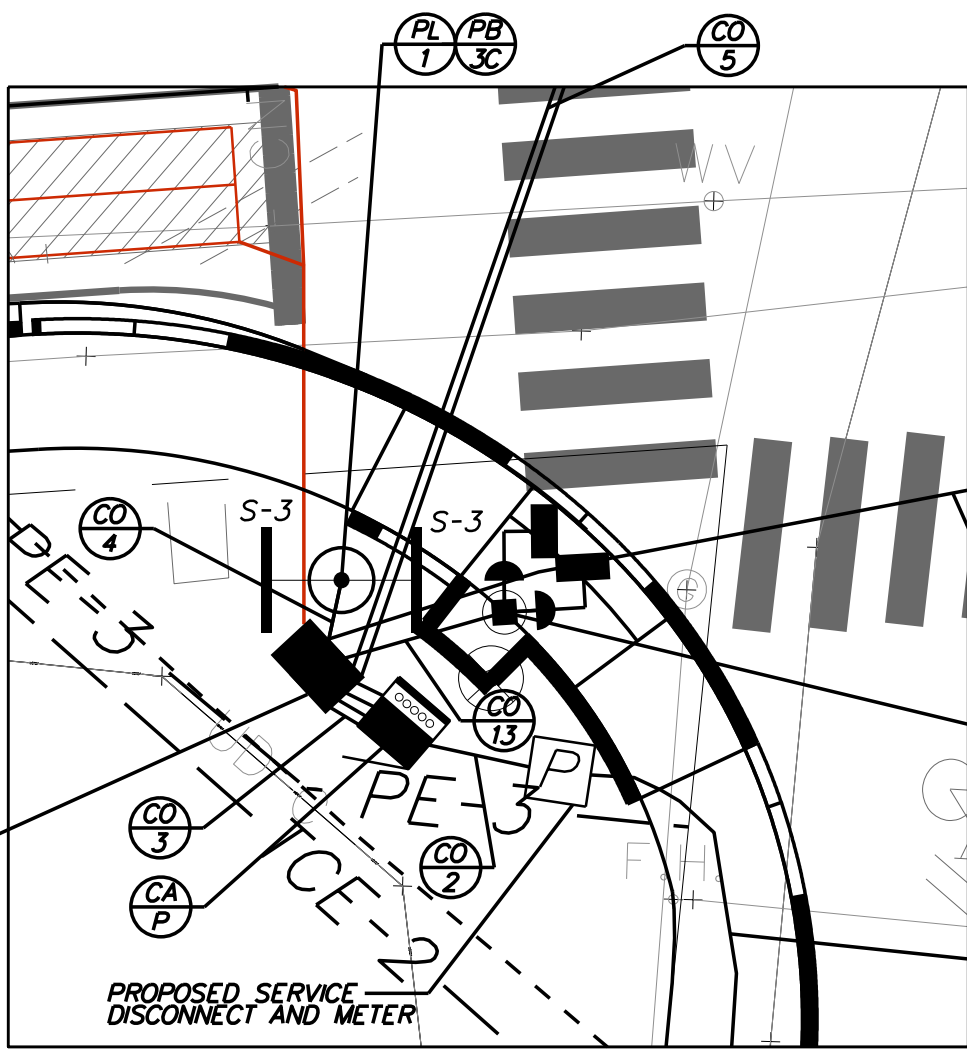
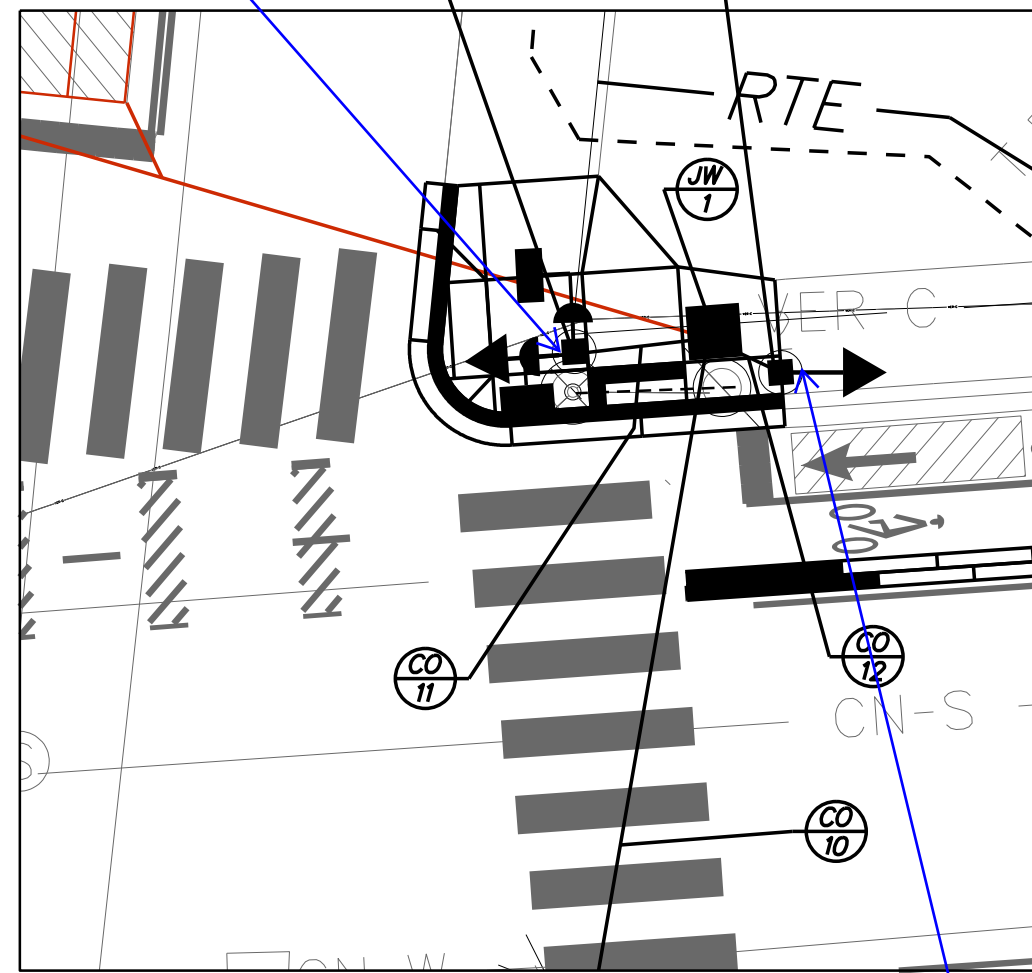
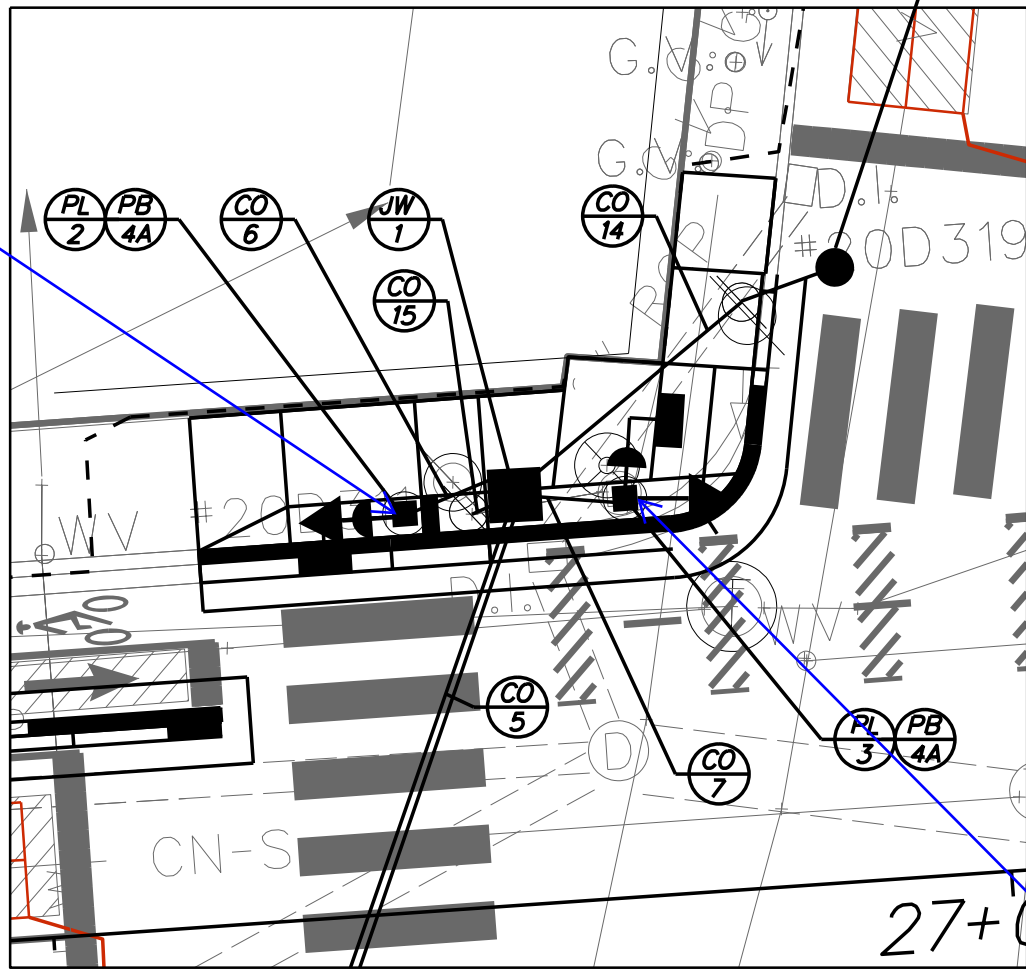
ULTIMATE

Install Bike Information Sign

Install Bike Information Sign

Install Bike Information Sign

Install Bike Information Sign



PREPARED BY JMT <i>James D. Smith Jr.</i>		03/07/2021	
THIS SEAL APPLIES TO ONLY THIS SHEET		DATE	

CONDUIT RUN SCHEDULE						
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE	
1	1 (GALV. STEEL)	2"	70'	T	(3)#8 THHN	
2	1 (GALV. STEEL)	2"	10'	T	(1)2/#8 U.F. W/GROUND	
3	4	4"	10'	T	(2)16/#14, (4)4/#18, GRIDSMART CAT5E CABLE, (4)9/#14, (8)5/#14 (4)1/#6 GRD, (6)2/#14	
4	1	3"	10'	T	(2)16/#14, (3)4/#18, GRIDSMART CAT5E CABLE, (1)1/#6 GRD	
5	1 (SCHD 80 HDPE)	4"	60'	B	(2)9/#14, (2)5/#14 (1)4/#18, (1)1/#6 GRD	
6	1	2.5"	10'	T	DELDOT FORCES	
7	1	2.5"	10'	T	(1)9/#14, (1)5/#14, (1)1/#6 GRD	
8	1 (SCHD 80 HDPE)	4"	80'	B	(2)9/#14, (4)5/#14, (1)1/#6 GRD, (4)2/#14	
9	1	2.5"	15'	T	(1)5/#14, (1)1/#6 GRD	
10	1 (SCHD 80 HDPE)	4"	45'	B	(2)9/#14, (2)5/#14, (1)1/#6 GRD, (2)2/#14	
11	1	2.5"	10'	T	(1)9/#14, (2)5/#14, (1)1/#6 GRD	
12	1	2.5"	5'	T	(1)9/#14, (1)1/#6 GRD	
13	1	2.5"	10'	T	(2)5/#14, (1)1/#6 GRD	
14	1	3"	35'	T	(1)4/#18, (1)1/#6 GRD	
15	1	4"	5'	T	DELDOT FORCES	
16	1	2.5"	10'	T	(1)5/#14, (1)1/#6 GRD	
17	1	4"	25'	T	(2)2/#14, (1)1/#6 GRD	

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
 B=BORE, T=TRENCH, O=OPEN CUT

NOTE: FOR CLARITY OF THE DETAILS MAST ARMS ARE NOT SHOWN. ONLY THE POLE LOCATON IS SHOWN ON THE DETAIL.

SIGNAL PHASING

PHASING NOTES

- PHASES ASSOCIATED BY A SOLID LINE WILL NOT OPERATE CONCURRENTLY.
- PHASES ASSOCIATED BY A DASHED LINE WILL OPERATE CONCURRENTLY.

SIGNAL HEAD DIAGRAM

FLASHING YELLOW ARROW
 FLASHING YELLOW BIKE SIGNAL

LEGEND

EXISTING SYMBOL	PROPOSED SYMBOL
AB	ABANDON
CA	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)
CA	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)
CO	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)
CO	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)
JW	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)
JW	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)
MA	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)
MA	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)
OH	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
OH	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
PB	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
PB	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
PL	EXISTING POLE IDENTIFIER (* OF POLE)
PL	PROPOSED POLE IDENTIFIER (* OF POLE)
RM	REMOVE BY CONTRACTOR
RM	REMOVE BY OTHERS
RM	REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	⬢	⬢
MAST ARM	➤	➤
MICROWAVE DETECTION	⬢	⬢
OPTICOM RECEIVER	○	○
OVERHEAD SIGNING	⬢	⬢
PEDESTRIAN POLE/BASE	⊙	⊙
PEDESTRIAN PUSHBUTTON	➤	➤
PEDESTRIAN SIGNAL HEAD	⬢	⬢
RIGHT-OF-WAY	---	R/W
SERVICE PEDESTAL	⬢	⬢
SIGNAL CABINET	⬢	⬢
SIGNAL HEAD	➤	➤
SIGNAL POLE/BASE	⊙	⊙
CCTV CAMERA	⬢	⬢
SPAN WIRE	XX	XX
UTILITY POLE	⊙	⊙
VIDEO DETECTION	⬢	⬢

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY
JMT

Mir A. Wahed

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

SEAL

CONCURRENCE FOR INSTALLATION

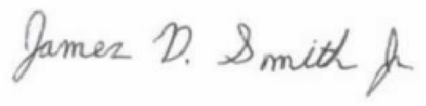
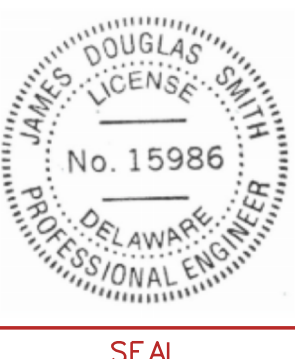
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 CHIEF TRAFFIC ENGINEER DATE

ADDITIONAL GENERAL SIGNAL NOTES

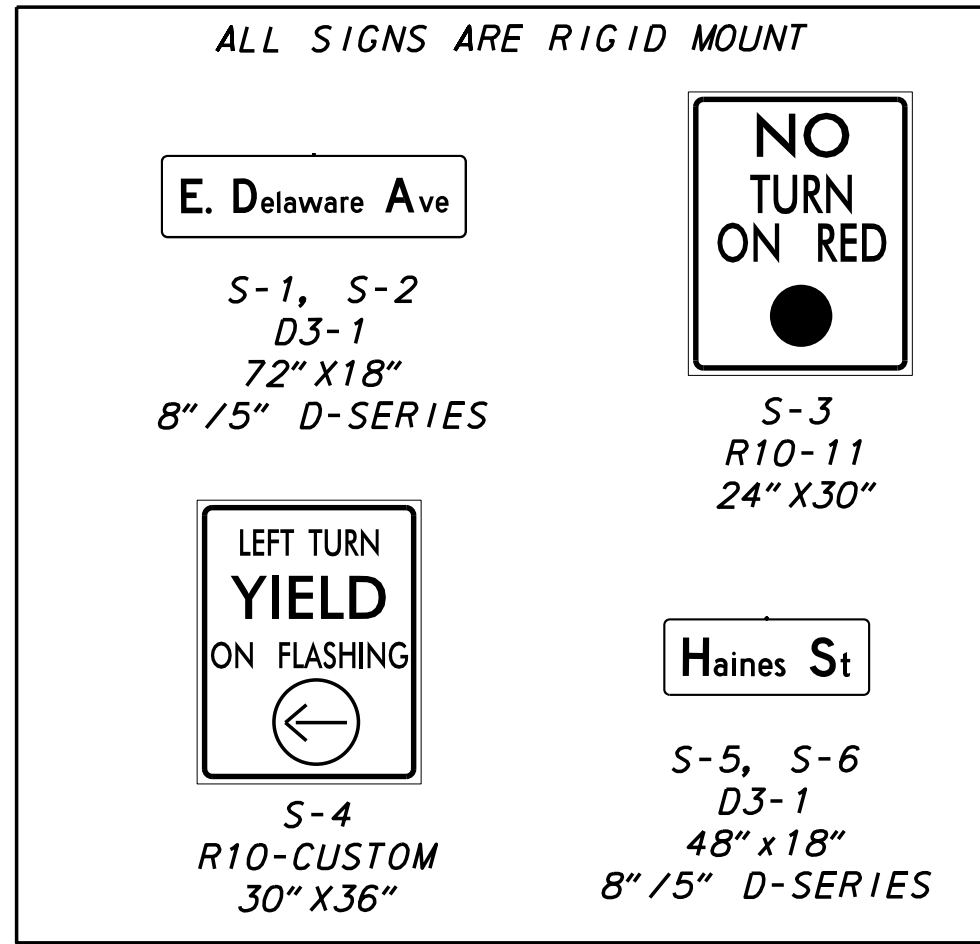
- ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.
- PROPOSED POLE BASES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50% OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42 TO 48 INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.
- THE FISHEYE VIDEO DETECTOR SHALL BE PLACED ON THE MAST ARM TO ACHIEVE OPTIMUM DETECTION PER MANUFACTURER'S SPECIFICATION. THE DESIRED DETECTION ZONES SHALL BE SET AND APPROVED BY DELDOT.
- THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR VEHICLES AND BICYCLES, AS WELL AS PROVIDE INTERSECTION BIKE COUNTS.
- THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BICYCLES ONLY AND WILL PROVIDE INTERSECTION BIKE COUNTS. LOOP DETECTORS WILL PROVIDE DETECTION FOR VEHICLES ONLY.
- R10-3E SIGNS SHALL BE INSTALLED ON POLES WITH ONE PEDESTRIAN SIGNAL HEAD. R10-3E-DE SIGNS SHALL BE INSTALLED ON POLES WITH TWO PEDESTRIAN SIGNAL HEADS. INSTALL R10-3E LEFT SIGNS ON PEDESTRIAN POLE #4. INSTALL R10-3E RIGHT SIGNS ON SIGNAL POLE #1.
- BACKPLATES WILL BE INSTALLED FOR ALL SIGNAL HEADS. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATES.
- ALL PEDESTRIAN POLES WITH PEDESTRIAN SIGNALS ARE 10' AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.
- INSTALL TYPE 4 ANCHOR BOLTS ON TYPE 6 POLE BASE FOUNDATIONS.
- FOR ANY PEDESTAL POLE WITH BIKE AND PEDESTRIAN SIGNALS, BIKE SIGNALS SHOULD BE INSTALLED HIGH ENOUGH FOR A PEDESTRIAN SIGNAL AND PUSHBUTTON TO BE INSTALLED BELOW THE BIKE SIGNALS.
- ITMS CONDUITS #10 AND 15 SHALL BE RUN FROM UTILITY POLE #20E313 TO THE TYPE 4 JUNCTION WELL NEXT TO THE CABINET TO CONNECT THE SIGNAL TO THE ITMS CABLES.
- INSTALL PROPOSED MAST ARM POLE BASES FOR POLE #1 AND #5 FLUSH WITH THE ADJACENT SIDEWALK.
- THE SIGNAL EQUIPMENT ON THE SOUTHEAST CORNER SHALL BE INSTALLED AS TO AVOID BLOCKING OR DISTURBING THE EXISTING TSONAS SIGN.
- THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM #837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.

ADDITIONAL GENERAL SIGNAL NOTES

- INSTALL TYPE P TRAFFIC CONTROLLER CABINET ON A 56" X58" CABINET BASE.
- ON THE SOUTHWEST CORNER OF THE INTERSECTION, INSTALL THE SERVICE DISCONNECT AND METER AS WELL AS THE ASSOCIATED TYPE 1 JUNCTION WELL, TYPE P CONTROLLER CABINET, TYPE 4 JUNCTION WELL, AND POLE #3 WITHIN THE CONCRETE AREA ADJACENT TO THE PROPOSED SIDEWALK.

PREPARED BY JMT 		03/07/2021 DATE
THIS SEAL APPLIES TO ONLY THIS SHEET		SEAL 

OVERHEAD SIGN DETAILS

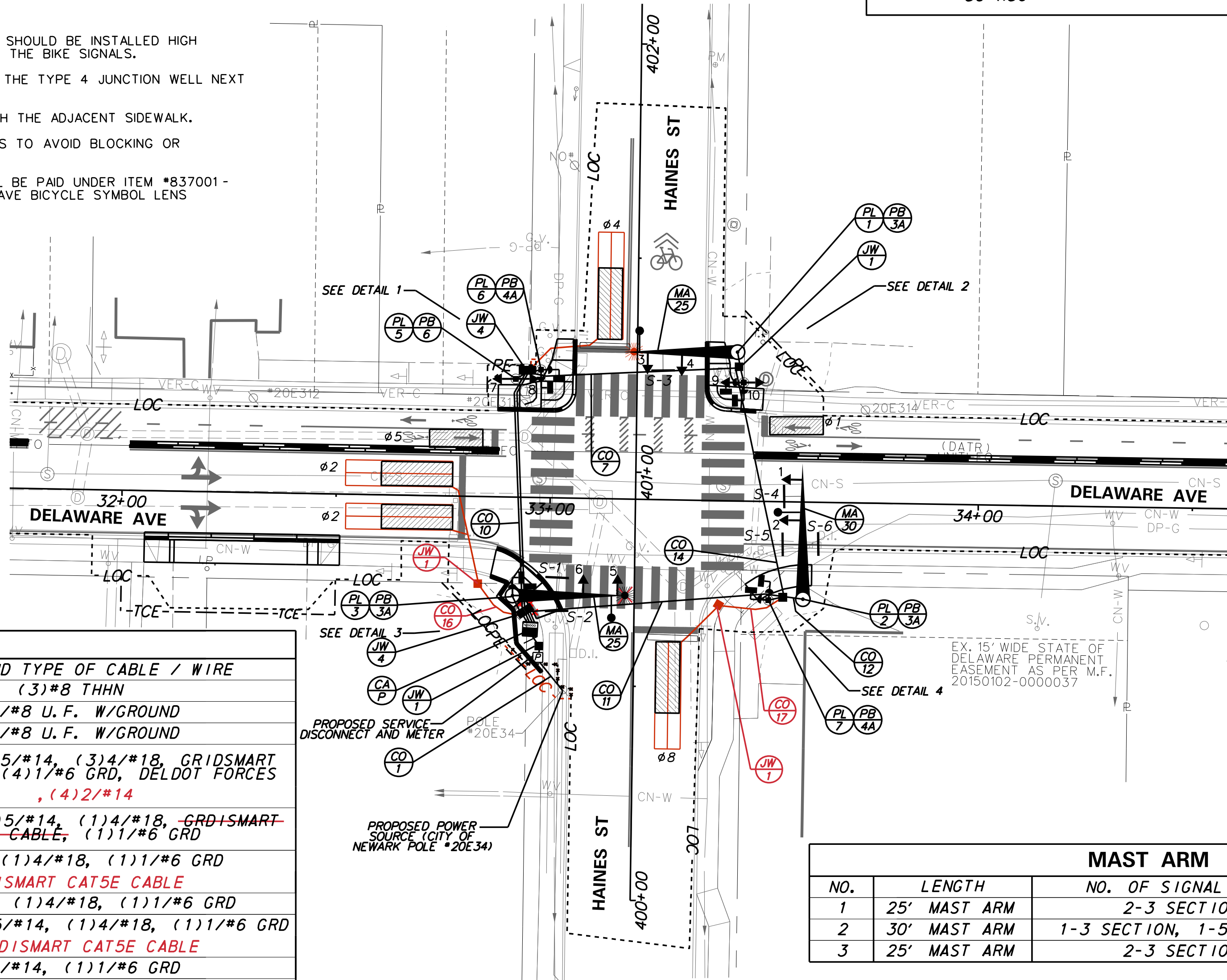


ULTIMATE



ADDITIONAL LEGEND

- MINIMUM DETECTION ZONE
- FISHEYE VIDEO DETECTION



CONDUIT RUN SCHEDULE

CO#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
1	1 (GALV. STEEL)	2"	10'	B/T	(3)*#8 THHN
2A	1 (GALV. STEEL)	2"	5'	T	(1)2/*8 U.F. W/GROUND
2B	1 (GALV. STEEL)	2"	5'	T	(1)2/*8 U.F. W/GROUND
3	4	4"	5'	T	(7)9/*14, (8)5/*14, (3)4/*18, GRIDSMART CAT5E CABLE, (4)1/*6 GRD, DELDOT FORCES, (4)2/*14
4	1	3"	5'	T	(1)9/*14, (2)5/*14, (1)4/*18, GRD+SMART-CAT5E CABLE, (1)1/*6 GRD
5	1	2.5"	5'	T	(1)9/*14, (1)4/*18, (1)1/*6 GRD, GRDISMART CAT5E CABLE
6	1	3"	5'	T	(1)9/*14, (1)4/*18, (1)1/*6 GRD
7	1 (SCH 80 HDPE)	4"	50'	B	(3)9/*14, (2)5/*14, (1)4/*18, (1)1/*6 GRD, GRDISMART CAT5E CABLE
8	1	2.5"	5'	T	(2)5/*14, (1)1/*6 GRD
9	1	2.5"	5'	T	(2)9/*14, (1)1/*6 GRD
10	1 (SCH 80 HDPE)	4"	60'	B	(5)9/*14, (4)5/*14, (1)4/*18, (1)1/*6 GRD, GRDISMART CAT5E CABLE, (1)2/*14
	1 (SCH 80 HDPE)	4"	60'	B	DELDOT FORCES
11	1 (SCH 80 HDPE)	4"	60'	B	(1)9/*14, (2)5/*14, (1)4/*18, (1)1/*6 GRD, (1)2/*14
12	1	3"	5'	T	(1)9/*14, (1)4/*18, (1)1/*6 GRD
13	1	2.5"	5'	T	(2)5/*14, (1)1/*6 GRD
14	1 (SCH 80 HDPE)	4"	55'	B	EMPTY
15	1	4"	10'	T	DELDOT FORCES
16	1	4"	15'	T	(2)2/*14, (1)1/*6 GRD
17	1 (SCH 80 HDPE)	4"	20'	B	(1)2/*14, (1)1/*6 GRD

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT

MAST ARM SCHEDULE

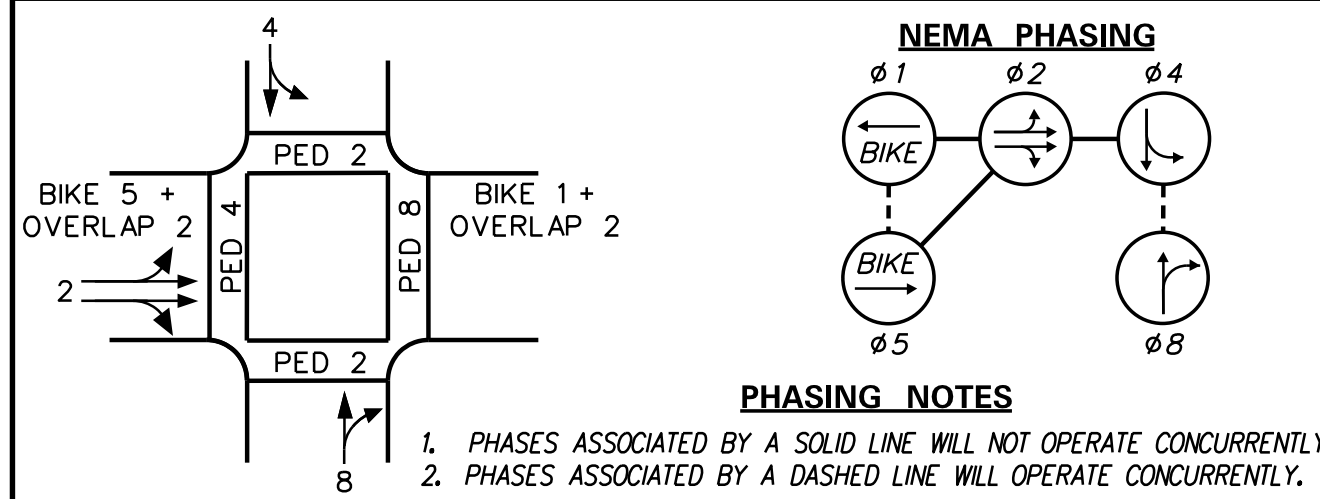
NO.	LENGTH	NO. OF SIGNAL HEADS	S.F. OF SIGNAGE
1	25' MAST ARM	2-3 SECTION	5 SF
2	30' MAST ARM	1-3 SECTION, 1-5 SECTION	13.25 SF
3	25' MAST ARM	2-3 SECTION	18 SF

POLE SCHEDULE

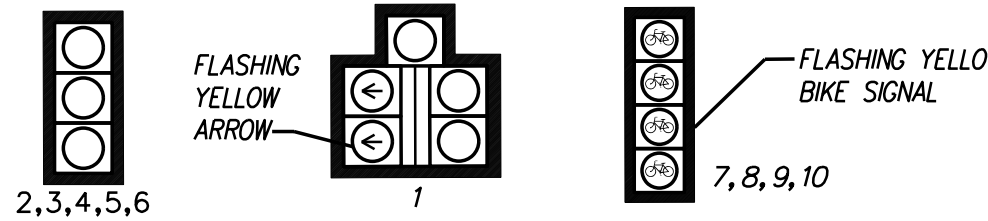
NO.	TYPE	STATION	OFFSET
1	25' MAST ARM W/ TYPE 3A BASE	33+43	35' LT
2	30' MAST ARM W/ TYPE 3A BASE	33+59	23' RT
3	25' MAST ARM W/ TYPE 3A BASE	32+92	23' RT
4	14' PEDESTAL POLE W/ TYPE 6 BASE	33+45	28' LT
5	14' PEDESTAL POLE W/ TYPE 6 BASE	32+91	28' LT
6	10' PEDESTAL POLE W/ TYPE 4A BASE	32+97	30' LT
7	10' PEDESTAL POLE W/ TYPE 4A BASE	33+51	23' RT

THE 10' AND 14' PEDESTAL POLES SHALL BE SPUN ALUMINUM, SCHEDULE 80.

SIGNAL PHASING



SIGNAL HEAD DIAGRAM



LEGEND

EXISTING SYMBOL	PROPOSED SYMBOL
AB	ABANDON
CA	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)
CA	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)
CO	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)
CO	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)
JW	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)
JW	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)
MA	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)
MA	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)
OH	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
OH	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
PB	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
PB	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
PL	EXISTING POLE IDENTIFIER (* OF POLE)
PL	PROPOSED POLE IDENTIFIER (* OF POLE)
RM	REMOVE BY CONTRACTOR
RM	REMOVE BY OTHERS
RM	REMOVE BY TRAFFIC CONTRACTOR

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY

JMT



Mir A. Wahed

12/03/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

SEAL

CONCURRENCE FOR INSTALLATION

CHIEF OF TRAFFIC ENGINEERING

12/6/2021

DATE

DELAWARE AVENUE SEPARATED BIKEWAY- ORCHARD ROAD TO LIBRARY AVENUE

CONTRACT

T201801501

COUNTY

NEW CASTLE

PERMIT NO.

N856

DESIGNED BY: K. REHM/J. SMITH

CHECKED BY: M. WAHED

SIGNAL PLAN DELAWARE AVE AND HAINES ST

SECTION

JMT

SHEET NO.

182 A

ADDENDA / REVISIONS

REPLACES SHEET 182
ADDED LOOP DETECTORS

12-03-2021 (JDS)
03-07-2022 (JDS)



ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.

4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.

5. POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP OR SIDEWALK, IN ACCORDANCE WITH CURRENT ADA BEST PRACTICES. THE PEDESTRIAN PUSHBUTTONS SHALL BE INSTALLED AT A MINIMUM HEIGHT OF 15 INCHES AND A MAXIMUM HEIGHT OF 48 INCHES ABOVE THE LANDING AREA/SIDEWALK, BUT SHOULD BE INSTALLED AT A HEIGHT OF 42 INCHES TO 48 INCHES. THE PEDESTRIAN PUSHBUTTON SHALL BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE LANDING AREA TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING, INCLUDING BRACKETS, NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE GROUND. ALL PEDESTRIAN SIGNAL HEADS WILL BE COUNTDOWN SIGNALS.

6. FISHEYE VIDEO DETECTOR SHALL BE PLACED ON THE MAST ARM TO ACHIEVE OPTIMUM DETECTION PER MANUFACTURER'S SPECIFICATION. THE DESIRED DETECTION ZONES SHALL BE SET AND APPROVED BY DELDOT.

7. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BOTH VEHICLES AND BICYCLES, AS WELL AS PROVIDE INTERSECTION BIKE COUNTS.

7. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BICYCLES ONLY AND WILL PROVIDE INTERSECTION BIKE COUNTS. LOOP DETECTORS WILL PROVIDE DETECTION FOR VEHICLES ONLY.

8. REMOVAL OF POLE BASES SHALL BE PAID BY ITEM *211000.

9. R10-3E SIGNS SHALL BE INSTALLED ON POLES WITH ONE PEDESTRIAN SIGNAL HEAD. R10-3E-DE SIGNS SHALL BE INSTALLED ON POLES WITH TWO PEDESTRIAN SIGNAL HEADS.

10. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10' AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.

11. BACKPLATES WILL BE INSTALLED FOR SIGNAL HEADS *7-10. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATE.

12. THE TRAFFIC CONTRACTOR SHALL INSTALL THE PROPOSED APS STATION ON POLE *9, *10, AND *12, ROTATED SO THAT THE FACE OF THE PUSHBUTTON IS PARALLEL WITH THE CROSSING.

13. INSTALL TYPE 4 ANCHOR BOLTS ON TYPE 6 POLE BASE FOUNDATIONS.

14. THE TRAFFIC CONTRACTOR SHALL REPLACE ALL OF THE APS STATIONS WITH THE NEWEST VERSION.

15. REPLACE THE EXISTING JUNCTION WELLS ON THE SOUTHWEST CORNER AND IN THE ISLAND ON THE SOUTHEAST CORNER.

16. INSTALL PLASTIC LINING INSIDE CABINET TO PROTECT THE ELECTRICAL COMPONENTS FROM SPRINKLERS.

ADDITIONAL GENERAL SIGNAL NOTES

17. COORDINATE WITH BHADRESH PATEL (302-366-7000) AT THE CITY OF NEWARK TO ADJUST THE SECONDARY ELECTRICAL CABLES THAT ARE TOUCHING MAST ARMS *1 AND *2.

18. A SEPERATE ITMS CONDUIT SHALL BE RUN FROM UTILITY POLE *20F36 TO THE EXISTING TYPE 4 JUNCTION WELL NEXT TO THE CABINET TO CONNECT THE SIGNAL TO THE ITMS CABLES.

19. THE NEW CABINET SHOULD BE CONSTRUCTED IN THE DELDOT CONTROLLER SHOP AND THEN INSTALLED OVERNIGHT.

20. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO ANY CHANGES TO THE TRAFFIC SIGNAL OPERATION. THE CONTRACTOR SHALL HAVE CORRESPONDING SIGNAL TIMESHEET ISSUED VIA DELDOT TRAFFIC/TMC PRIOR TO MODIFYING THE TRAFFIC SIGNAL OPERATION.

21. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM *837001- FURNISH AND INSTALL 8" SIGNAL HEAD RIDIGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.

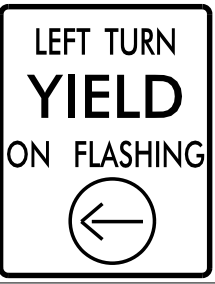
ULTIMATE

SIGN DETAILS

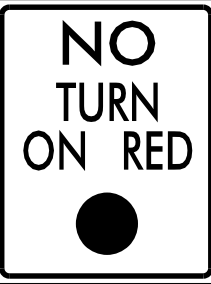
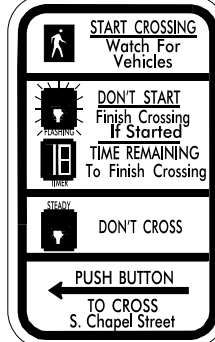
ALL SIGNS ARE RIGID MOUNT

S Chapel St

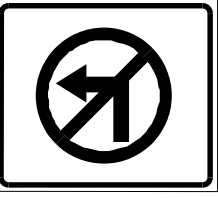
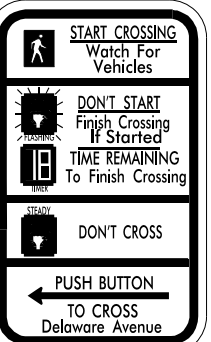
S-1
D3-1
66"x18"
8" / 5" D-SERIES



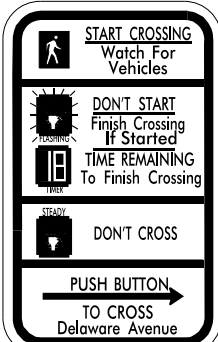
S-2
R10-CUSTOM
30" X36"



S-3
R10-11
24" X30"

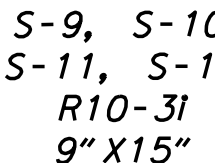


S-4
R3-2
36" X36"



E. Delaware Ave

S-5, S-6,
S-7, S-8
D3-1
72"x18"
8" / 5" D-SERIES



S-9, S-10,
S-11, S-12
R10-31
9" X15"



S-13
R10-31
9" X15"



S-14, S-15,
S-16
R10-31
9" X15"



PREPARED BY
JMT

James D. Smith Jr.

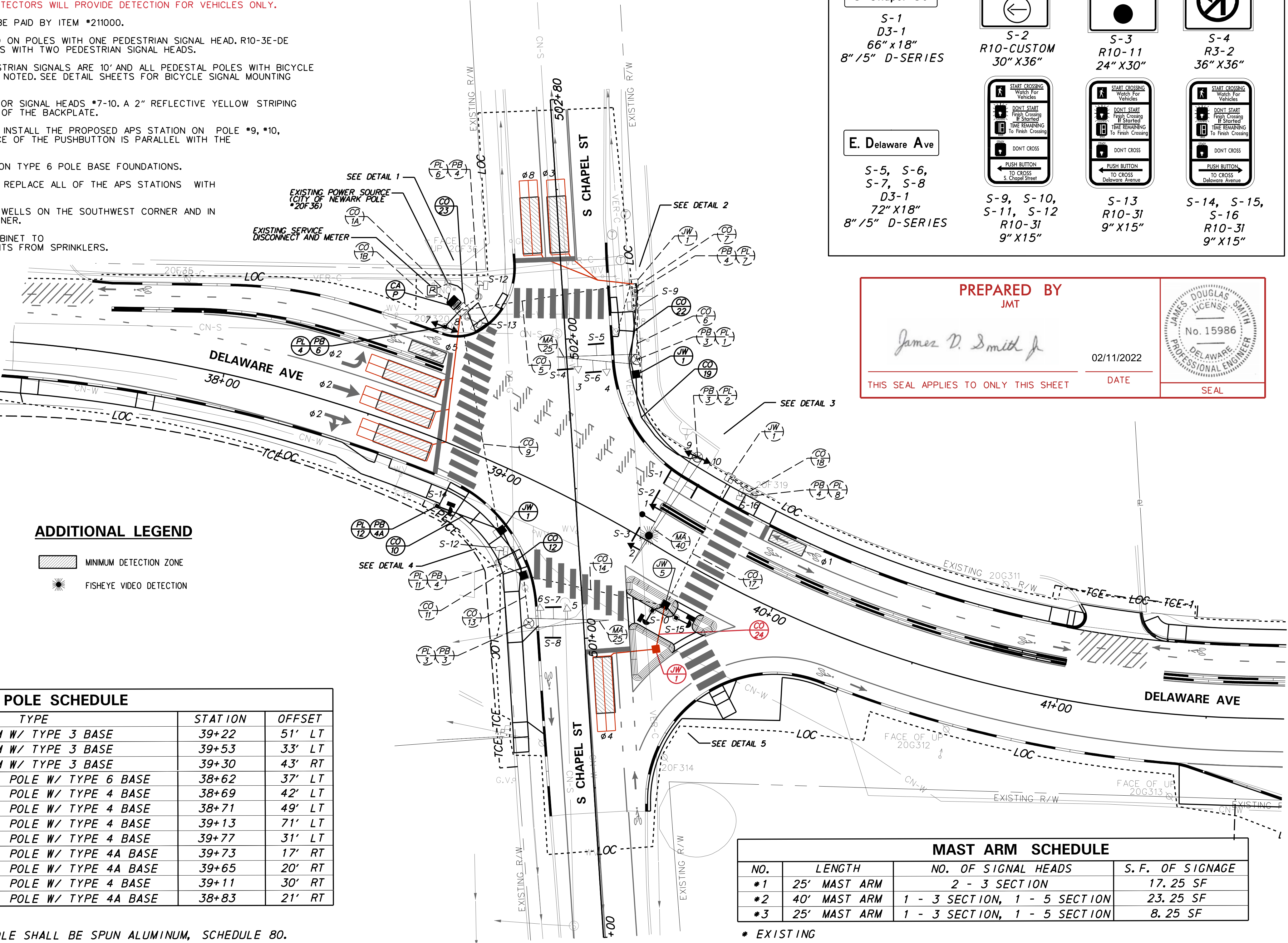
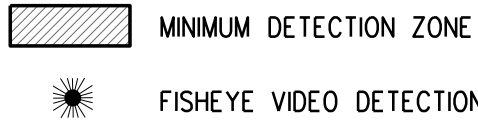
02/11/2022

THIS SEAL APPLIES TO ONLY THIS SHEET

DATE



ADDITIONAL LEGEND



POLE SCHEDULE

NO.	TYPE	STATION	OFFSET
*1	25' MAST ARM W/ TYPE 3 BASE	39+22	51' LT
*2	40' MAST ARM W/ TYPE 3 BASE	39+53	33' LT
*3	25' MAST ARM W/ TYPE 3 BASE	39+30	43' RT
4	14' PEDESTAL POLE W/ TYPE 6 BASE	38+62	37' LT
*5	10' PEDESTAL POLE W/ TYPE 4 BASE	38+69	42' LT
*6	10' PEDESTAL POLE W/ TYPE 4 BASE	38+71	49' LT
*7	10' PEDESTAL POLE W/ TYPE 4 BASE	39+13	71' LT
*8	10' PEDESTAL POLE W/ TYPE 4 BASE	39+77	31' LT
9	10' PEDESTAL POLE W/ TYPE 4A BASE	39+73	17' RT
10	10' PEDESTAL POLE W/ TYPE 4A BASE	39+65	20' RT
*11	10' PEDESTAL POLE W/ TYPE 4 BASE	39+11	30' RT
12	10' PEDESTAL POLE W/ TYPE 4A BASE	38+83	21' RT

* EXISTING
THE 14' PEDESTAL POLE SHALL BE SPUN ALUMINUM, SCHEDULE 80.

MAST ARM SCHEDULE

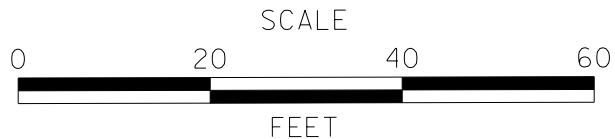
NO.	LENGTH	NO. OF SIGNAL HEADS	S. F. OF SIGNAGE
*1	25' MAST ARM	2 - 3 SECTION	17.25 SF
*2	40' MAST ARM	1 - 3 SECTION, 1 - 5 SECTION	23.25 SF
*3	25' MAST ARM	1 - 3 SECTION, 1 - 5 SECTION	8.25 SF

* EXISTING

ADDED LOOP DETECTORS

02-11-2022 (JDS)

ADDENDA / REVISIONS



DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT

T201801501

COUNTY

NEW CASTLE

PERMIT NO.

N430T

DESIGNED BY:

J.SMITH

CHECKED BY:

M.WAHED

SIGNAL PLAN
DELAWARE AVENUE
@ SOUTH CHAPEL STREET

SECTION

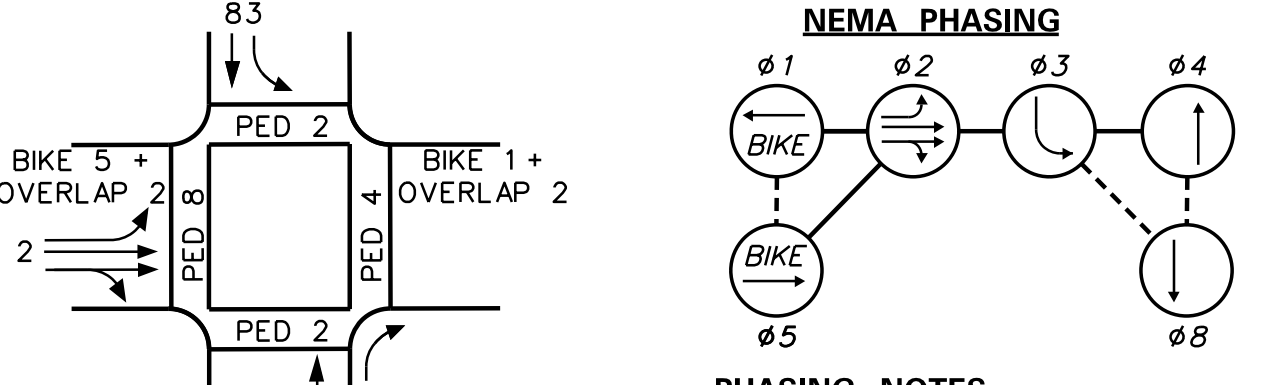
JMT

SHEET NO.

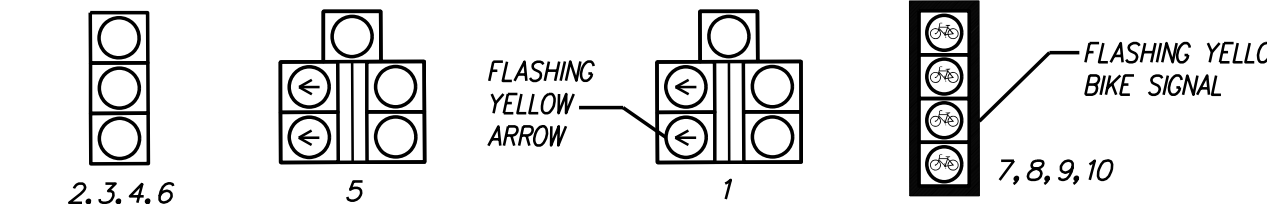
188

SIGNAL PHASING

NEMA PHASING



SIGNAL HEAD DIAGRAM



LEGEND

EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.
LOOP DETECTOR, TYPE 1	
LOOP DETECTOR, TYPE 2	
LUMINAIRE	
MAST ARM	
MICROWAVE DETECTION	
OPTICOM RECEIVER	
OVERHEAD SIGNING	
PEDESTRIAN POLE/BASE	
PEDESTRIAN PUSHBUTTON	
PEDESTRIAN SIGNAL HEAD	
RIGHT-OF-WAY	R/W
SERVICE PEDESTAL	
SIGNAL CABINET	
SIGNAL HEAD	
SIGNAL POLE/BASE	
CCTV CAMERA	
SPAN WIRE	
UTILITY POLE	
VIDEO DETECTION	

GENERAL SIGNAL NOTES

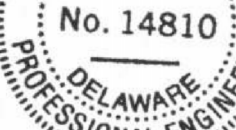
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PREPARED BY

JMT

PREPARED BY

JMT



07/20/2021

THIS SEAL APPLIES TO ALL SHEETS
BEARING THE "JMT" SECTION DESIGNATION.

DATE

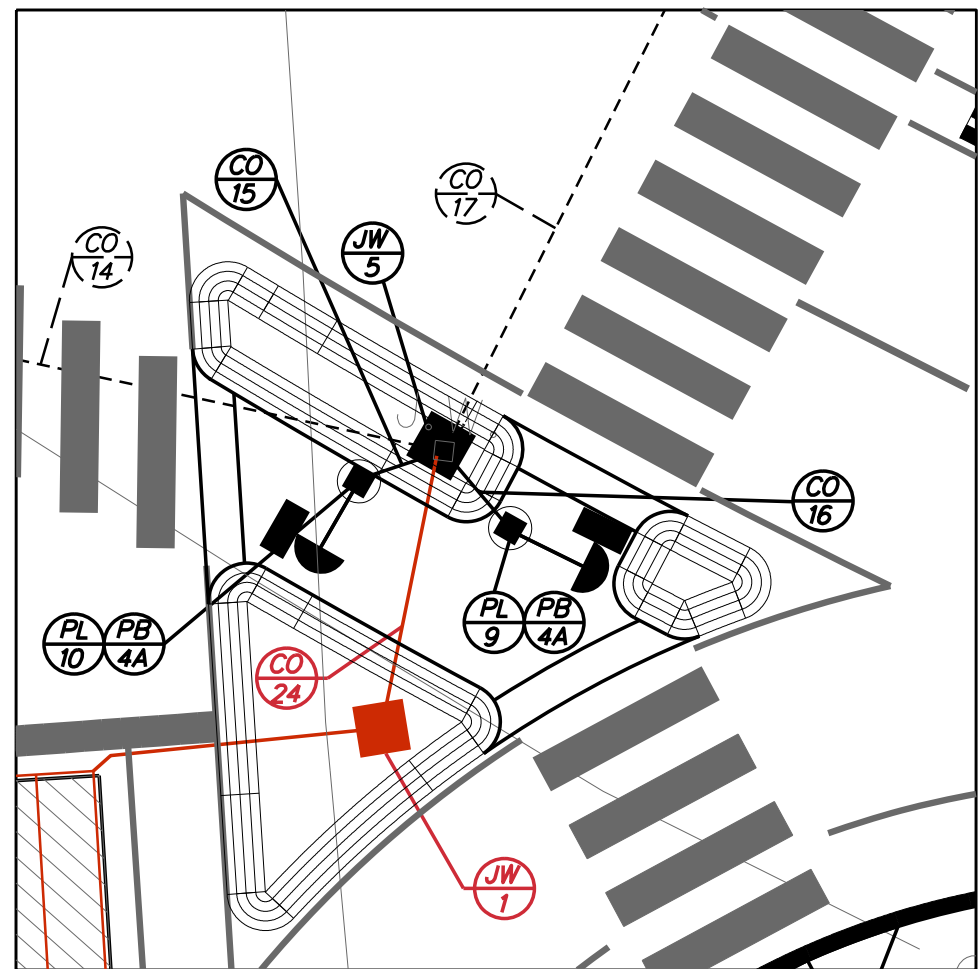
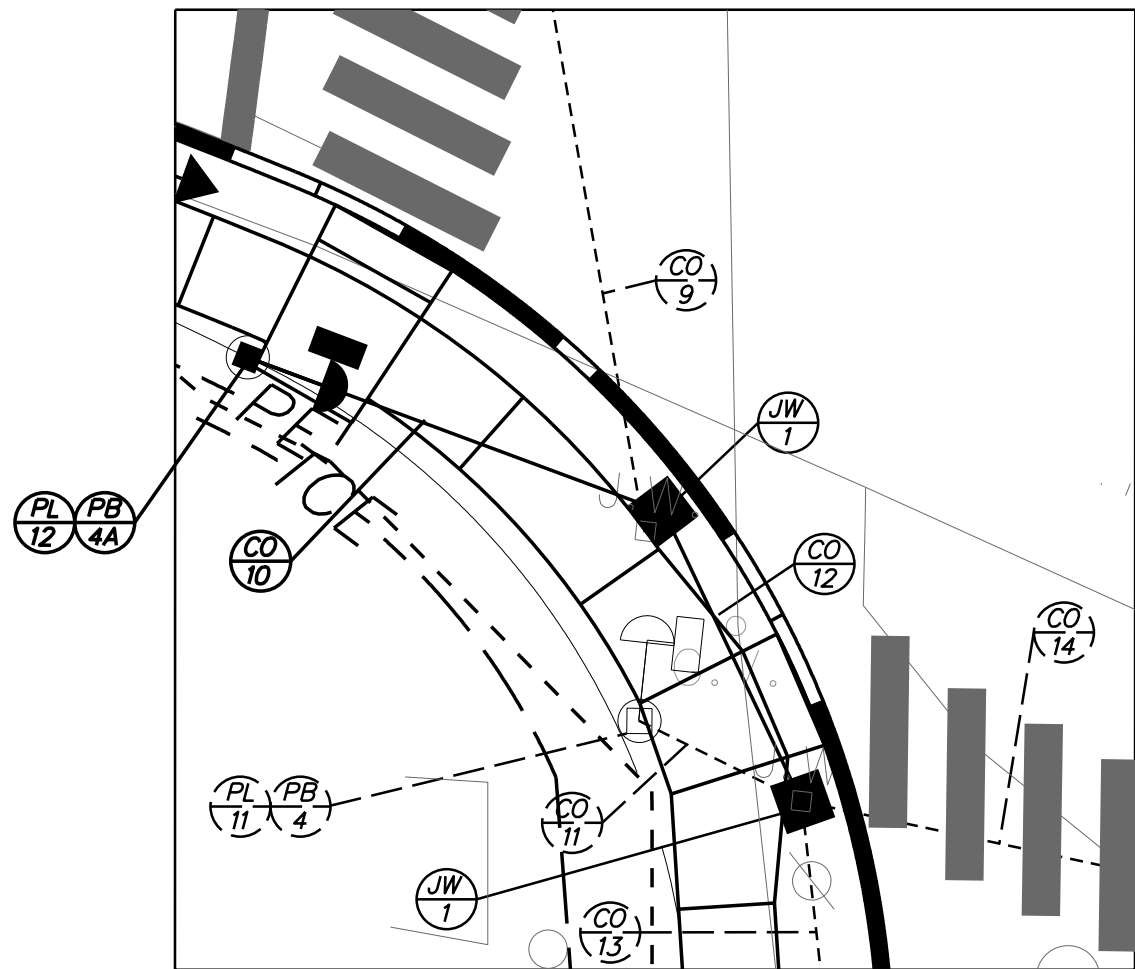
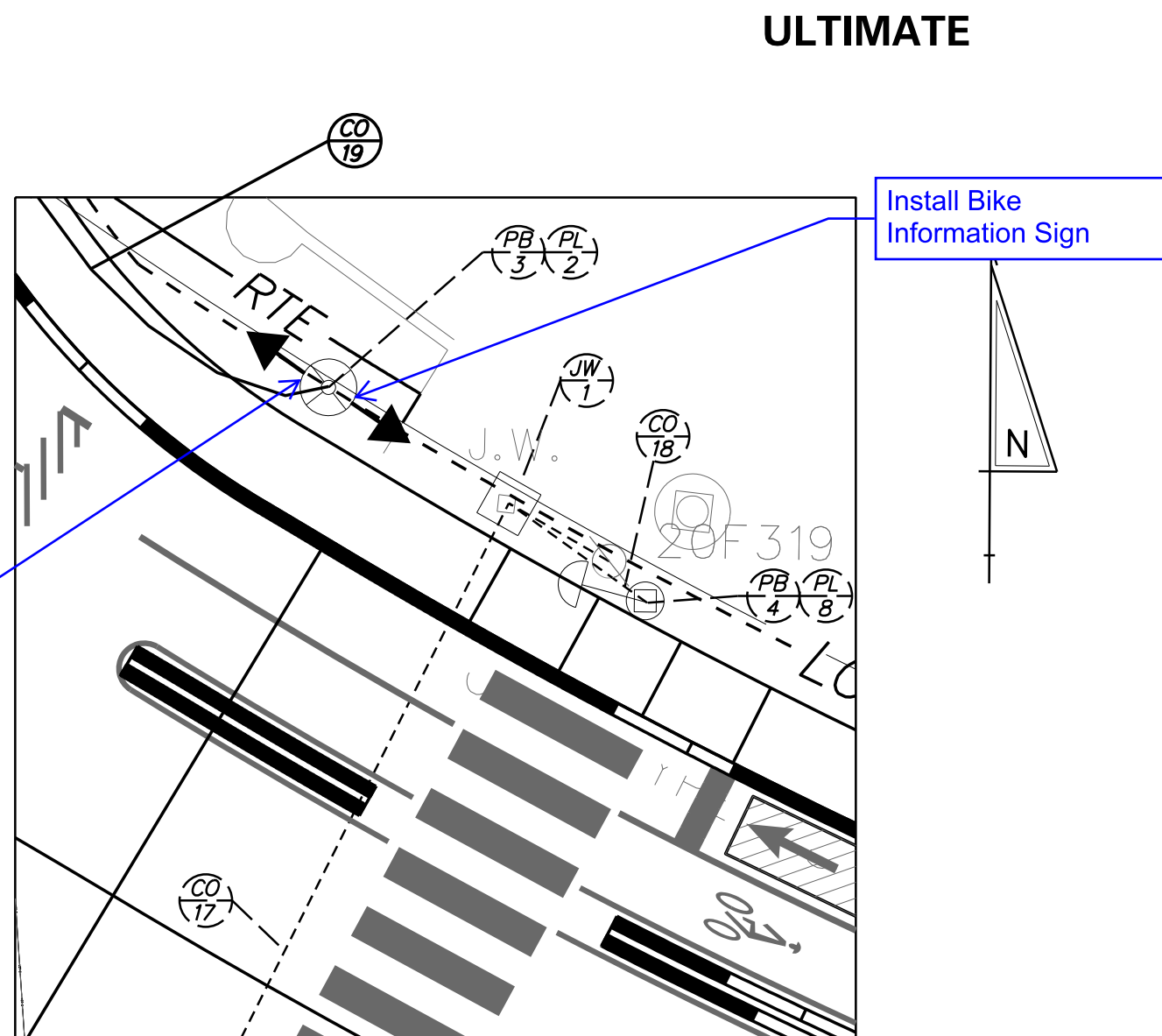
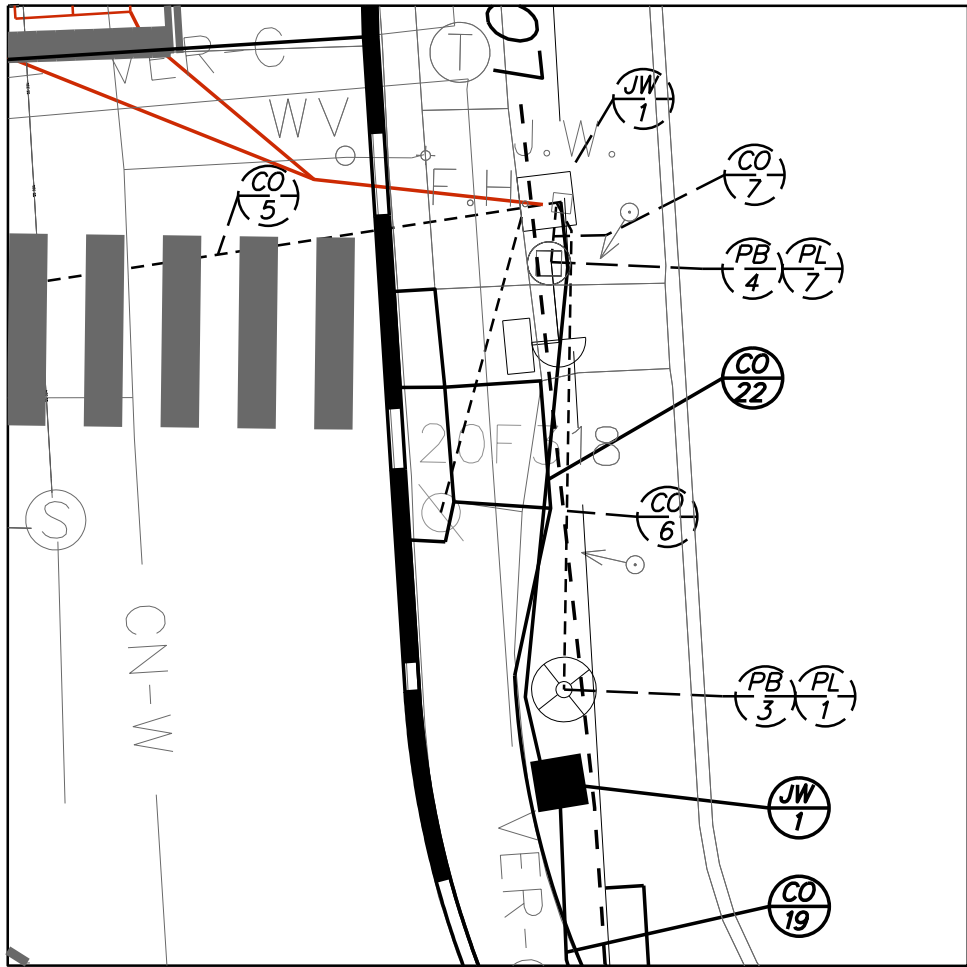
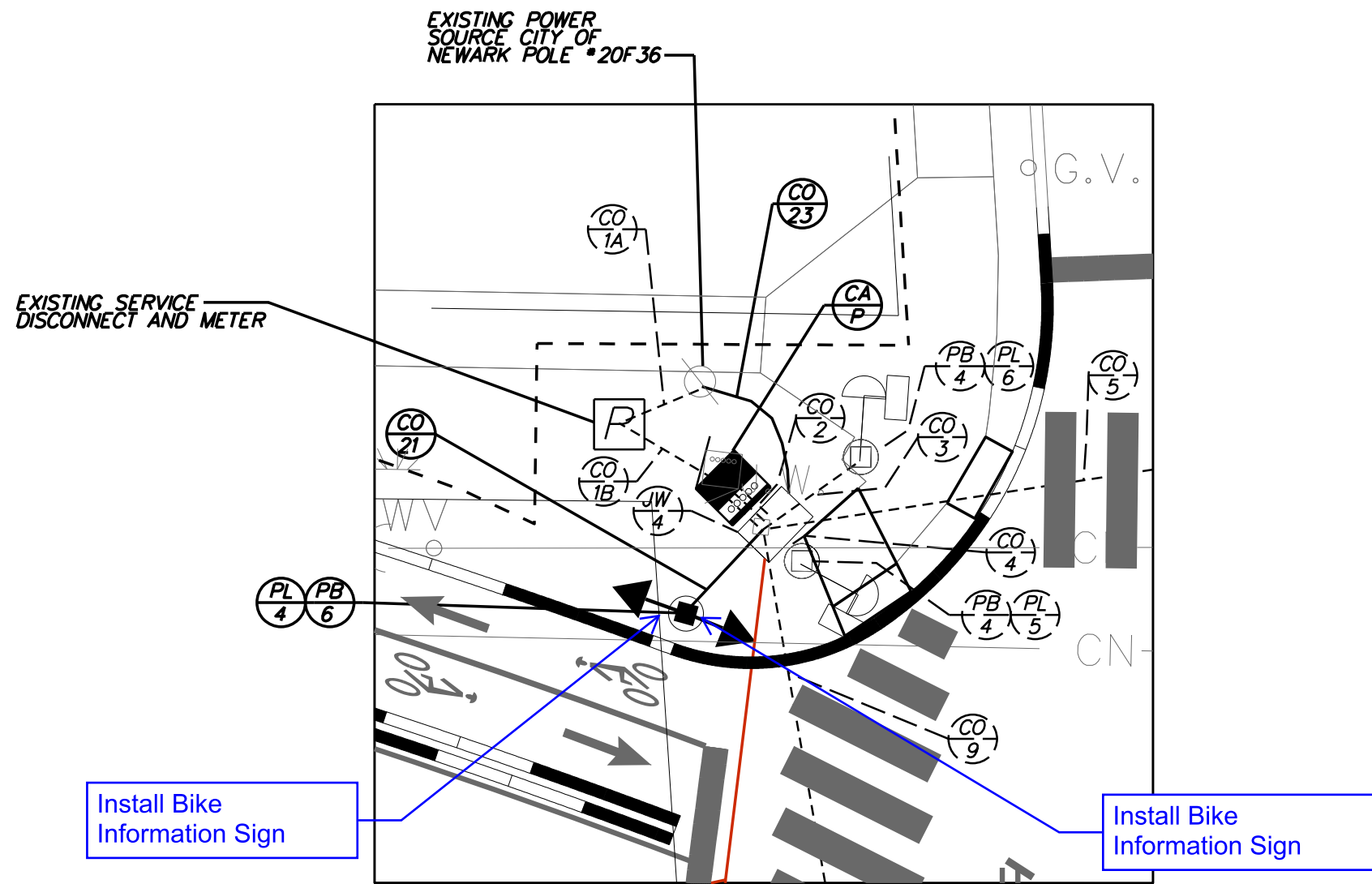
SEAL

CONCURRENCE FOR INSTALLATION

CHIEF TRAFFIC ENGINEER

7/28/2021

DATE



NOTE: FOR CLARITY OF THE DETAILS MAST ARMS ARE NOT SHOWN, ONLY THE POLE LOCATION IS SHOWN ON THE DETAIL.

CONDUIT RUN SCHEDULE					
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
*1A	1 (GALV. STEEL)	2. 0"	10'	-	EX. [(3)*8 THHN]
*1B	1 (GALV. STEEL)	2. 0"	10'	-	EX. [(1)2/*8 U.F. W/ GROUND]
*2	3	2. 5"	5'	-	NEW [(5)9/*14, (2)4/*18, (1)16/*14, (8)5/*14, GRIDSMART CAT5E CABLE, (3)1/*6 GRD], (6)2/*14], EX. [(1)9/*14, (1)4/*18]
*3	1	2. 5"	10'	-	NEW [(1)5/*14, (1)1/*6 GRD]
*4	1	2. 5"	5'	-	NEW [(1)5/*14, (1)1/*6 GRD]
*5	1	2. 5"	60'	-	NEW [(2)9/*14, (1)5/*14, (1)16/*14, (1)4/*18, GRIDSMART CAT5E CABLE, (1)1/*6 GRD], (2)2/*14], EX. [(1)9/*14, (1)4/*18]
*6	1	2. 5"	25'	-	NEW [(1)1/*6 GRD], EX. [(1)9/*14, (1)4/*18]
*7	1	2. 5"	5'	-	NEW [(1)5/*14, (1)1/*6 GRD]
*9	1	2. 5"	75'	-	NEW [(1)9/*14, (1)4/*18, (5)5/*14, (1)1/*6 GRD], (1)2/*14]
10	1	2. 5"	25'	T	NEW [(1)5/*14, (1)1/*6 GRD]
*11	1	2. 5"	10'	-	NEW [(1)5/*14, (1)1/*6 GRD]
12	1	4"	20'	T	NEW [(1)9/*14, (1)4/*18, (4)5/*14, (1)1/*6 GRD], (1)2/*14]
*13	1	2. 5"	20'	-	NEW [(1)9/*14, (1)4/*18, (1)1/*6 GRD]
*14	1	2. 5"	50'	-	NEW [(3)5/*14, (1)1/*6 GRD], (1)2/*14]
15	1	2. 5"	5'	T	NEW [(1)5/*14, (1)1/*6 GRD]
16	1	2. 5"	10'	T	NEW [(1)5/*14, (1)1/*6 GRD]
*17	1	2. 5"	50'	-	NEW [(1)5/*14, (1)1/*6 GRD]
*18	1	2. 5"	15'	-	NEW [(1)5/*14, (1)1/*6 GRD]
19	1	3"	40'	T	NEW [(1)16/*14, (1)4/*18, (2)9/*14, GRIDSMART CAT5E CABLE, (1)1/*6 GRD]
21	1	3"	10'	T	NEW [(2)9/*14, (1)1/*6 GRD]
22	1	4"	30'	T	NEW [(1)16/*14, (1)4/*18, (2)9/*14, GRIDSMART CAT5E CABLE, (1)1/*6 GRD]
23	1	4"	25'	T	NEW [DELDOT FORCES]
24	1	4"	15'	T	NEW [(1)2/*14]

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT

PREPARED BY JMT	
<i>James D. Smith</i>	02/11/2022
THIS SEAL APPLIES TO ONLY THIS SHEET	DATE
SEAL	

JAMES DOUGLAS
LICENSE
No. 15986
DELAWARE
PROFESSIONAL ENGINEER

SIGNAL PHASING

NEMA PHASING

PHASING NOTES

- PHASES ASSOCIATED BY A SOLID LINE WILL NOT OPERATE CONCURRENTLY.
- PHASES ASSOCIATED BY A DASHED LINE WILL OPERATE CONCURRENTLY.

SIGNAL HEAD DIAGRAM

FLASHING YELLOW ARROW

FLASHING YELLOW BIKE SIGNAL

LEGEND

EXISTING SYMBOL	PROPOSED SYMBOL
AB	ABANDON
CA	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)
CA	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)
CO	EXISTING CONDUIT RUN IDENTIFIER (# OF CONDUIT RUN)
CO	PROPOSED CONDUIT RUN IDENTIFIER (# OF CONDUIT RUN)
JW	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)
JW	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)
MA	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)
MA	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)
OR	EXISTING OVERHEAD RUN IDENTIFIER (# OF OVERHEAD RUN)
OR	PROPOSED OVERHEAD RUN IDENTIFIER (# OF OVERHEAD RUN)
PB	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
PB	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
PL	EXISTING POLE IDENTIFIER (# OF POLE)
PL	PROPOSED POLE IDENTIFIER (# OF POLE)
RM	REMOVE BY CONTRACTOR
RO	REMOVE BY OTHERS
RM	REMOVE BY TRAFFIC CONTRACTOR

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY JMT	07/20/2021	MIR A WAHED LICENSE No. 14810 DELAWARE PROFESSIONAL ENGINEER
THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.	DATE	

CONCURRENCE FOR INSTALLATION

CHIEF TRAFFIC ENGINEER

7/28/2021

DATE

CONTRACT T201801501	PERMIT NO. N430T	SIGNAL PLAN DELAWARE AVENUE @ SOUTH CHAPEL STREET	SECTION JMT SHEET NO. 189
COUNTY NEW CASTLE	DESIGNED BY: J.SMITH		
CHECKED BY: M.WAHED			

ADDENDA / REVISIONS

ADDED LOOP DETECTORS	02-11-2022 (JDS)
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N.T.S.

DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT
T201801501

PERMIT NO.
N430T

COUNTY
NEW CASTLE

DESIGNED BY: J.SMITH

CHECKED BY: M.WAHED

SIGNAL PLAN
DELAWARE AVENUE
@ SOUTH CHAPEL STREET

SECTION
JMT
SHEET NO.
189

ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.
4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.
5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42 TO 48 INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.
6. THE BICYCLE SIGNALS (*X, 4 AND 5) WILL BE IN PHASE WITH THE PEDESTRIAN SIGNALS, BOTH THE BICYCLE SIGNALS AND PEDESTRIAN SIGNALS WILL BE ACTIVATED WHEN THERE IS A CALL AT THE PEDESTRIAN PUSH BUTTONS OR BICYCLE PUSH BUTTONS.
7. REPLACE EXISTING PEDESTRIAN PUSHBUTTON SIGNS WITH R10-3E SIGNS ON POLE *2 AND *3.
8. REPLACE THE EXISTING PEDESTRIAN SIGNAL HEADS WITH A NEW COUNTDOWN PEDESTRIAN SIGNAL HEADS.
9. NEW CABLES WILL BE INSTALLED TO REPLACE THE EXISTING CABLES. THE EXISTING CABINET, POLES, AND FOUNDATIONS WILL REMAIN AS IS.
10. BACKPLATES WILL BE INSTALLED FOR SIGNAL HEADS *X, *4, AND *5. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATE.
11. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10', AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.
12. INSTALL R10-26 SIGN WITH RIGHT ARROW ON PEDESTAL POLE *4 FOR THE BIKE PUSH BUTTON.
13. A NEW SERVICE DISCONNECT SHALL BE INSTALLED; HOWEVER, THE EXISTING POWER SOURCE (CITY OF NEWARK POLE *20G33) SHALL BE MAINTAINED.
14. AN ITMS CONDUIT SHALL BE RUN FROM UTILITY POLE *20G33 TO THE TYPE 1 JUNCTION WELL NEXT TO THE POLE. A SEPARATE ITMS CONDUIT SHALL BE RUN FROM THE TYPE 1 JUNCTION WELL TO THE TYPE 4 JUNCTION WELL NEXT TO THE CABINET TO CONNECT THE SIGNAL TO THE ITMS CABLES.
15. INSTALL CABINET BASE EXTENSION ON EXISTING CABINET.
16. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO ANY CHANGES TO THE TRAFFIC SIGNAL OPERATION. THE CONTRACTOR SHALL HAVE CORRESPONDING SIGNAL TIMESHEET ISSUED VIA DELDOT TRAFFIC/TMC PRIOR TO MODIFYING THE TRAFFIC SIGNAL OPERATION.
17. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM *837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.
18. INSTALL BICYCLE PUSHBUTTON ON POLE #3 AT A MAXIMUM HEIGHT OF 46".

REVISION PREPARED BY
JMT
James D. Smith Jr.
THIS SEAL APPLIES TO THIS SHEET

02/02/23
DATE

SEAL
JAMES DOUGLAS SMITH
LICENSE
No. 15986
DELAWARE
PROFESSIONAL ENGINEER

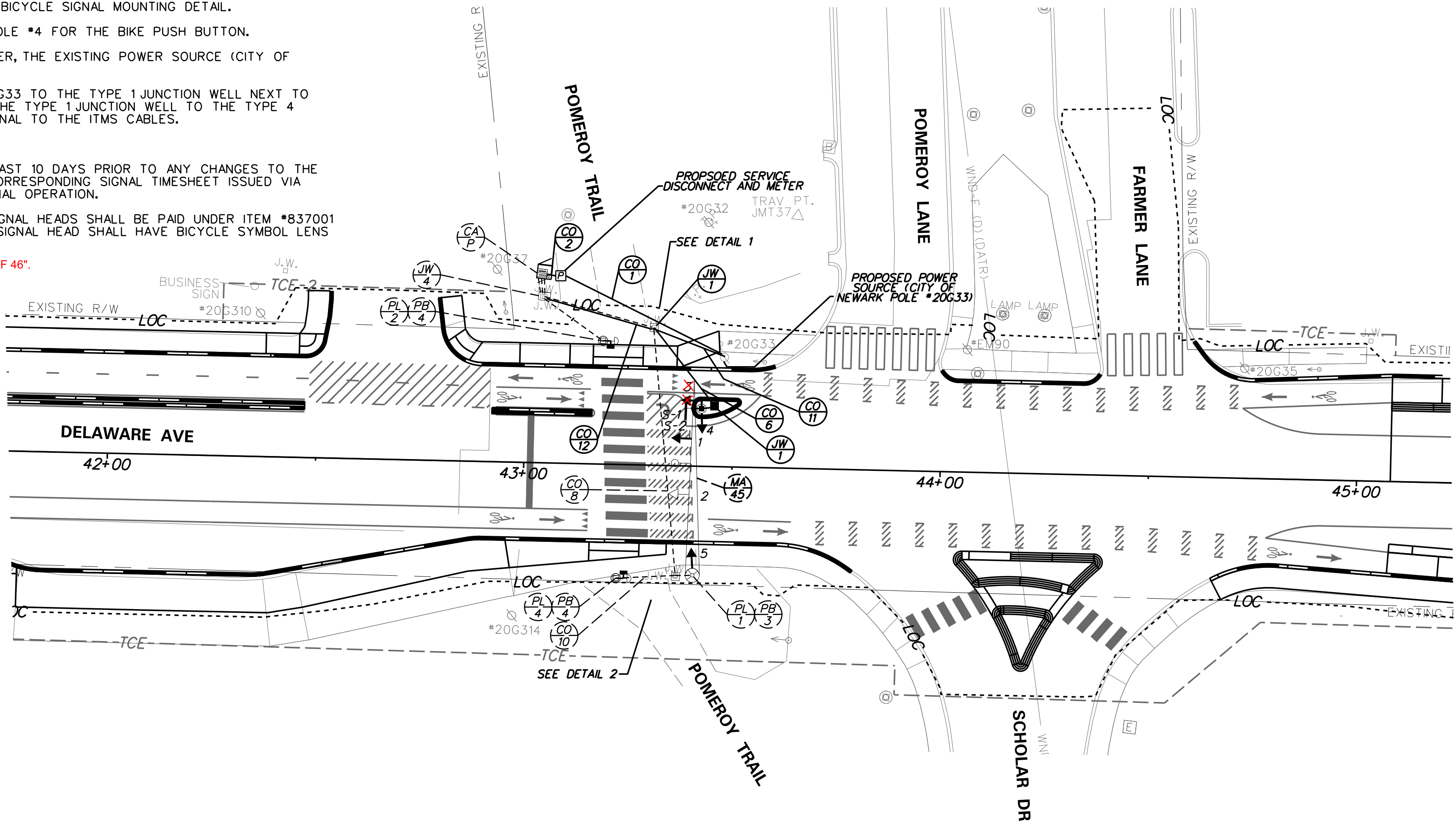
SIGN DETAILS

ALL SIGNS ARE RIGID MOUNT

ALL 
→
S-1
CUSTOM
18" X 12"


S-2
R10-26
9" X 15"

ULTIMATE



CONDUIT RUN SCHEDULE					
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
1	1 (GALV. STEEL)	2"	45'	T	(3)*8 THHN
2	1 (GALV. STEEL)	2"	5'	T	(1)2/*8 U.F. W/ GROUND
*3	3	2.5"	5'	-	(2)9/*14, (4)5/*14, (1)4/*18, (3)1/*6 GRD
*4	1	2.5"	20'	-	(1)5/*14, (1)1/*6 GRD
*5	1	2.5"	30'	-	(2)9/*14, (3)5/*14, (1)4/*18, (1)1/*6 GRD
6	1 (SCH 80 HDPE)	4"	25'	B	(2)5/*14, (1)1/*6 GRD
7	1	2.5"	5'	O	(2)5/*14, (1)1/*6 GRD
*8	1 (SCH 80 HDPE)	2.5"	60'	-	(2)9/*14, (1)5/*14, (1)4/*18, (1)1/*6 GRD
*9	1	2.5"	5'	-	(2)9/*14, (1)4/*18, (1)1/*6 GRD
*10	1	2.5"	15'	-	(1)5/*14, (1)1/*6 GRD
11	1	4"	20'	T	DELDOT FORCES
12	1	4"	30'	T	DELDOT FORCES

* EXISTING
NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT

MAST ARM SCHEDULE			
NO.	LENGTH	NO. OF SIGNAL HEADS	S.F. OF SIGNAGE
*1	45' MAST ARM	2 - 3 SECTION	0 SF

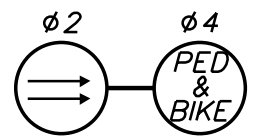
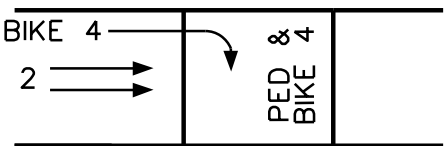
* EXISTING

POLE SCHEDULE			
NO.	TYPE	STATION	OFFSET
*1	45' MAST ARM W/ TYPE 3 BASE	43+41	26' RT
*2	10' PEDESTAL POLE W/ TYPE 4 BASE	43+18	30' LT
3	14' PEDESTAL POLE W/ TYPE 6 BASE	43+42	14' LT
*4	10' PEDESTAL POLE W/ TYPE 4 BASE	43+23	27' RT

* EXISTING
THE 14' PEDESTAL POLE SHALL BE SPUN ALUMINUM, SCHEDULE 80.

SIGNAL PHASING

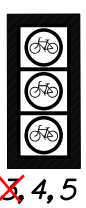
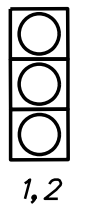
NEMA PHASING



PHASING NOTES

1. PHASES ASSOCIATED BY A SOLID LINE WILL NOT OPERATE CONCURRENTLY.
2. PHASES ASSOCIATED BY A DASHED LINE WILL OPERATE CONCURRENTLY.

SIGNAL HEAD DIAGRAM



LEGEND

(AB)	ABANDON	(OH)	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	(OH)	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	(PB)	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PB)	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PL)	EXISTING POLE IDENTIFIER (* OF POLE)
(JW)	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(PL)	PROPOSED POLE IDENTIFIER (* OF POLE)
(JW)	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(RM)	REMOVE BY CONTRACTOR
(MA)	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY OTHERS
(MA)	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	◊	◊
MAST ARM	▶	▶
MICROWAVE DETECTION	◊	◊
OPTICOM RECEIVER	◊	◊
OVERHEAD SIGNING	◊	◊
PEDESTRIAN POLE/BASE	⊙	⊙
PEDESTRIAN PUSHBUTTON	→	→
PEDESTRIAN SIGNAL HEAD	→	→
RIGHT-OF-WAY	---	R/W
SERVICE PEDESTAL	□	□
SIGNAL CABINET	□	□
SIGNAL HEAD	→	→
SIGNAL POLE/BASE	⊙	⊙
CCTV CAMERA	◊	◊
SPAN WIRE	XX	XX
UTILITY POLE	⊙	⊙
VIDEO DETECTION	◊	◊

GENERAL SIGNAL NOTES

1. ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
2. ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY

JMT

Mir A. Wahed

THIS SEAL APPLIES TO ALL SHEETS
BEARING THE "JMT" SECTION DESIGNATION.

07/20/2021

DATE



SEAL

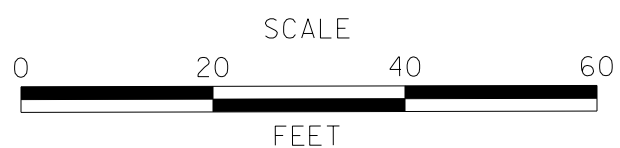
CONCURRENCE FOR INSTALLATION

[Signature]
CHIEF TRAFFIC ENGINEER

7/28/2021
DATE

ADDENDA / REVISIONS

REMOVE BICYCLE SIGNAL HEAD 2/2/23 (J.Smith)



DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT

T201801501

COUNTY

NEW CASTLE

PERMIT NO.

N609T

DESIGNED BY: J.SMITH

CHECKED BY: M.WAHED

SIGNAL PLAN
DELAWARE AVE AND
POMEROY TRAIL

SECTION

JMT

SHEET NO.

192

ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.

4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.

5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (501 OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42 TO 48 INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.

6. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BOTH VEHICLES AND BICYCLES, AS WELL AS PROVIDE INTERSECTION BIKE COUNTS.

6. THE FISHEYE CAMERA SYSTEM WILL PROVIDE DETECTION FOR BICYCLES ONLY AND WILL PROVIDE INTERSECTION BIKE COUNTS. LOOP DETECTORS WILL PROVIDE DETECTION FOR VEHICLES ONLY.

7. THE FISHEYE DETECTOR SHALL BE PLACED ON THE MAST ARM TO ACHIEVE OPTIMUM DETECTION PER MANUFACTURER'S SPECIFICATIONS. THE DESIRED DETECTION ZONES SHALL BE SET AND APPROVED BY DELDOT.

8. THE EXISTING CONTROLLER CABINET SHALL BE MAINTAINED UNTIL THE PROPOSED CONTROLLER CABINET IS OPERATIONAL.

9. REMOVAL OF POLE BASES SHALL BE PAID FOR BY *211000.

10. ALL EXISTING LOOP DETECTORS SHALL BE ABANDONED.

11. R10-3E SIGNS SHALL BE INSTALLED ON POLES WITH ONE PEDESTRIAN SIGNAL HEAD. R10-3E-DE SIGNS SHALL BE INSTALLED ON POLES WITH TWO PEDESTRIAN SIGNAL HEADS. INSTALL A R10-3E RIGHT SIGN ON PEDESTRIAN POLE *5.

12. BACKPLATES WILL BE INSTALLED FOR SIGNAL HEAD *S 7, 8, 9, 10, 11, AND 12. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATE.

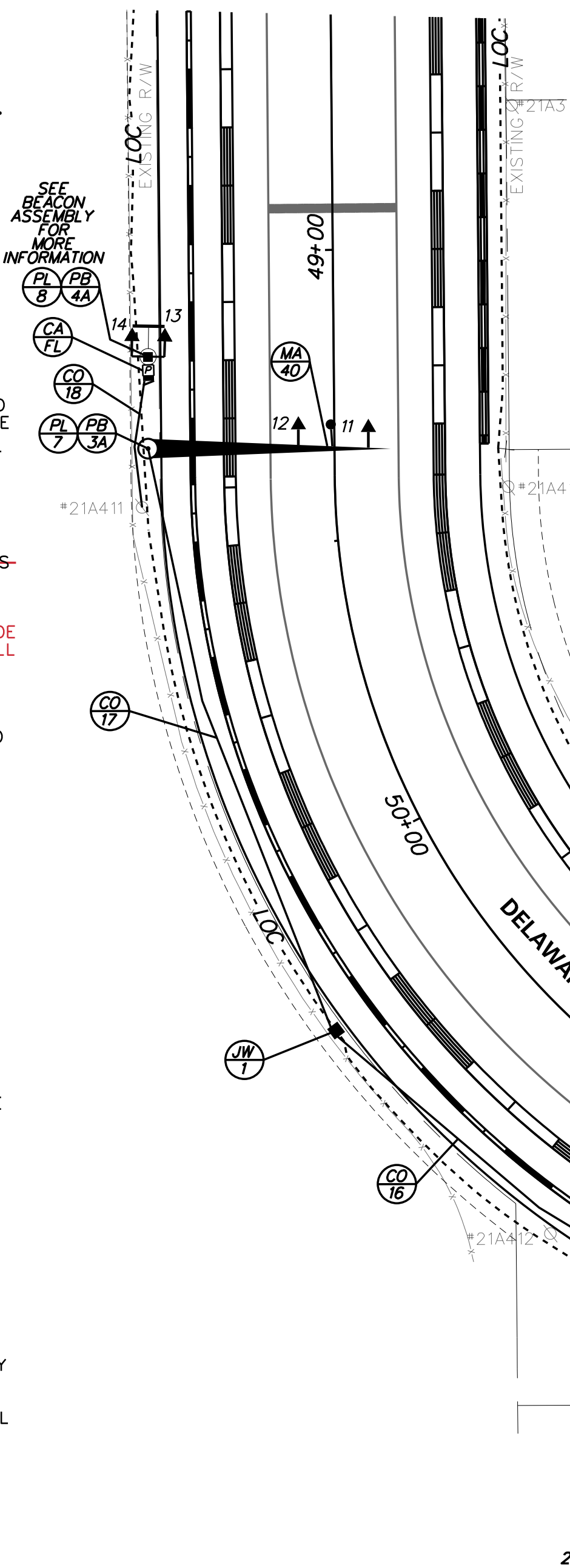
13. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10', AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.

14. THE EXISTING SPAN WIRE AT THE INTERSECTION AND AT THE ADVANCE SIGNAL SHALL BE REMOVED. THE UTILITY POLES SHALL REMAIN.

15. AN ITMS CONDUIT SHALL BE RUN FROM UTILITY POLE *21A42 TO THE TYPE 4 JUNCTION WELL NORTH OF POLE. A SEPARATE ITMS CONDUIT SHALL BE RUN FROM THE TYPE 4 JUNCTION WELL TO THE TYPE 4 JUNCTION WELL NEXT TO THE CABINET TO CONNECT THE SIGNAL TO THE ITMS CABLES.

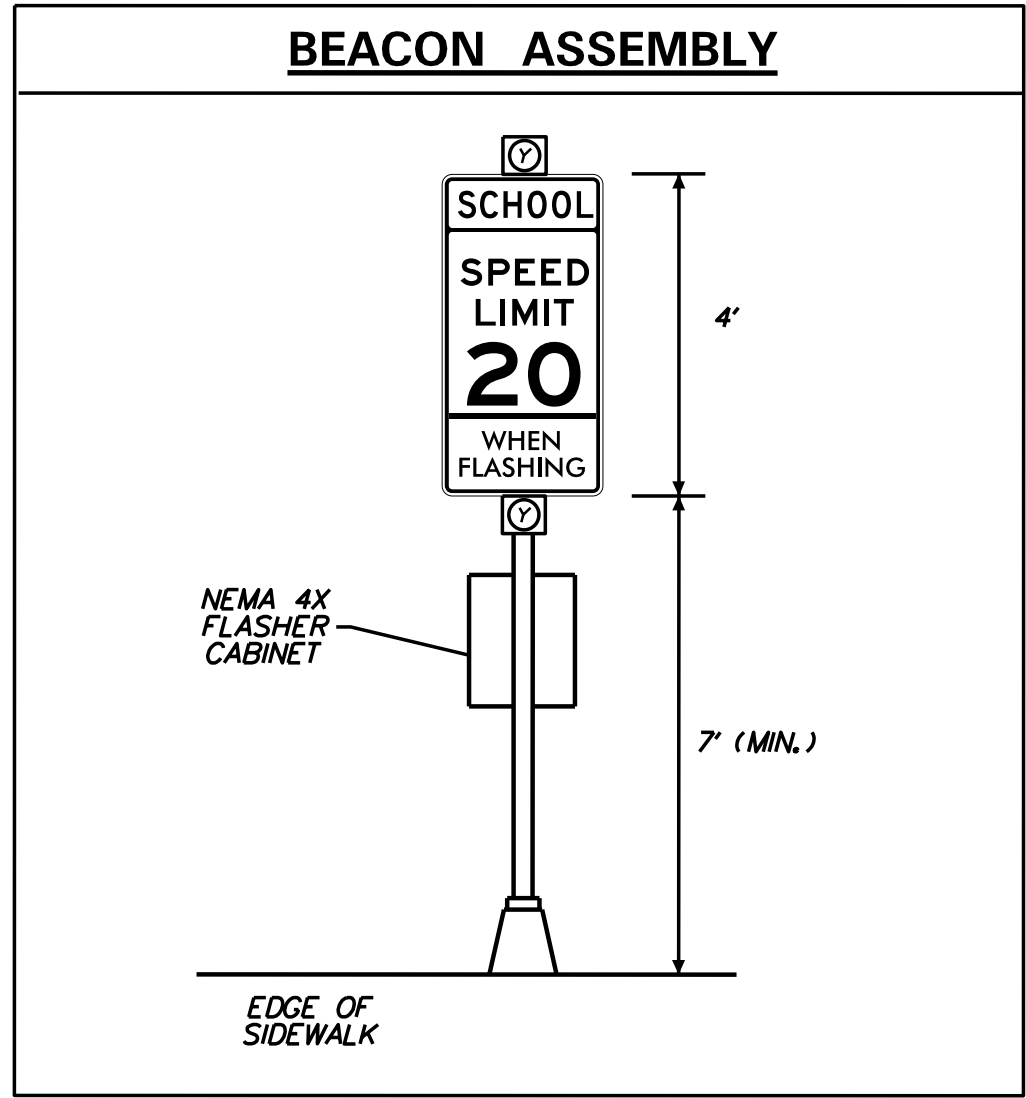
16. INSTALL TYPE 4 ANCHOR BOLTS ON TYPE 6 POLE BASE FOUNDATIONS.

17. MAINTAIN THE EXISTING SERVICE DISCONNECT FOR THE TYRE AVENUE SIGNAL. THE NEW SIGNAL CABINET WILL BE POWERED THROUGH THE EXISTING DISCONNECT.

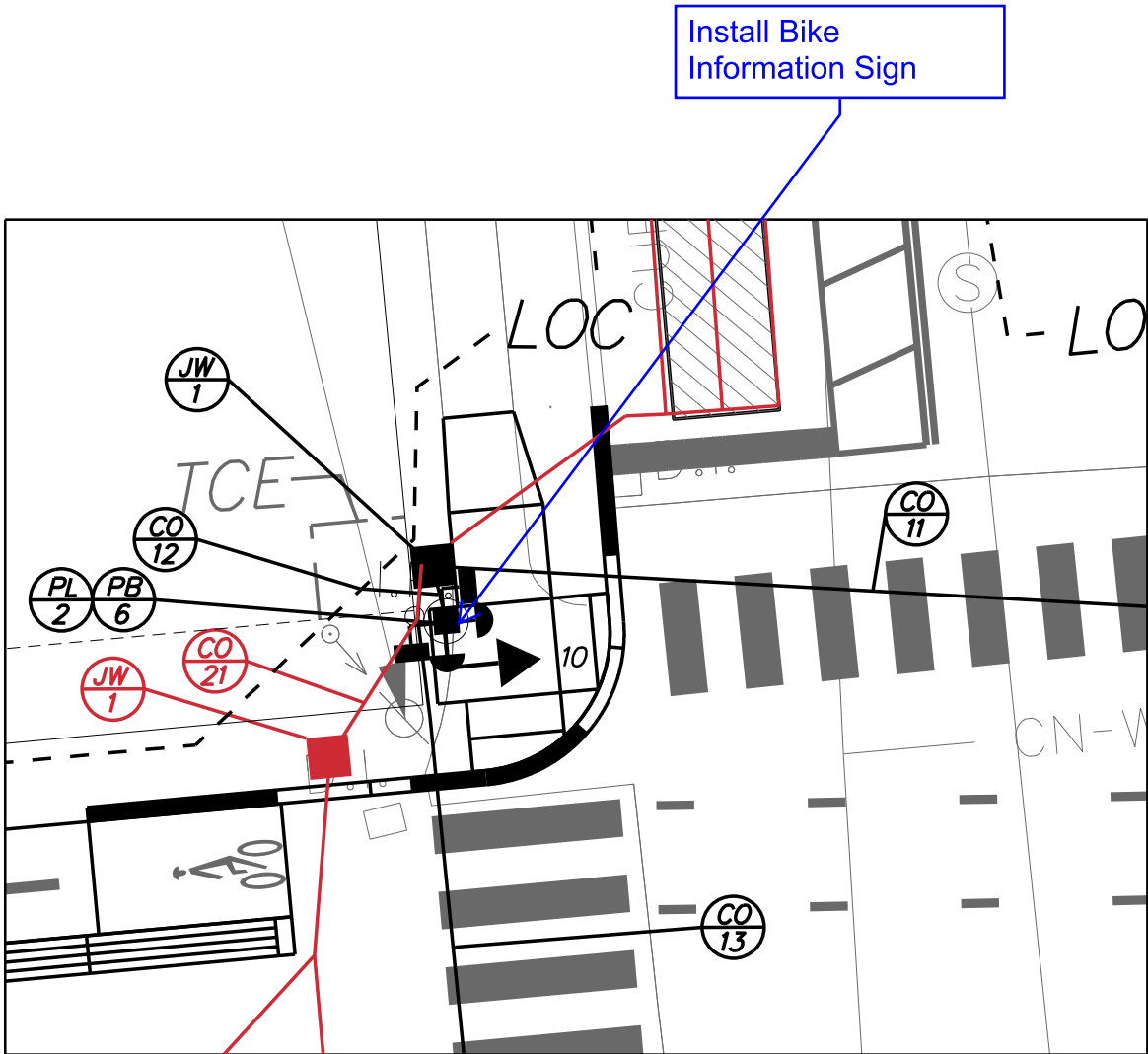


CONDUIT RUN SCHEDULE					
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
*1A	1 (GALV. STEEL)	2"	5'	-	(3)#8 THHN
1B	1 (GALV. STEEL)	2"	5'	T	(1)2/#8 U.F. W/ GROUND
2	1 (GALV. STEEL)	2"	50'	B	(1)2/#8 U.F. W/ GROUND
3	1 (GALV. STEEL)	2"	10'	T	(1)2/#8 U.F. W/ GROUND
4	4	4"	10'	T	(6)5/#14, (1)/16/#14, (5)9/#14, (4)4/#18, GRIDSMA RT CAT5E CABLE, (4)1/#6 GRD, (4)2/#14
5	1	3"	10'	T	(1)16/#14, (3)4/#18, GRIDSMA RT CAT5E CABLES, (1)1/#6 GRD
6	1 (SCH 80 HDPE)	4"	60'	B	(1)5/#14, (1)9/#14, (1)1/#6 GRD, (1)2/#14
7	1	2.5"	10'	T	DELDOT FORCES
8	1 (SCH 80 HDPE)	4"	65'	B	(2)5/#14, (1)1/#6 GRD
9	1	2.5"	10'	T	EMPTY
10	1	2.5"	10'	T	(1)9/#14, (1)1/#6 GRD
11	1 (SCH 80 HDPE)	4"	60'	B	(1)5/#14, (1)9/#14, (1)1/#6 GRD
12	1	2.5"	5'	T	(1)5/#14, (1)1/#6 GRD
13	1 (SCH 80 HDPE)	4"	85'	B	(2)5/#14, (1)9/#14, (1)1/#6 GRD, (3)2/#14
14	1	2.5"	15'	T	(2)5/#14, (1)9/#14, (1)1/#6 GRD
15	1	4"	110'	T	(1)9/#14, (1)4/#18, (1)1/#6 GRD
16	1	4"	85'	T	(1)9/#14, (1)4/#18, (1)1/#6 GRD
17	1	3"	110'	T	(1)9/#14, (1)4/#18, (1)1/#6 GRD
18A	1 (GALV. STEEL)	3"	20'	T	(1)2/#8 U.F. W/ GROUND
18B	1 (GALV. STEEL)	3"	10'	T	(1)2/#8 U.F. W/ GROUND, (1)2/#14
19	1	4"	10'	T	DELDOT FORCES
20	1	4"	20'	T	(1)2/#14
21	1	4"	15'	T	(2)2/#14
22	1	3"	10'	T	(1)2/#14

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT



- SCHOOL SPEED LIMIT BEACON ASSEMBLY SHALL CONSIST OF A PAIR OF ALTERNATELY FLASHING BEACONS AND A 'SCHOOL SPEED LIMIT 20 WHEN FLASHING' SIGN (S5-1 24"x48" MOUNTED ON A PEDESTAL POLE.
- THE BEACONS SHALL BE 12-INCH YELLOW LED ONE-SECTION TRAFFIC SIGNAL HEADS.
- THE BEACONS SHALL OPERATE ONLY DURING ACTIVE SCHOOL HOURS.
- THE FLASHING RATE SHALL NOT BE LESS THAN 50 OR MORE THAN 60 TIMES PER MINUTE.
- THE ILLUMINATED PERIOD OF EACH FLASH SHALL NOT BE LESS THAN 1/2 NOR MORE THAN 2/3 OF THE FLAST CYCLE.
- SIGNFIX (OR SIMILAR) SHALL BE INSTALLED ON THE BACK OF THE PROPOSED SIGN FOR MOUNTING SUPPORT.
- TO INSTALL THE PROPOSED SIGN AND BEACONS AT THE REQUIRED MOUNTING HEIGHTS, 4-INCH RIGID ALUMINUM CONDUIT SHALL BE THREADED TO THE TOP OF THE PEDESTAL POLE.
- THE PEDESTAL POLE SHALL BE SPUN ALUMINUM, SCHEDULE 80.



ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.
4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.
5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42 TO 48 INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.
6. BACKPLATES WILL BE INSTALLED FOR ALL SIGNAL HEADS. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE NEW BACKPLATES.
7. THE EXISTING CONTROLLER CABINET SHALL BE MAINTAINED UNTIL THE PROPOSED CONTROLLER CABINET IS OPERATIONAL.
8. INSTALL CDMA IN THE PROPOSED CABINET PRIOR TO DISRUPTING THE EXISTING ITMS FIBER CONNECTION.
9. THE BICYCLE SIGNALS SHALL BE BAGGED UNTIL THE SEPARATED BIKEWAY IS OPENED AT THE END OF PROJECT CONSTRUCTION.
10. THE CORNERS OF THE INTERSECTION WILL BE CONSTRUCTED ONE AT A TIME TO MAINTAIN PEDESTRIAN ACCESS. WHEN A PEDESTRIAN MOVEMENT IS PROHIBITED, THE CORRESPONDING PEDESTRIAN SIGNAL HEADS SHALL BE BAGGED.
11. REMOVAL OF POLE BASES SHALL BE PAID BY ITEM #211000.
12. R10-3E SIGNS SHALL BE INSTALLED ON POLES WITH ONE PEDESTRIAN SIGNAL HEAD.
13. INSTALL PROPOSED TYPE 4 JUNCTION WELL AT STATION L14+36 OFFSET 43' RIGHT OVER EXISTING CONDUIT #34.
14. EXISTING LUMINAIRES MOUNTED ON EXISTING STRAIN SIGNAL POLES SHALL BE REMOVED BY TRAFFIC CONTRACTOR.

ADDITIONAL GENERAL SIGNAL NOTES

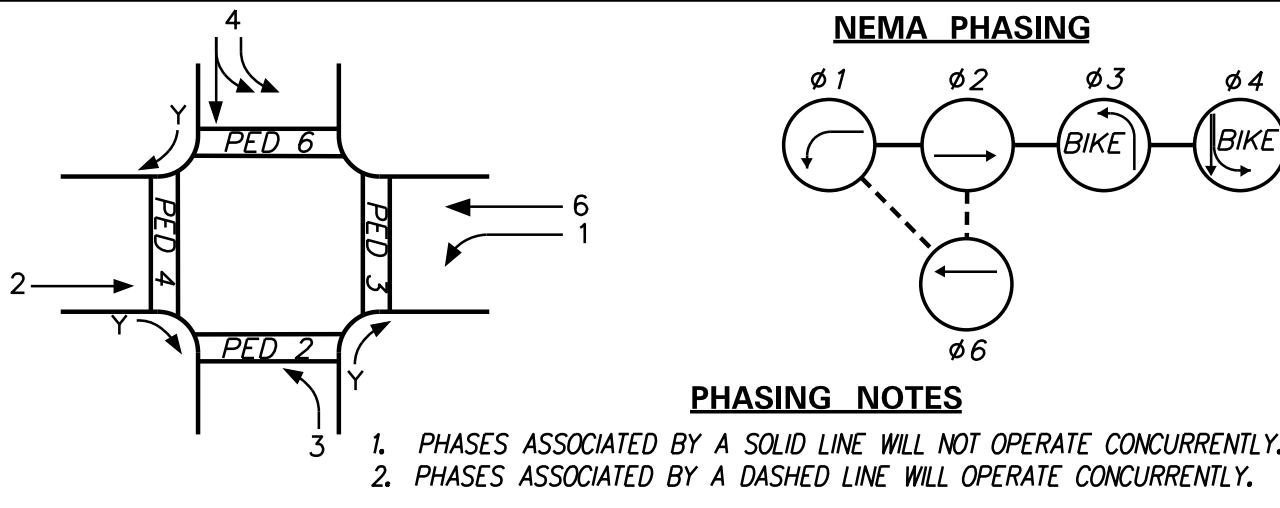
15. BICYCLES WILL BE DETECTED VIA TYPE "D" LOOP DETECTORS ALONG EB DELAWARE AVENUE. BICYCLES WILL BE DETECTED VIA PUSHBUTTONS ALONG WESTBOUND DELAWARE AVENUE.
16. THE BICYCLE SIGNALS (*8 and *11) WILL BE IN PHASE WITH THE PEDESTRIAN SIGNALS. BOTH THE BICYCLE AND SIGNALS AND THE PEDESTRIAN SIGNALS WILL BE ACTIVATED WHEN THERE IS A CALL AT THE PEDESTRIAN PUSH BUTTONS OR THE BICYCLE PUSHBUTTONS OR BICYCLE LOOP DETECTORS.
17. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10' TALL, AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14' TALL, UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.
18. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO ANY CHANGES TO THE TRAFFIC SIGNAL OPERATION, AND WHEN THE DETECTION LOOPS ARE CUT. THE CONTRACTOR SHALL HAVE CORRESPONDING SIGNAL TIMESHEET ISSUED VIA DELDOT TRAFFIC/TMC PRIOR TO MODIFYING THE TRAFFIC SIGNAL OPERATION.
19. CONNECT EXISTING CONDUIT #13 AND PROPOSED CONDUITS #12, #14, AND #38 INTO EXISTING JUNCTION WELL AT STATION L12+88 OFFSET 5' LEFT.

ADDITIONAL GENERAL SIGNAL NOTES

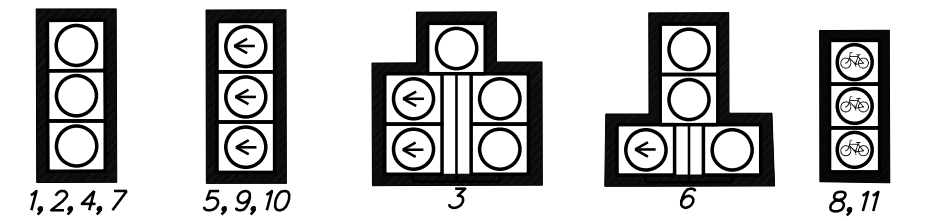
20. CONNECT EXISTING CONDUIT #22 AND PROPOSED CONDUITS #21, #23, #24, AND #25 INTO EXISTING JUNCTION WELL AT STATION L13+58 OFFSET 46' LEFT.
21. REPLACE EXISTING TYPE 1 JUNCTION WELL AT STATION 13+02 OFFSET 58' LEFT WITH A NEW TYPE 1 JUNCTION WELL. MAINTAIN ALL EXISTING AND PROPOSED CONDUIT CONNECTIONS AS NEEDED.
22. INSTALL TEMPORARY CONDUIT DURING SIGNAL CONSTRUCTION FROM PROPOSED TYPE 4 JUNCTION WELL ON SOUTHWEST CORNER TO EXISTING STRAIN POLE AT STATION L12+58 OFFSET 88' LEFT.
23. CONTRACTOR SHALL REPLACE AND RECONNECT THE LOOP DETECTORS PRIOR TO STARTING WORK AT ANOTHER INTERSECTION / LOCATION AS PART OF THIS PROJECT.
24. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM #837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIDIG MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.
25. IF A PEDESTRIAN CROSSING IS CLOSED, BAG THE CORRESPONDING SIGNAL HEADS.
26. WHEN THE DELAWARE AVENUE APPROACH IS CLOSED, THE CONTRACTOR SHALL BAG THE CORRESPONDING SIGNAL HEADS AND THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO THE CHANGE TO THE SIGNAL OPERATION.
27. INSTALL ADVANCE OPTICOM SENSOR AT STATION 53+77 OFFSET 17' RIGHT SHOWN ON SHEET 203A OF THIS SIGNAL PLAN. THE OPTICOM SENSOR SHALL BE OPERATIONAL UPON THE RE-OPENING OF THE SECTION OF DELAWARE AVENUE BETWEEN TYRE AVENUE AND LIBRARY AVENUE. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO THE RE-OPENING OF THIS SEGMENT.

ULTIMATE

SIGNAL PHASING



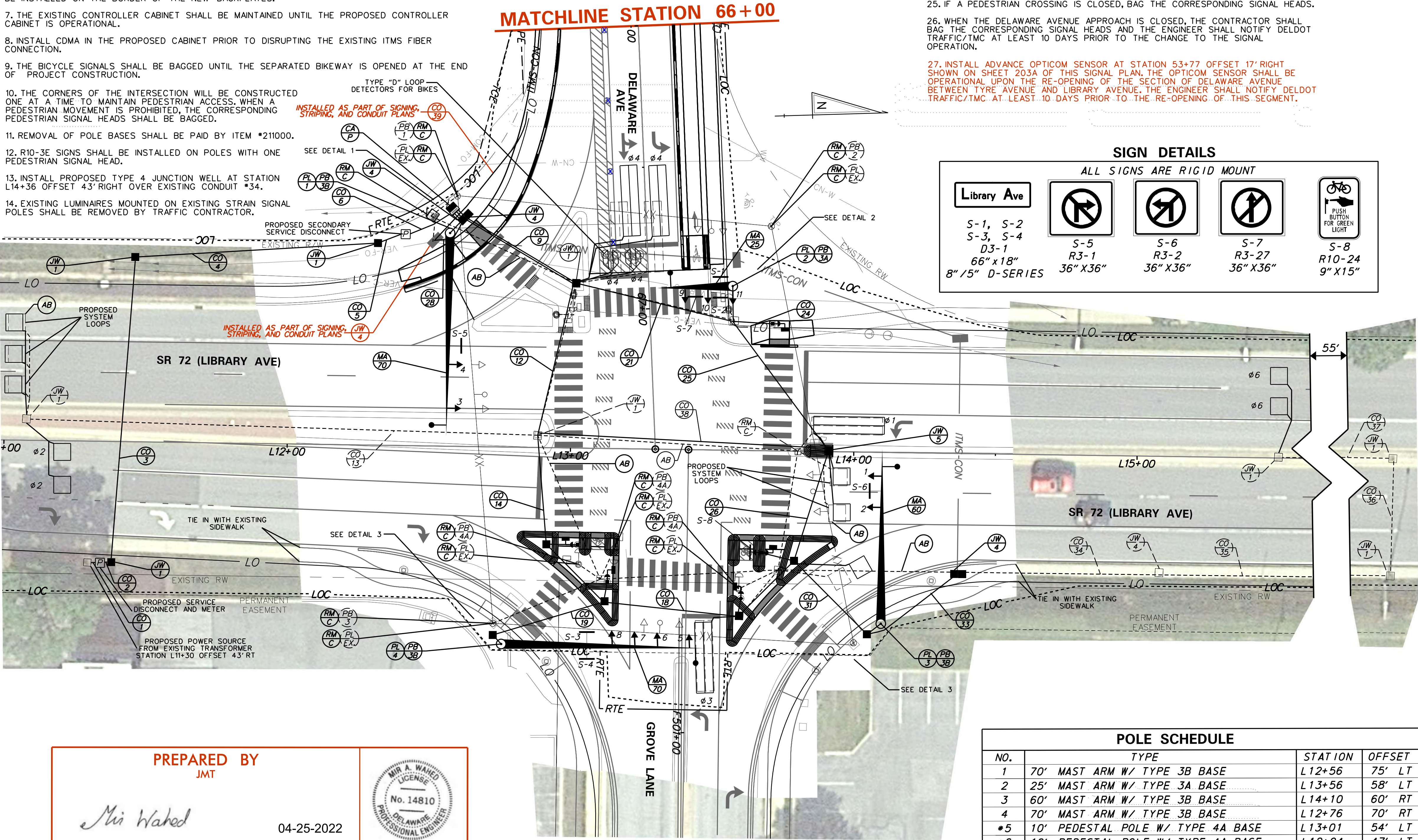
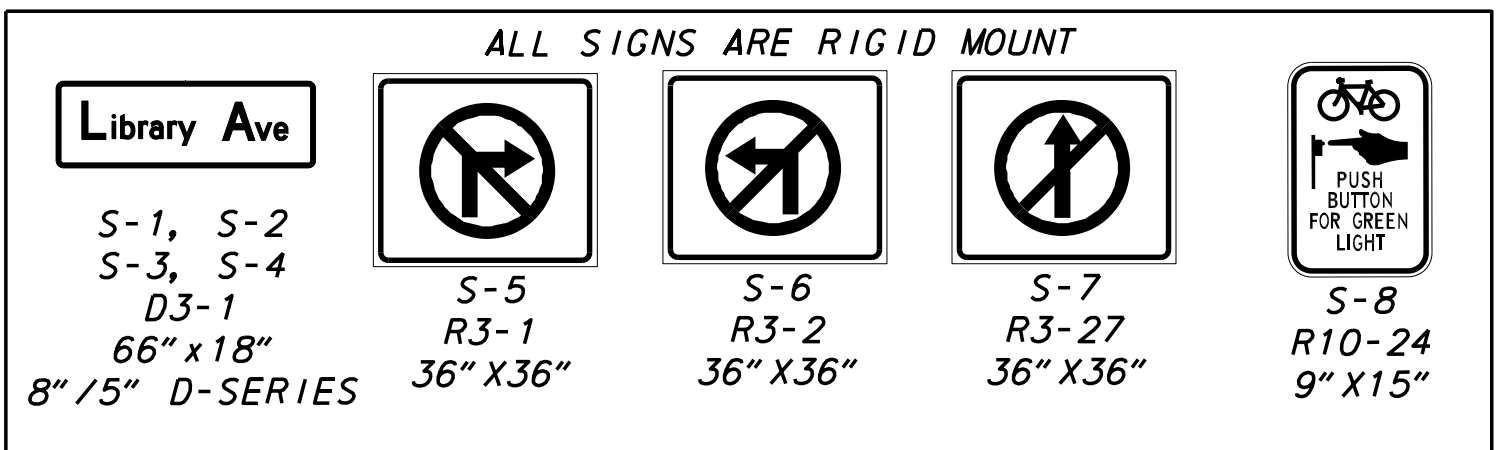
SIGNAL HEAD DIAGRAM



LEGEND

AB	ABANDON	OH	EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
CA	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	OH	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
CA	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	PB	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
CO	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	PB	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
CO	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	PL	EXISTING POLE IDENTIFIER (* OF POLE)
JW	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	PL	PROPOSED POLE IDENTIFIER (* OF POLE)
JW	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	RM	REMOVE BY CONTRACTOR
MA	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	RM	REMOVE BY OTHERS
MA	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	RM	REMOVE BY TRAFFIC CONTRACTOR

SIGN DETAILS



MAST ARM SCHEDULE

NO.	LENGTH	NO. OF SIGNAL HEADS	S. F. OF SIGNAGE
1	70' MAST ARM	1-3 SECTION, 1-5 SECTION	9 SF
2	25' MAST ARM	3-3 SECTION W/ 1 ON POLE	17.25 SF
3	60' MAST ARM	2-3 SECTION	9 SF
4	70' MAST ARM	1-4 SECTION, 3-3 SECTION	8.25 SF

POLE SCHEDULE

NO.	TYPE	STATION	OFFSET
1	70' MAST ARM W/ TYPE 3B BASE	L12+56	75' LT
2	25' MAST ARM W/ TYPE 3A BASE	L13+56	58' LT
3	60' MAST ARM W/ TYPE 3B BASE	L14+10	60' RT
4	70' MAST ARM W/ TYPE 3B BASE	L12+76	70' RT
*5	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+01	54' LT
*6	10' PEDESTAL POLE W/ TYPE 4A BASE	L12+94	47' LT
*7	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+55	50' LT
8	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+71	45' LT
9	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+68	36' RT
10	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+61	52' RT
11	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+14	47' RT
12	10' PEDESTAL POLE W/ TYPE 4A BASE	L13+03	34' RT

* EXISTING

PREPARED BY

JMT

Mir Waked

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

CONCURRENCE FOR INSTALLATION

CHIEF TRAFFIC ENGINEER

7/28/2021

DATE

DELAWARE AVENUE SEPARATED BIKEWAY- ORCHARD ROAD TO LIBRARY AVENUE

CONTRACT

T201801501

COUNTY

NEW CASTLE

PERMIT NO.

N424T

DESIGNED BY: S. BUNITSKY/J. SMITH

CHECKED BY: M. WAHED

SIGNAL PLAN

SR 72 (LIBRARY AVE)

AND DELAWARE AVE

SECTION

JMT

SHEET NO.

202

ADDENDA / REVISIONS

ADVANCE OPTICOM 4-25-2022 (MAW)
Remove tubular markers, per construction (DelDOT) - 04/26/2023

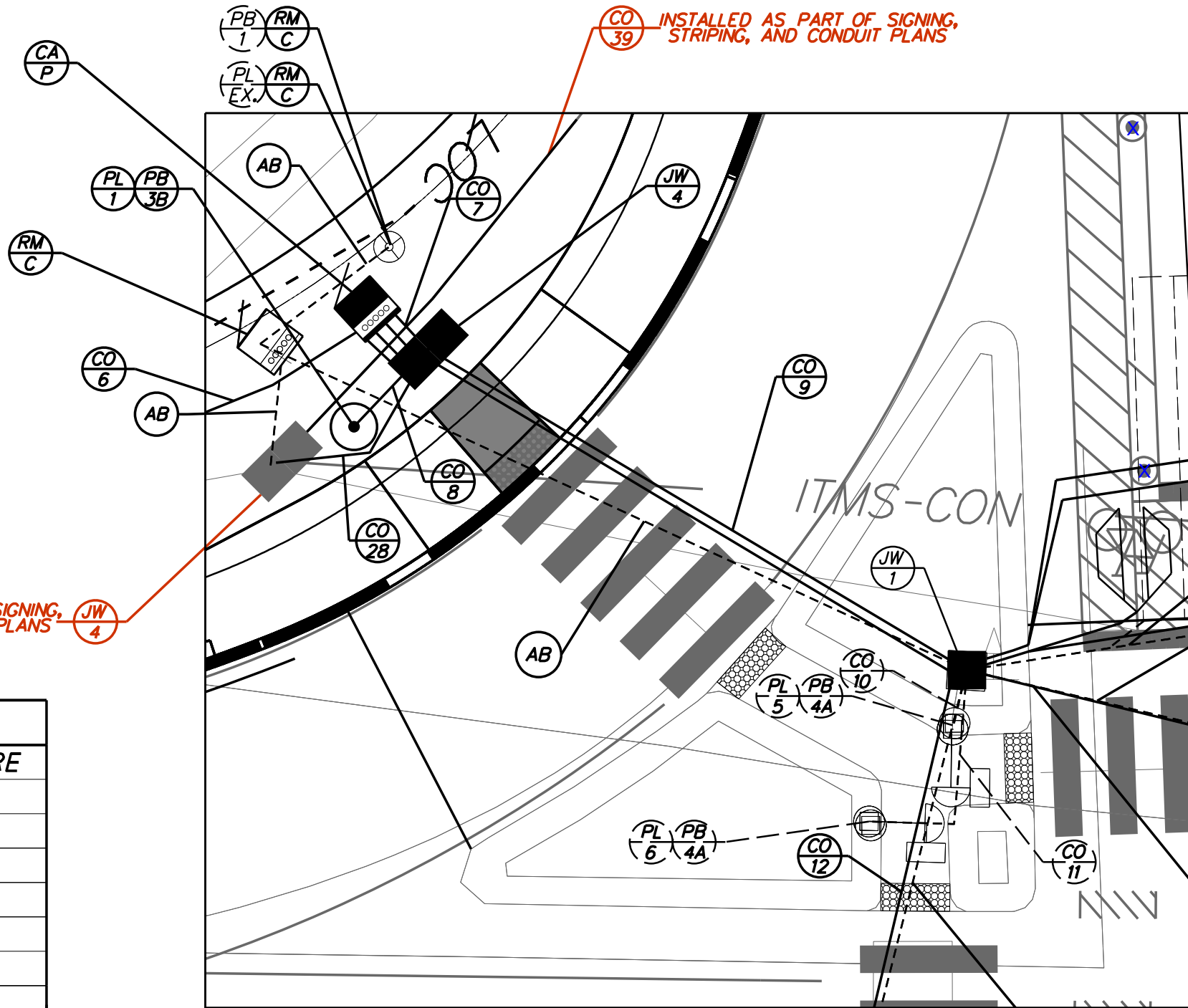


PREPARED BY
JMT

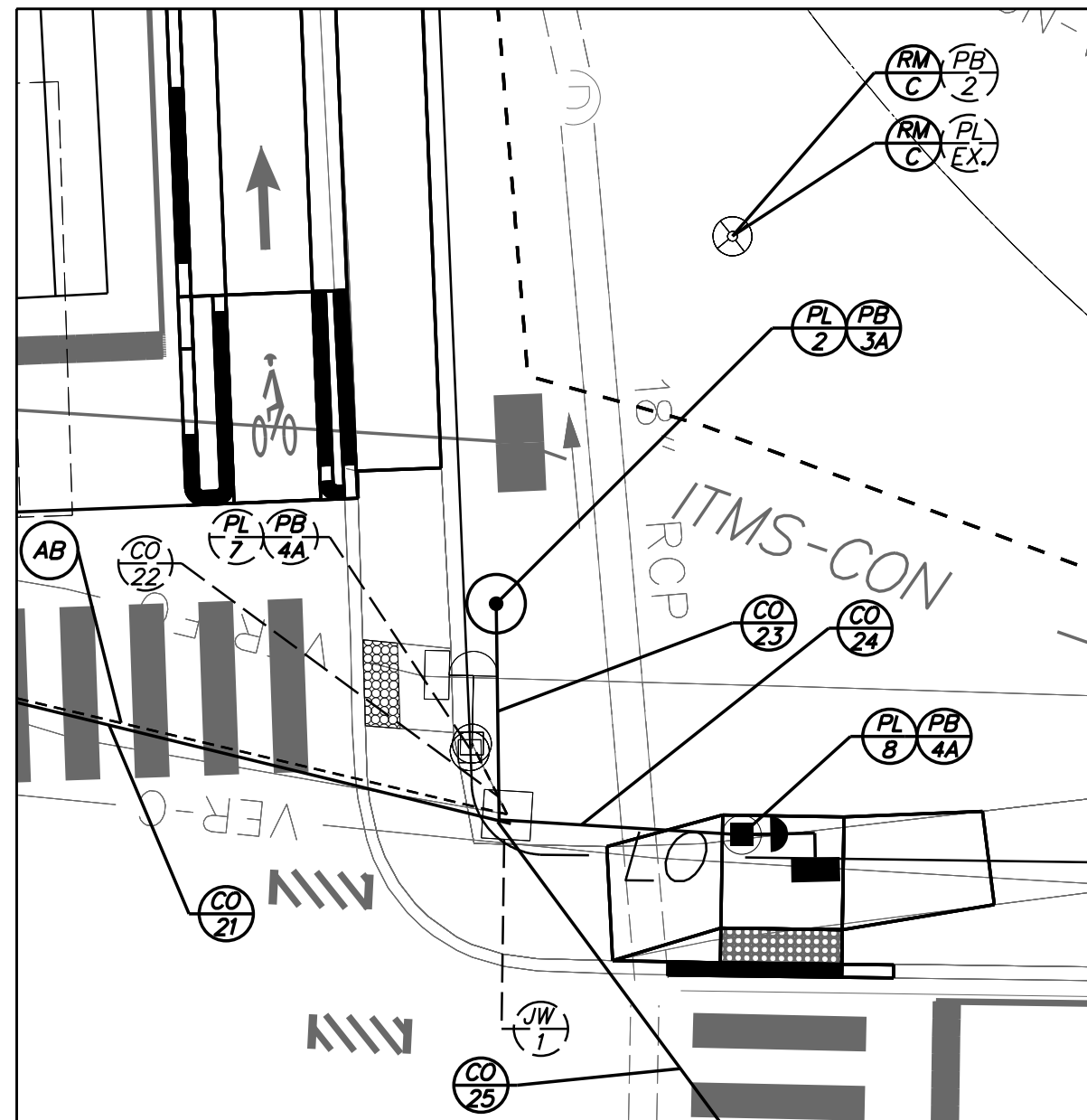
Mu Waked

THIS SEAL APPLIES TO ONLY THIS SHEET

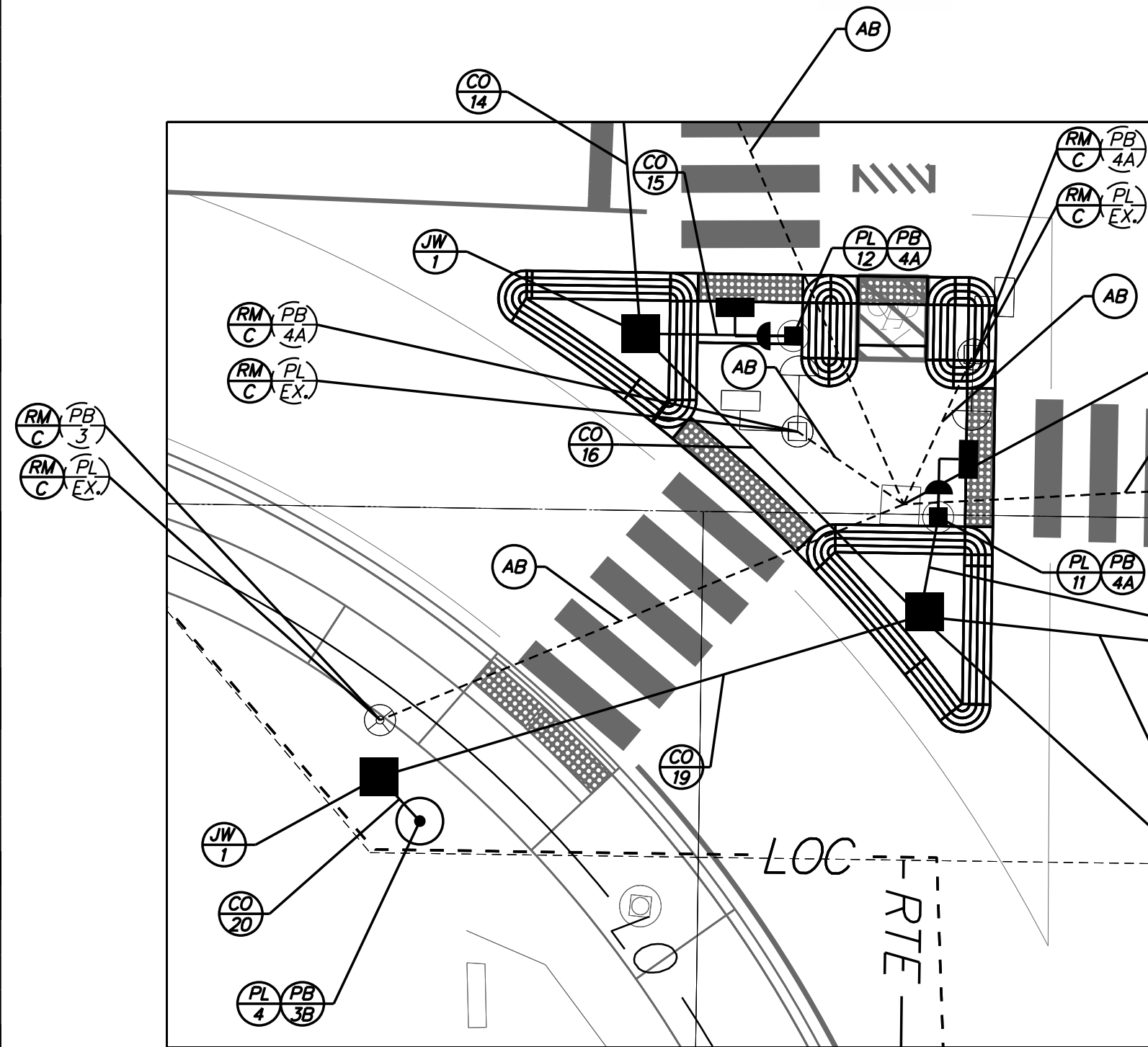
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DATE
SEAL



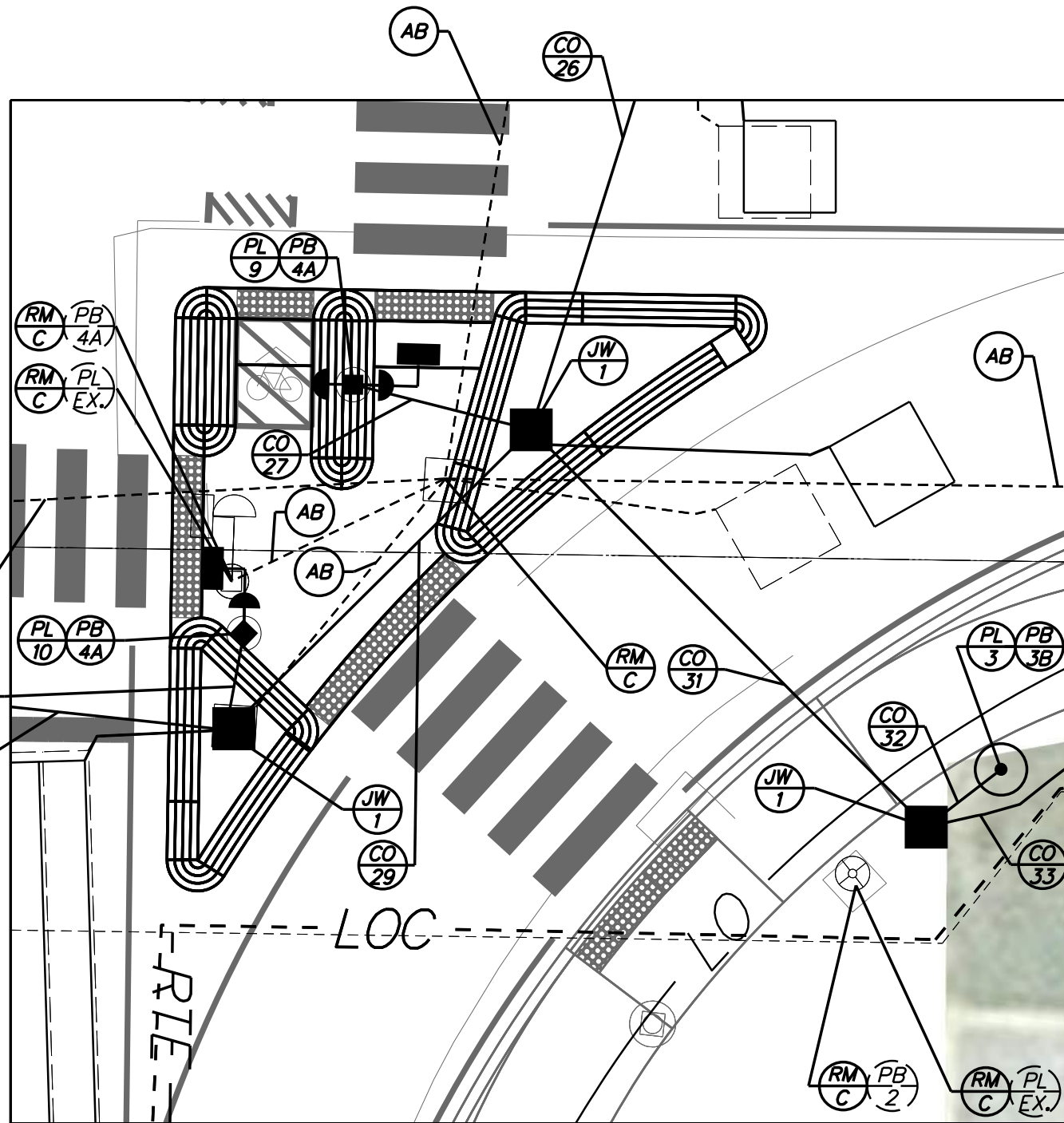
DETAIL 1 - SOUTHWEST CORNER
SCALE: 1"=10'



DETAIL 2 - NORTHWEST CORNER
SCALE: 1"=10'



DETAIL 3 - SOUTHEAST CORNER
SCALE: 1"=10'



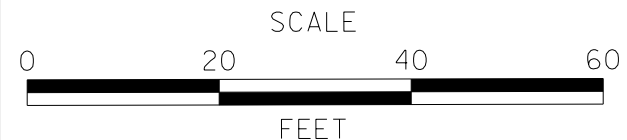
DETAIL 4 - NORTHEAST CORNER
SCALE: 1"=10'

CONDUIT RUN SCHEDULE					
CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
1	1 (GALV. STEEL)	2"	5'	T	(3)2/#8 THHN
2	1 (GALV. STEEL)	2"	5'	T	(1)2/#8 U.F. W/ GROUND
3	1 (GALV. STEEL)	2"	110'	B	(1)2/#8 U.F. W/ GROUND
4	1 (GALV. STEEL)	2"	90'	T	(1)2/#8 U.F. W/ GROUND
5	1 (GALV. STEEL)	2"	10'	T	(1)2/#8 U.F. W/ GROUND
6	1 (GALV. STEEL)	2"	20'	T	(1)2/#8 U.F. W/ GROUND
7	4	4"	5'	T	(6)9/#14, (9)5/#14, (16)2/#14, (4)5/#18, ITMS FIBER, (4)1/#6 GRD
8	1	3"	10'	T	(1)9/#14, (1)4/#18, (1)1/#6 GRD
9	2 (SCH. 80 HDPE)	4"	50'	B	(5)9/#14, (9)5/#14, (16)2/#14, (3)4/#18, (1)1/#6 GRD
*10	1	2.5"	5'	-	(1)5/#14, (1)1/#6 GRD
*11	1	2.5"	20'	-	(1)5/#14, (1)1/#6 GRD
12	1 (SCH. 80 HDPE)	4"	60'	B	(2)9/#14, (2)5/#14, (5)2/#14, (1)4/#18, (1)1/#6 GRD
*13	1	2.5"	180'	-	(5)2/#14, (1)1/#6 GND
14	1 (SCH. 80 HDPE)	4"	40'	B	(2)9/#14, (2)5/#14, (1)4/#18, (1)1/#6 GND
15	1	2.5"	15'	T	(1)5/#14, (1)1/#6 GRD
16	1	4"	30'	T	(2)9/#14, (1)5/#14, (1)4/#18, (1)1/#6 GND
17	1	2.5"	10'	T	(1)5/#14, (1)1/#6 GRD
18	1 (SCH. 80 HDPE)	4"	50'	B	EMPTY
19	1 (SCH. 80 HDPE)	4"	40'	B	(2)9/#14, (1)4/#18, (1)1/#6 GRD
20	1	3"	10'	T	(2)9/#14, (1)4/#18, (1)1/#6 GRD
21	1 (SCH. 80 HDPE)	4"	60'	B	(3)9/#14, (5)5/#14, (7)2/#14, (2)4/#18, (1)1/#6 GRD
*22	1	2.5"	5'	-	(1)5/#14, (1)1/#6 GRD
23	1	3"	15'	T	(2)9/#14, (1)4/#18, (1)1/#6 GRD
24	1	2.5"	15'	T	(1)5/#14, (1)1/#6 GRD
25	1 (SCH. 80 HDPE)	4"	60'	B	(1)9/#14, (3)5/#14, (7)2/#14, (1)4/#18, (1)1/#6 GRD
26	1 (SCH. 80 HDPE)	4"	45'	B	(1)9/#14, (3)5/#14, (4)2/#14, (1)4/#18, (1)1/#6 GRD
27	1	2.5"	15'	T	(2)5/#14, (1)1/#6 GRD
28	1	4"	15'	T	ITMS FIBER, (1)4/#18, (1)1/#6 GRD
29	1	4"	30'	T	(1)5/#14, (1)2/#14, (1)1/#6 GRD
30	1	2.5"	10'	T	(1)5/#14, (1)1/#6 GRD
31	1 (SCH. 80 HDPE)	4"	40'	B	(1)9/#14, (2)2/#14, (1)4/#18, (1)1/#6 GRD
32	1	3"	10'	T	(1)9/#14, (1)4/#18, (1)1/#6 GRD
33	1	4"	40'	T	(2)2/#14, (1)1/#6 GRD
*34	1	2.5"	75'	-	(2)2/#14, (1)1/#6 GRD
*35	1	2.5"	85'	-	(2)2/#14, (1)1/#6 GRD
*36	1	2.5"	45'	-	(2)2/#14, (1)1/#6 GRD
*37	1	2.5"	45'	-	(2)2/#14, (1)1/#6 GRD
38	1 (SCH. 80 HDPE)	4"	105'	B	EMPTY

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
B=BORE, T=TRENCH, O=OPEN CUT
* = EXISTING
** = INSTALLED AS PART OF SIGNING, STRIPING AND CONDUIT PLANS

ADDENDA / REVISIONS

ADVANCE OPTICOM 4-25-2022 (MAW)
Remove tubular markers, per construction (DelDOT) - 04/26/2023

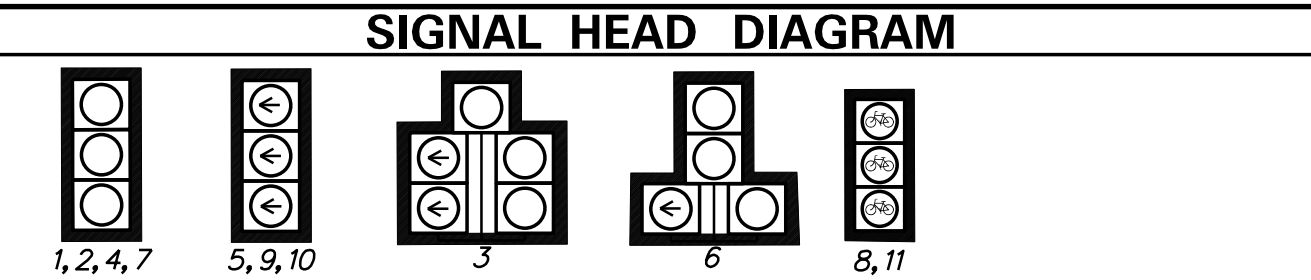
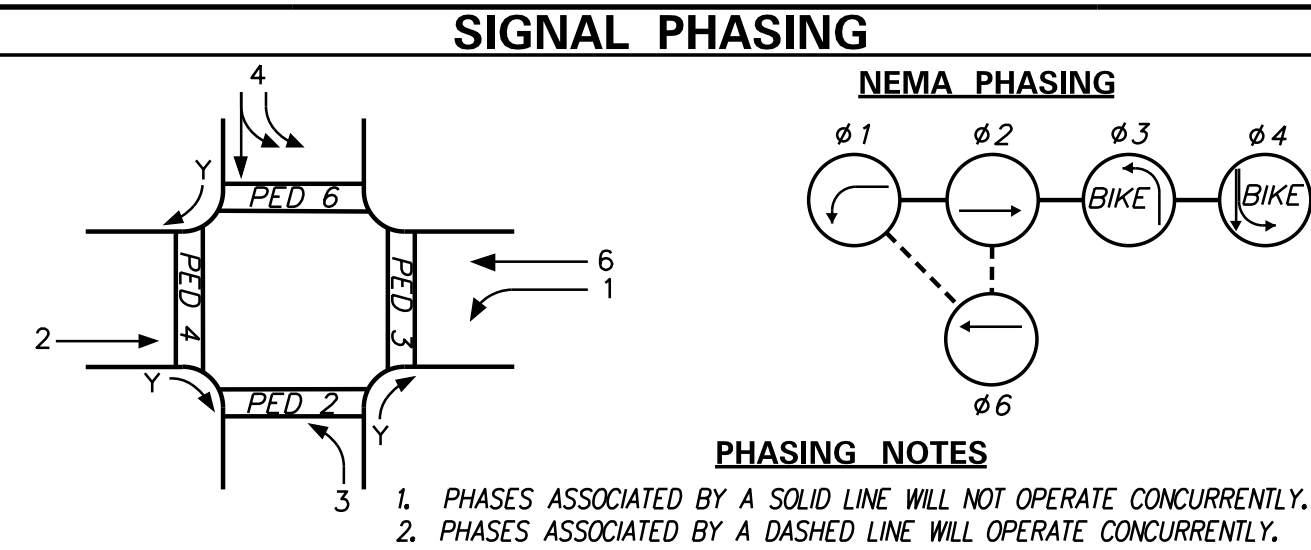


DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT
T201801501
COUNTY
NEW CASTLE
PERMIT NO.
N424T
DESIGNED BY: S. BUNITSKY/J. SMITH
CHECKED BY: M. WAHED

SIGNAL PLAN
SR 72 (LIBRARY AVE)
AND DELAWARE AVE

SECTION
JMT
SHEET NO.
203



LEGEND			
	ABANDON		EXISTING OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)		PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)		EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)		PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)		EXISTING POLE IDENTIFIER (* OF POLE)
	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)		PROPOSED POLE IDENTIFIER (* OF POLE)
	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)		REMOVE BY CONTRACTOR
	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)		REMOVE BY OTHERS
	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)		REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	
LOOP DETECTOR, TYPE 1		
LOOP DETECTOR, TYPE 2		
LUMINAIRE		
MAST ARM		
MICROWAVE DETECTION		
OPTICOM RECEIVER		
OVERHEAD SIGNING		
PEDESTRIAN POLE/BASE		
PEDESTRIAN PUSHBUTTON		
PEDESTRIAN SIGNAL HEAD		
RIGHT-OF-WAY		
SERVICE PEDESTAL		
SIGNAL CABINET		
SIGNAL HEAD		
SIGNAL POLE/BASE		
CCTV CAMERA		
SPAN WIRE		
UTILITY POLE		
VIDEO DETECTION		

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

PREPARED BY
JMT

Mu Waked

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

07/20/2021
DATE
SEAL

CONCURRENCE FOR INSTALLATION

Mu Waked
CHIEF TRAFFIC ENGINEER

7/28/2021
DATE

ADDITIONAL GENERAL SIGNAL NOTES

3. POLE BASES, CABINET BASES, AND CONDUIT JUNCTION WELLS TO BE REMOVED IN ACCORDANCE WITH SECTION 211 OF THE STANDARD SPECIFICATIONS OR AS DIRECTED BY THE ENGINEER. EXISTING CONDUIT IS TO BE ABANDONED.

4. ALL GALVANIZED CONDUIT (GRC) SHALL BE REAMED AND THREADED. ALL GRC SHALL BE THREADED TOGETHER WITH APPROVED COUPLINGS. SET SCREW, BOLTED, AND COMPRESSION FITTING ARE NOT ACCEPTABLE.

5. PROPOSED POLE BASES SUPPORTING POLES WITH PEDESTRIAN PUSHBUTTONS SHOULD BE CONSTRUCTED IMMEDIATELY ADJACENT TO THE FLAT (50:1 OR FLATTER) LANDING AREA OF THE CURB RAMP WHEN POSSIBLE. WHERE THERE ARE PHYSICAL CONSTRAINTS THAT MAKE IT IMPRACTICAL TO PLACE THE PEDESTRIAN PUSHBUTTON ADJACENT TO A LEVEL ALL-WEATHER SURFACE, THE SURFACE SHOULD BE AS LEVEL AS FEASIBLE. THE PEDESTRIAN PUSHBUTTON SHOULD BE INSTALLED AT A HEIGHT OF 42" TO 48" INCHES ABOVE THE ADJOINING SIDEWALK, AND SHOULD BE LOCATED SUCH THAT THE MAXIMUM REACH DISTANCE DOES NOT EXCEED 10 INCHES FROM THE ADJOINING SIDEWALK TO THE FACE OF THE PUSHBUTTON. PEDESTRIAN SIGNAL HEADS SHALL BE MOUNTED WITH THE BOTTOM OF THE SIGNAL HOUSING INCLUDING BRACKETS NOT LESS THAN 7 FEET OR MORE THAN 10 FEET ABOVE.

6. THE BICYCLE SIGNALS (*7, 8 AND 9) WILL BE IN PHASE WITH THE PEDESTRIAN SIGNALS. BOTH THE BICYCLE SIGNALS AND PEDESTRIAN SIGNALS WILL BE ACTIVATED WHEN THERE IS A CALL AT THE PEDESTRIAN PUSH BUTTONS OR BICYCLE PUSH BUTTONS.

7. BACKPLATES WILL BE INSTALLED FOR SIGNAL HEADS *7-9. A 2" REFLECTIVE YELLOW STRIPING WILL BE INSTALLED ON THE BORDER OF THE BACKPLATE.

8. ALL PEDESTAL POLES WITH PEDESTRIAN SIGNALS ARE 10', AND ALL PEDESTAL POLES WITH BICYCLE SIGNALS ARE 14', UNLESS OTHERWISE NOTED. SEE DETAIL SHEETS FOR BICYCLE SIGNAL MOUNTING DETAIL.

9. THE PEDESTRIAN CROSSING ACROSS AMSTEL AVENUE SHALL BE MAINTAINED AT ALL TIMES.

10. UPGRADE THE AUDIBLE PEDESTRIAN PUSH BUTTON STATIONS (APS) WITH THE LATEST VERSION ON THE NORTHWEST CORNER, SOUTHWEST CORNER, AND SOUTHEAST CORNER.

11. FOR ANY PEDESTAL POLE WITH BIKE AND PEDESTRIAN SIGNALS, BIKE SIGNALS SHOULD BE INSTALLED HIGH ENOUGH FOR A PEDESTRIAN SIGNAL AND PUSHBUTTON TO BE INSTALLED BELOW BIKE SIGNALS.

12. THE ENGINEER SHALL NOTIFY DELDOT TRAFFIC/TMC AT LEAST 10 DAYS PRIOR TO ANY CHANGES TO THE TRAFFIC SIGNAL OPERATION. THE CONTRACTOR SHALL HAVE CORRESPONDING SIGNAL TIMESHEET ISSUED VIA DELDOT TRAFFIC/TMC PRIOR TO MODIFYING THE TRAFFIC SIGNAL OPERATION.

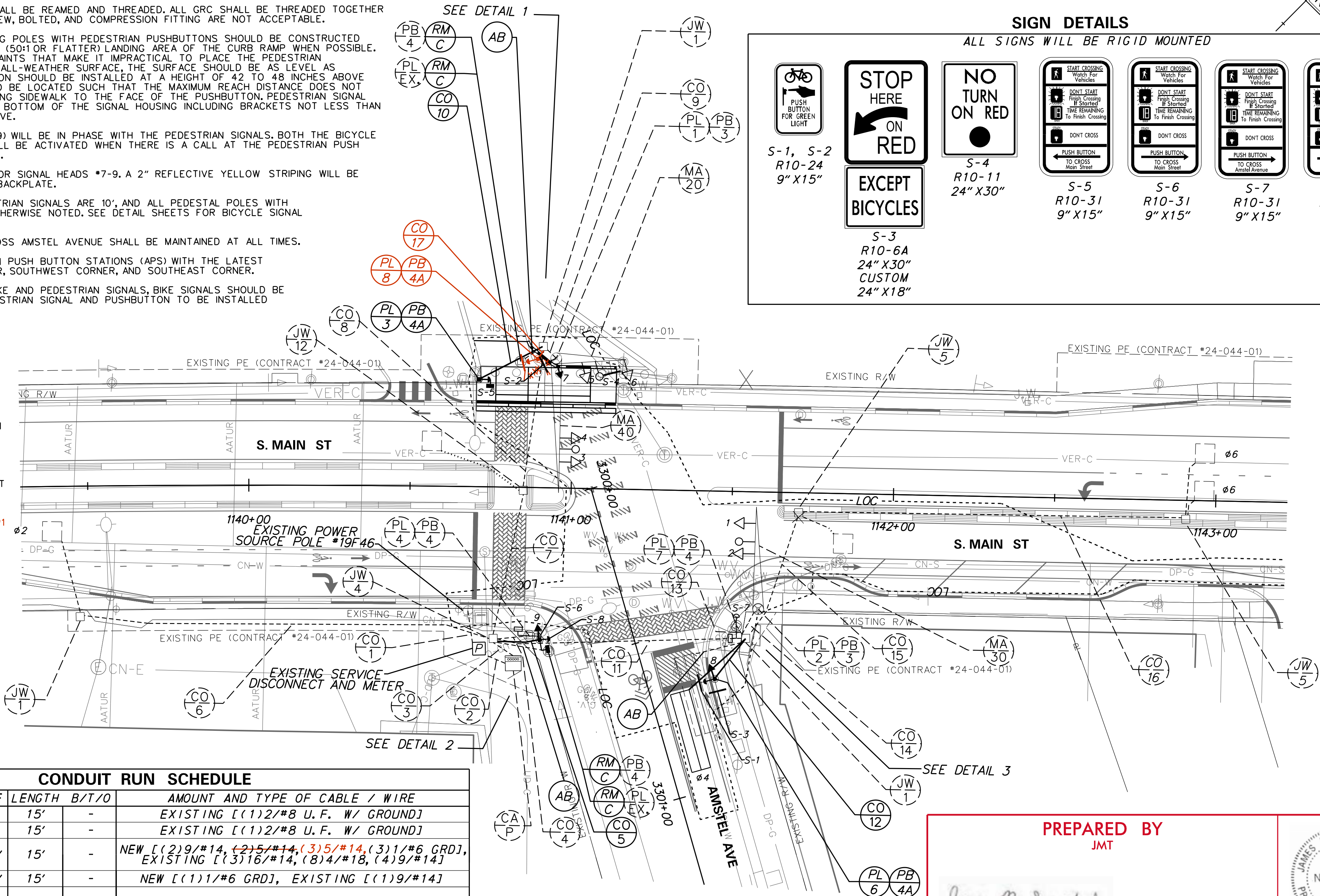
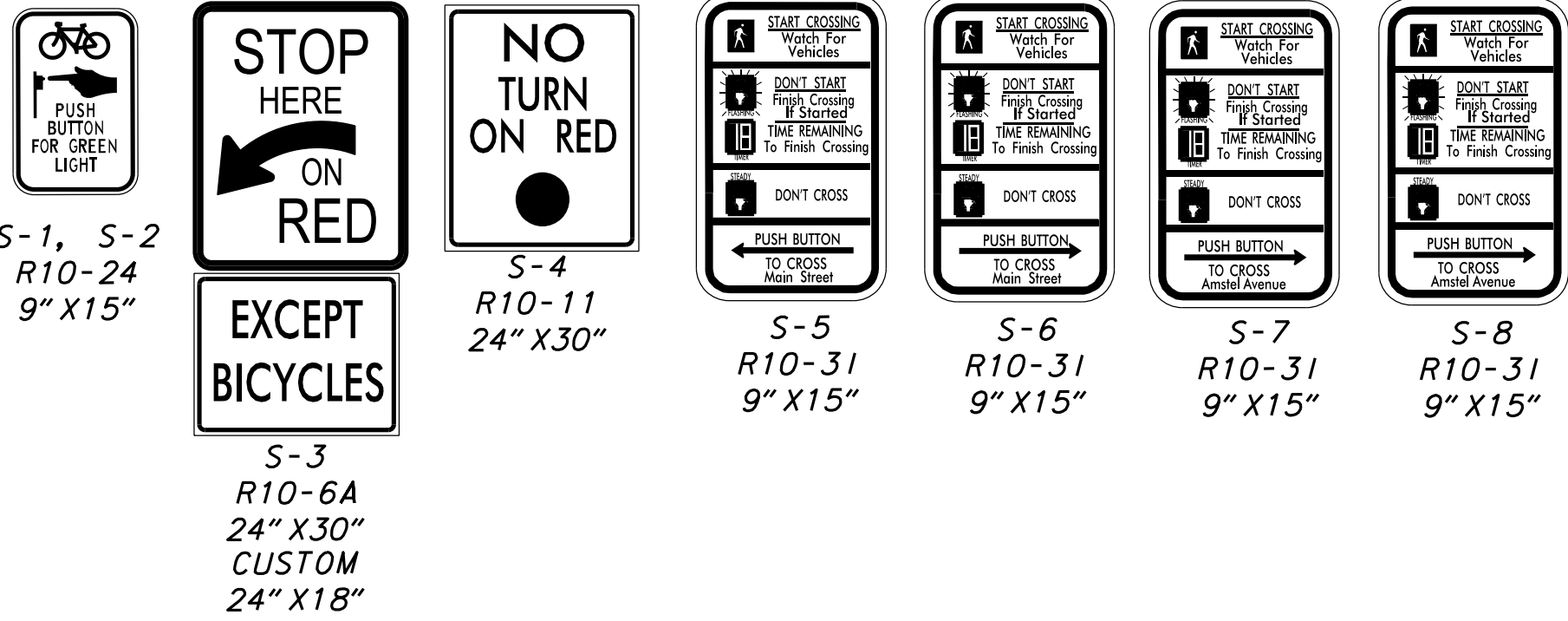
13. THE FURNISHING AND INSTALLATION OF THE 8" BICYCLE SIGNAL HEADS SHALL BE PAID UNDER ITEM *837001 - FURNISH AND INSTALL 8" SIGNAL HEAD RIGID MOUNT. THE SIGNAL HEAD SHALL HAVE BICYCLE SYMBOL LENS INSTALLED AS PER REQUIREMENT OF THE PLANS.

15. BICYCLE PUSHBUTTON AND SIGN SHOULD BE RELOCATED FROM POLE *1 TO POLE *8. POLE *8 IS A 6' PEDESTAL POLE.

ULTIMATE

SIGN DETAILS

ALL SIGNS WILL BE RIGID MOUNTED



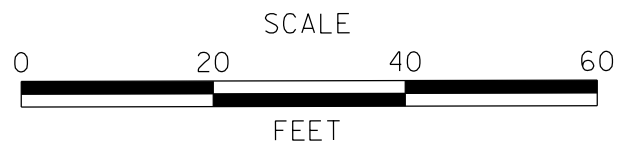
CONDUIT RUN SCHEDULE

CR#	# OF CONDUITS	SIZE	LENGTH	B/T/O	AMOUNT AND TYPE OF CABLE / WIRE
*1	1 (GALV. STEEL)	2"	15'	-	EXISTING [(1)2/#8 U.F. W/ GROUND]
*2	1 (GALV. STEEL)	2"	15'	-	EXISTING [(1)2/#8 U.F. W/ GROUND]
*3	3	2.5"	15'	-	NEW [(2)9/#14, (2)5/#14, (3)5/#14, (3)1/#6 GRD], EXISTING [(3)16/#14, (8)4/#18, (4)9/#14]
*4	1	2.5"	15'	-	NEW [(1)1/#6 GRD], EXISTING [(1)9/#14]
5	1	2.5"	15'	T	NEW [(1)9/#14, (1)5/#14, (1)1/#6 GRD],
*6	1	2.5"	130'	-	NEW [(1)1/#6 GRD], EXISTING [(1)4/#18]
*7	1	2.5"	50'	-	NEW [(1)9/#14, (1)5/#14, (2)5/#14, (1)1/#6 GRD], EXISTING [(2)16/#14, (3)4/#18], REMOVE [(1)9/#14]
*8	1	2.5"	45'	-	NEW [(1)9/#14, (1)5/#14, (2)5/#14, (1)1/#6 GRD], EXISTING [(2)16/#14, (2)4/#18], REMOVE [(1)9/#14]
*9	1	2.5"	10'	-	NEW [(1)9/#14, (1)1/#6 GRD], EXISTING [(2)16/#14, (2)4/#18]
10	1	2.5"	25'	O	NEW [(1)5/#14, (1)1/#6 GRD]
*11	1	2.5"	80'	-	NEW [(1)9/#14, (1)1/#6 GRD], EXISTING [(1)16/#14, (1)9/#14, (4)4/#18]
12	1	2.5"	20'	O	NEW [(1)9/#14, (1)1/#6 GRD]
*13	1	2.5"	5'	-	NEW [(1)1/#6 GRD], EXISTING [(1)9/#14]
*14	1	2.5"	10'	-	NEW [(1)1/#6 GRD], EXISTING [(1)16/#14, (1)4/#18]
*15	1	2.5"	125'	-	NEW [(1)1/#6 GRD], EXISTING [(2)4/#18]
*16	1	2.5"	125'	-	NEW [(1)1/#6 GRD], EXISTING [(1)4/#18]
17	1	2.5"	10'	T	NEW [(1)5/#14, (1)1/#6 GRD]

NOTE: ALL CONDUIT SHALL BE SCHEDULE 80 PVC UNLESS OTHERWISE NOTED.
* EXISTING OR CONSTRUCTED IN EARLIER PHASE
B=BORE, T=TRENCH, O=OPEN CUT

ADDENDA / REVISIONS

INSTALL 6' PEDESTAL POLE WITH PUSHBUTTON 2/2/23 (J. SMITH)



DELAWARE AVENUE SEPARATED
BIKEWAY- ORCHARD ROAD
TO LIBRARY AVENUE

CONTRACT

T201801501

COUNTY

NEW CASTLE

PERMIT NO.

N415T

DESIGNED BY:

J. SMITH

CHECKED BY:

M. WAHED

SIGNAL PLAN
S MAIN ST AND
AMSTEL AVE

SECTION

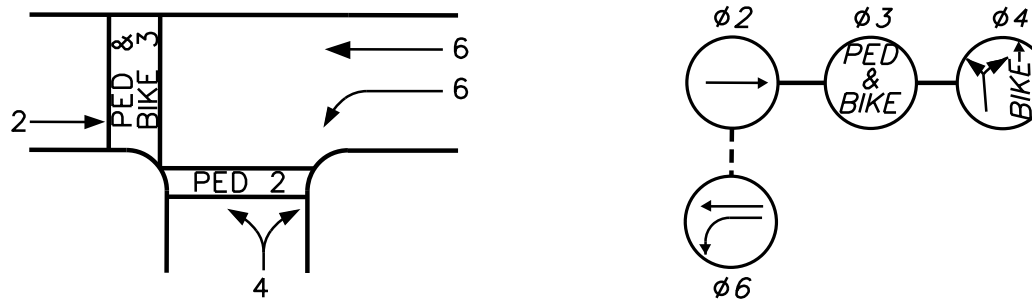
JMT

SHEET NO.

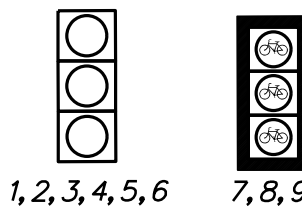
204

SIGNAL PHASING

NEMA PHASING



SIGNAL HEAD DIAGRAM



LEGEND

EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.
LOOP DETECTOR, TYPE 1	
LOOP DETECTOR, TYPE 2	
LUMINAIRE	
MAST ARM	
MICROWAVE DETECTION	
OPTICOM RECEIVER	
OVERHEAD SIGNING	
PEDESTRIAN POLE/BASE	
PEDESTRIAN PUSHBUTTON	
PEDESTRIAN SIGNAL HEAD	
RIGHT-OF-WAY	R/W
SERVICE PEDESTAL	
SIGNAL CABINET	
SIGNAL HEAD	
SIGNAL POLE/BASE	
CCTV CAMERA	
SPAN WIRE	
UTILITY POLE	
VIDEO DETECTION	

GENERAL SIGNAL NOTES

- ALL SIGNAL EQUIPMENT REMOVED FROM A PROJECT IS TO BE RETURNED TO DELDOT TRAFFIC - DOVER, DELAWARE.
- ALL UNDERGROUND AND OVERHEAD UTILITIES SHOWN ON THESE PLANS ARE SCHEMATIC ONLY AND MAY NOT BE COMPLETE. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING MISS UTILITY, AND/OR THE APPROPRIATE UTILITY PRIOR TO THE BEGINNING OF CONSTRUCTION FOR THE UTILITY MARKOUTS. IF THE CONTRACTOR PERCEIVES THAT A CONFLICT BETWEEN UTILITIES AND THE TRAFFIC SIGNAL WILL OCCUR, THE CONTRACTOR SHALL NOTIFY DELDOT TRAFFIC IMMEDIATELY BEFORE CONSTRUCTION.

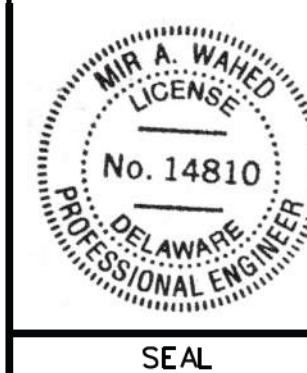
PREPARED BY

JMT

Mir A. Wahed

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

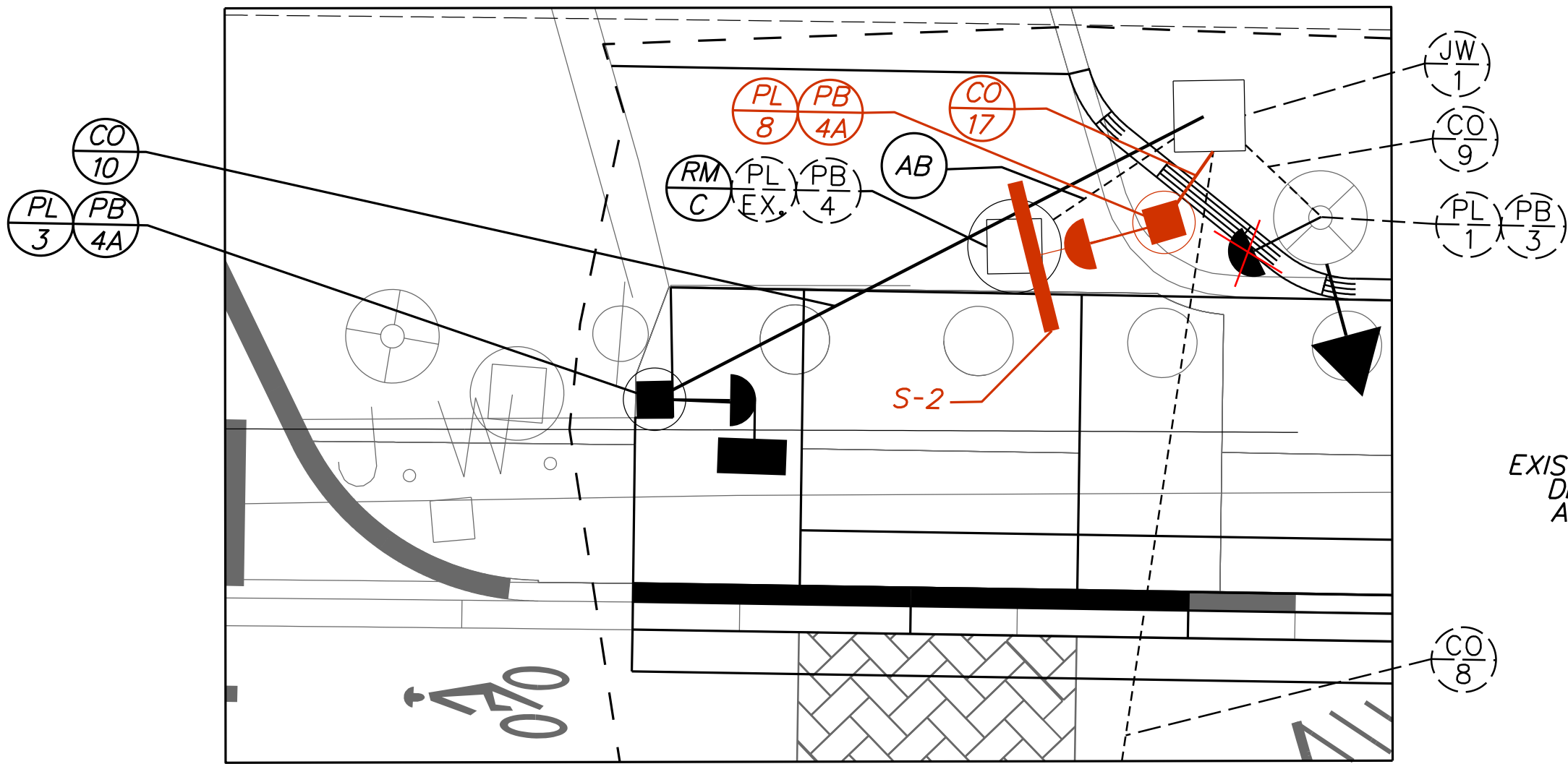


CONCURRENCE FOR INSTALLATION

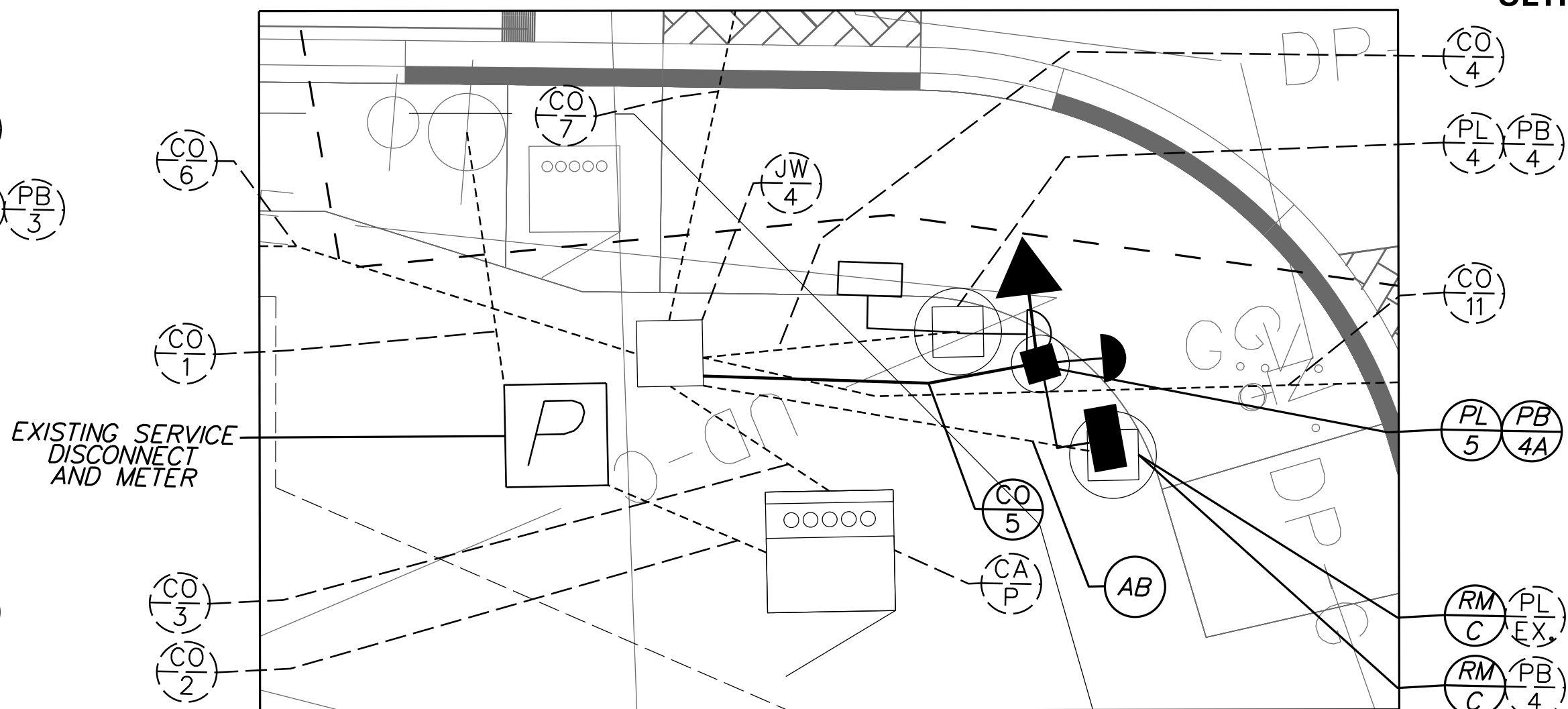
CHIEF TRAFFIC ENGINEER

7/28/2021

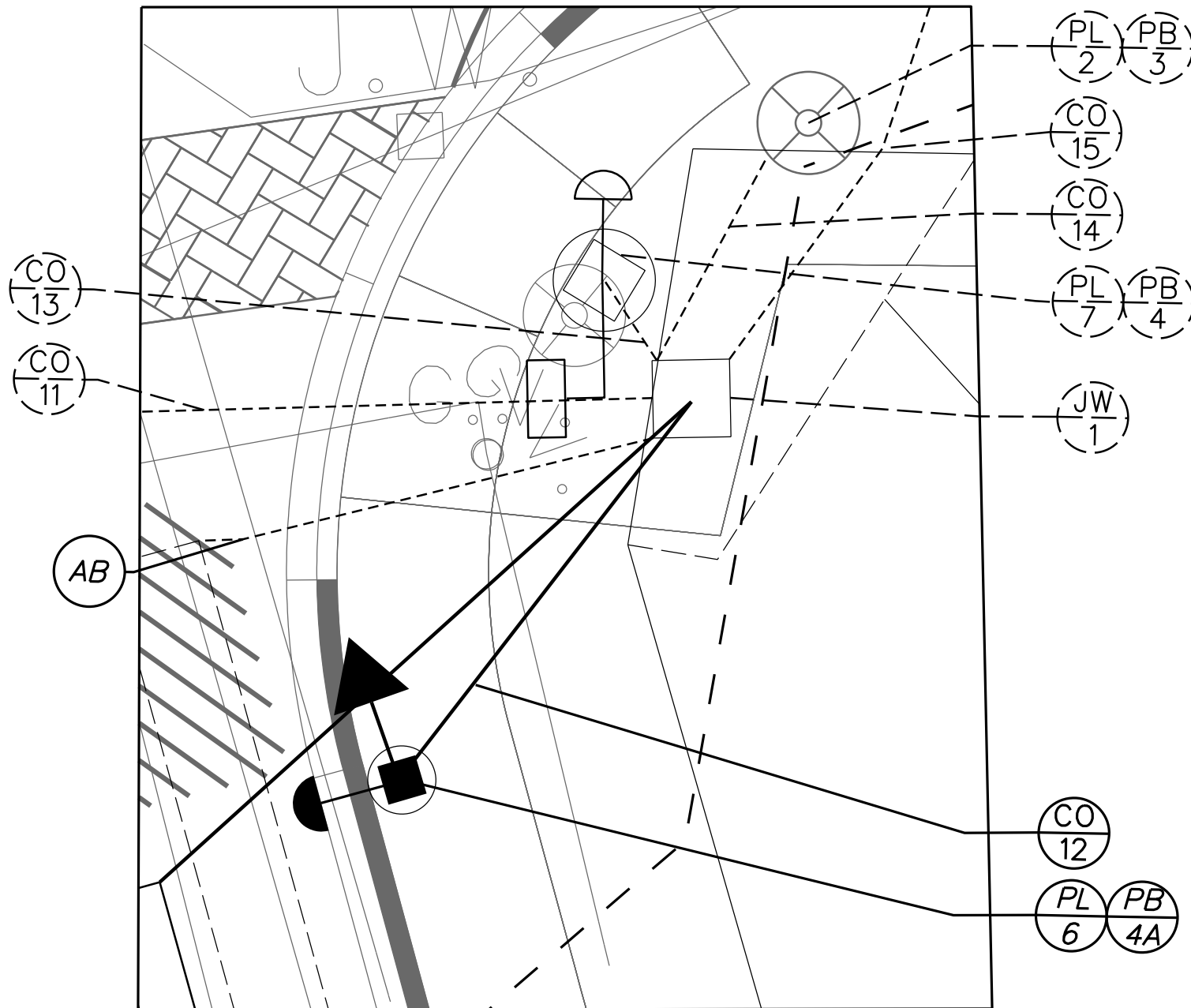
DATE



DETAIL 1 –
 SOUTHWEST CORNER
 SCALE: 1"=5'



DETAIL 2 –
 SOUTHEAST CORNER
 SCALE: 1"=5'



DETAIL 3–
 NORTHEAST CORNER
 SCALE: 1"=5'

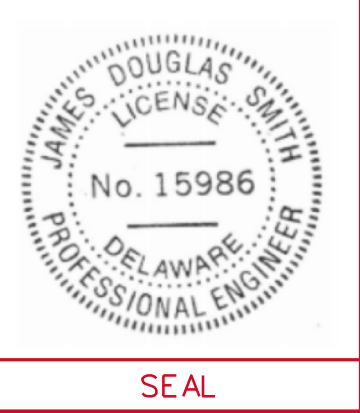
PREPARED BY
 JMT

James D. Smith, Jr.

02/02/23

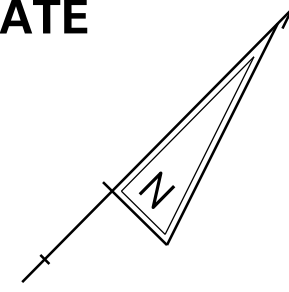
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THIS SEAL APPLIES TO ONLY THIS SHEET

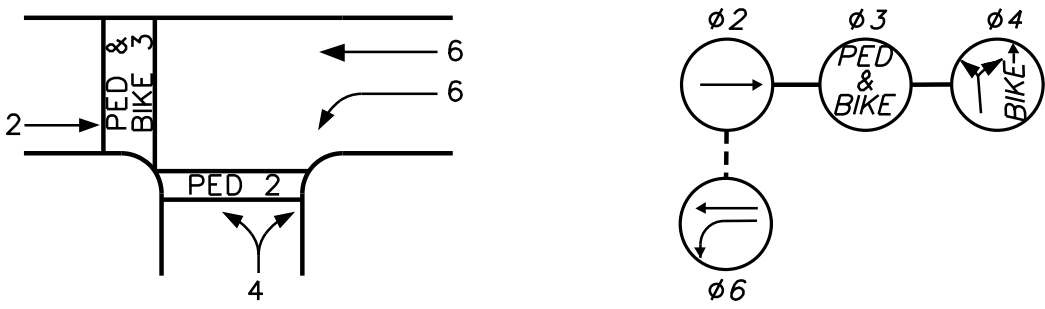


SEAL

ULTIMATE

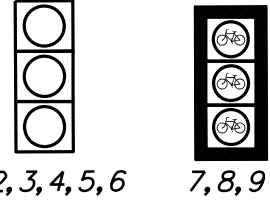


SIGNAL PHASING



- PHASING NOTES
1. PHASES ASSOCIATED BY A SOLID LINE WILL NOT OPERATE CONCURRENTLY.
 2. PHASES ASSOCIATED BY A DASHED LINE WILL OPERATE CONCURRENTLY.

SIGNAL HEAD DIAGRAM



LEGEND

(AB)	ABANDON	(PL)	EXISTING POLE IDENTIFIER (* OF POLE)
(CA)	EXISTING CABINET IDENTIFIER (TYPE OF CABINET)	(OP)	PROPOSED OVERHEAD RUN IDENTIFIER (* OF OVERHEAD RUN)
(CA)	PROPOSED CABINET IDENTIFIER (TYPE OF CABINET)	(PB)	EXISTING POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	EXISTING CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PB)	PROPOSED POLE BASE IDENTIFIER (TYPE OF POLE BASE)
(CO)	PROPOSED CONDUIT RUN IDENTIFIER (* OF CONDUIT RUN)	(PL)	EXISTING POLE IDENTIFIER (* OF POLE)
(JW)	EXISTING JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(PL)	PROPOSED POLE IDENTIFIER (* OF POLE)
(JW)	PROPOSED JUNCTION WELL IDENTIFIER (TYPE OF JUNCTION WELL)	(RM)	REMOVE BY CONTRACTOR
(MA)	EXISTING MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY OTHERS
(MA)	PROPOSED MAST ARM IDENTIFIER (LENGTH OF ARM)	(RM)	REMOVE BY TRAFFIC CONTRACTOR

	EXISTING SYMBOL	PROPOSED SYMBOL
JUNCTION WELL	J.W.	■
LOOP DETECTOR, TYPE 1	□	□
LOOP DETECTOR, TYPE 2	□	□
LUMINAIRE	⬢	⬢
MAST ARM	⬢	⬢
MICROWAVE DETECTION	⬢	⬢
OPTICOM RECEIVER	⬢	⬢
OVERHEAD SIGNING	⬢	⬢
PEDESTRIAN POLE/BASE	⬢	⬢
PEDESTRIAN PUSHBUTTON	⬢	⬢
PEDESTRIAN SIGNAL HEAD	⬢	⬢
RIGHT-OF-WAY	---	R/W
SERVICE PEDESTAL	⬢	⬢
SIGNAL CABINET	⬢	⬢
SIGNAL HEAD	⬢	⬢
SIGNAL POLE/BASE	⬢	⬢
CCTV CAMERA	⬢	⬢
SPAN WIRE	XX	XX
UTILITY POLE	⬢	⬢
VIDEO DETECTION	⬢	⬢

GENERAL SIGNAL NOTES

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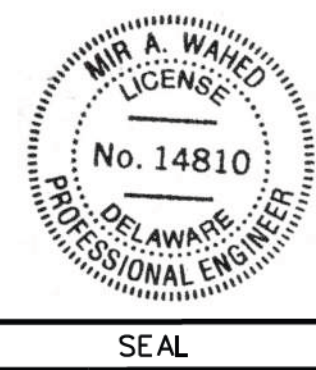
PREPARED BY
 JMT

Mir A. Wahed

07/20/2021

THIS SEAL APPLIES TO ALL SHEETS BEARING THE "JMT" SECTION DESIGNATION.

DATE



SEAL

CONCURRENCE FOR INSTALLATION

CHIEF TRAFFIC ENGINEER

7/28/2021

DATE

ADDENDA / REVISIONS

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SIGNAL PLAN
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 AMSTEL AVE

SECTION

JMT

SHEET NO.

205